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## MARKETS
**FREE FALL**

## BCI
**THE OIL RISK**

## NSE
**LISTING AND AFTER**

# INDIA'S SPACE BET

**A SLEW OF START-UPS HAS TURNED ISRO'S KNOWHOW INTO A NEW INDUSTRY. THE NEXT CHALLENGE: FINDING CUSTOMERS**

AI IMAGE

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Source: Global E-waste Monitor 2024

# A New Space Odyssey

For millennia, Indians have looked to the heavens, charting the movements of stars and planets. Through astrology, they have sought clues to what the future might bear.

Today, however, a new generation of Indians is looking skywards with a different ambition—not to divine the future, but to venture into the cosmos.

The successful launch of Vikram-1 by Skyroot Aerospace, India's first space-tech unicorn, from Sriharikota in July was a milestone for the country's private space industry. As the four-stage launch vehicle, built with an all-carbon-composite structure, blazed into the sky, it carried with it the ambition and can-do spirit of a new generation of founders.

This comes after six decades of public investment in space science and technology, and a series of sweeping reforms beginning in 2020 that opened the sector to private enterprise. The results are visible. India now has around 450 space start-ups, developing everything from rockets and satellites to advanced propulsion systems and space-based services. But these breakthroughs are only the beginning. As *Pradip K. Saha* writes in this issue's two-part cover package, the bigger challenge will be turning early successes into sustainable, profitable businesses.

Elsewhere, *Surabhi* examines India's increasingly worrying jobs challenge. The sad reality is that too many educated young Indians are struggling to find secure and reasonably well-paid work. Fresh graduates complain that entry-level jobs are scarce, and often pay so little that they live hand-to-mouth. Meanwhile, recruiters point to a dearth of skilled hands. Little wonder, then, that the lure of a *sarkari* job remains undiminished, especially as entry-level government jobs pay more than the private sector. Rapid advances in artificial intelligence and the ongoing US H-1B visa clampdown are only compounding the challenge.

Against this backdrop, question marks persist over India Inc's willingness to invest and hire. The quarterly *Business Today*-C Fore Business Confidence Index for October-December offers little reassurance that companies are ready to loosen their purse strings. Only 11% of the 500 C-Suite executives surveyed say they plan festive-season hiring, while a paltry 6% intend to increase capital spending.

Do note that the survey was conducted before the Reserve Bank of India raised interest rates for the first time in nearly four years. Crude oil above $100, rising global bond yields, and persistent selling by foreign investors have battered equities and the rupee. Despite unprecedented dollar inflows and central bank intervention, the Indian currency remains near record lows, while foreign exchange reserves have fallen by about $51 billion in just four weeks.

As we enter the festive season, consumer spending is expected to pick up. But the risks of higher inflation and slower growth continue to cloud the outlook for investment and hiring.

India's entrepreneurs are reaching for the stars, and policymakers need to help them soar even higher.

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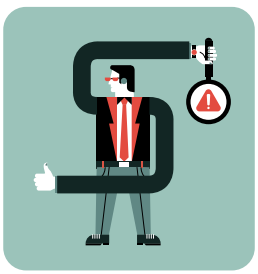
# Contents

OCTOBER 25, 2026 | VOLUME 35 | NUMBER 22


6
THE POINT

**India Inc Fires Up**

The private sector has emerged as the biggest contributor to India's new project pipeline



12
ECONOMY

**BCI: Cautious Confidence**

India Inc's confidence improves, but businesses remain on their guard

Q&A

16
**"Private sector capital is starting to move"**

Neelkanth Mishra, India's Executive Director at the World Bank, on his outlook for the Indian economy

18
**"No other democracy has grown as sustainably as India"**

Arvind Panagariya, Chairman, 16th Finance Commission, remains optimistic on the economy

IMAGE BY **NILANJAN DAS/AI**


COVER STORY
26

**INDIA'S SPACE BET**

India's private space companies have built the technology. Now they must find customers and achieve scale

36

**BEYOND ISRO'S BLUEPRINT**

Decades of state-led space exploration built the foundation for a private spacetech industry that must now build on it to create technology and businesses of its own

THE BT INTERVIEW | 8

**"HEALTH A STRUCTURAL, MULTI-DECADE OPPORTUNITY"**

RAVI LAMBAH OF TEMASEK ON INDIAN HEALTHCARE, AI, MEDTECH, AND MORE

COVER BY **HARDIK CHHABRA/ AI**

22
INVESTING

**After the Fall**

The Nifty's steep correction has shaken investors, but falling valuations and resilient economic growth are prompting fund managers to see opportunity amid the turmoil

46
MARKETS

**Options Beyond Options**

Following its blockbuster IPO, NSE is looking beyond its dominant derivatives business and expanding into commodities, electricity, gold, coal, and bonds to diversify revenue

54
ECONOMY

**The Young and the Restive**

India is growing fast and so are the aspirations of its youth. But there are not enough jobs for the educated young

62
ENERGY

**Cart Before the Horse?**

India's green hydrogen mission has built some capacity without creating enough demand, leaving subsidies unclaimed, projects stalled, and investors cautious due to high costs and weak export prospects

68
TECH

**Hi! My name is AI**

As digital teammates enter the workforce, they could fundamentally alter how IT services companies work and make money

76
INDUSTRY

**The Empty Cell**

Battery cell imports surge as India fails to build a component ecosystem. Can the new PLI change this?

82
DEFENCE

**Re-making of an Industry**

India's defence manufacturing ecosystem is undergoing a major transformation from import dependence towards self-reliance

86
THE GOOD LIFE: TRENDS

**Have Paws, Will Travel**

More pet parents are flying with their companions for the 'purr'fect vacation

90
BEST MANAGEMENT ADVICE

**"Leaders need to get comfortable making decisions with incomplete information"**

Industry veteran Rajesh Nambiar's piece of advice to executives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# INDIA INC FIRES UP

**The private sector has emerged as the biggest contributor to India's new project pipeline in recent years. *BT* explains:**

By **Ujjwal Thakur & Prince Tyagi**
Graphic by **Tanmoy Chakraborty**

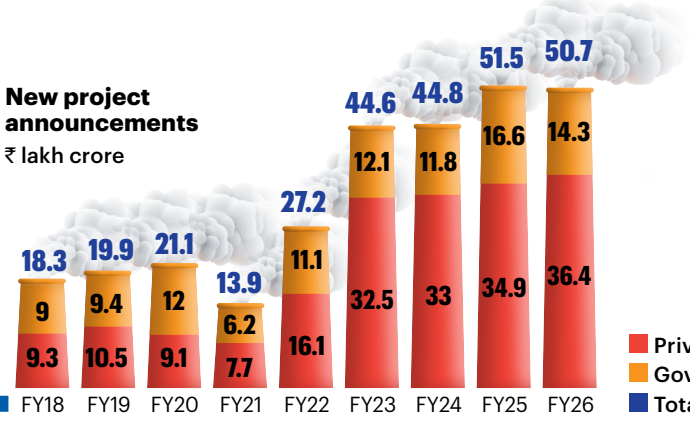
**₹4.4**
**LAKH CROORE**
Total corporate investment fuelled by private placements, FDI, and institutional bank financing

**20.4%**
Gross Fixed Capital Formation growth at current prices in Q1 FY27, surging from 5.4% in the same quarter a year ago

**₹155.4**
**LAKH CROORE**
Overall capex announcements in FY26, up from ₹95.2 lakh crore in FY23, translating into a strong 17.7% CAGR during FY23–26

## PRIVATE SECTOR DRIVES THE INVESTMENT PUSH

The private sector has emerged as the largest contributor to India's new project pipeline. Its share of project announcements rose to 71% in FY24–26, from 49% before Covid, while average annual announcements surged more than threefold to ₹34.8 lakh crore



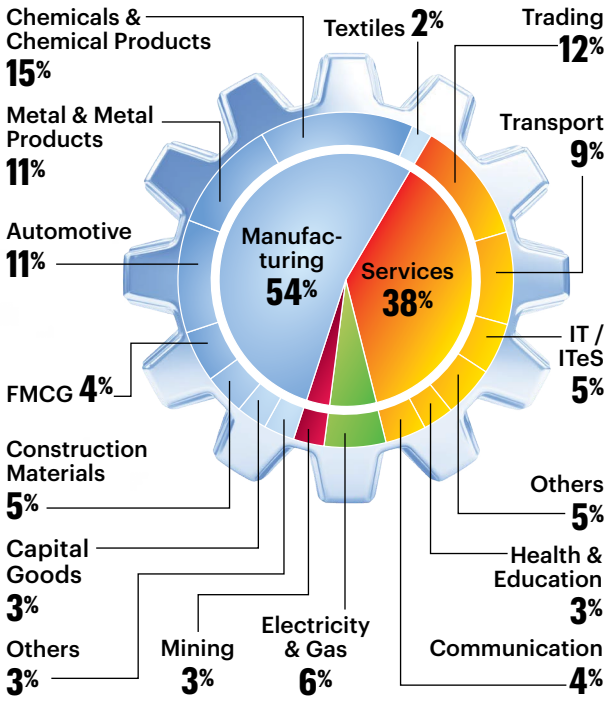
AI IMAGE

## MANUFACTURING LEADS PRIVATE INVESTMENT

Manufacturing remained the largest contributor to private capex in FY26, accounting for more than half of the total

**Private capex formation by sectors**

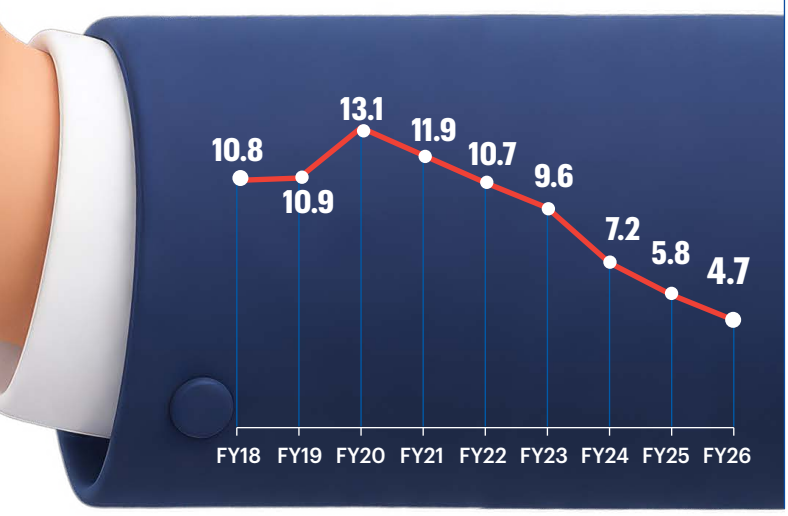
Sectoral Contribution—Addition to Total Net Fixed Assets in FY26 (%)




## FROM STALLING TO MOVING

The private investment cycle is gaining traction, with a much smaller share of projects now facing delays or stalling

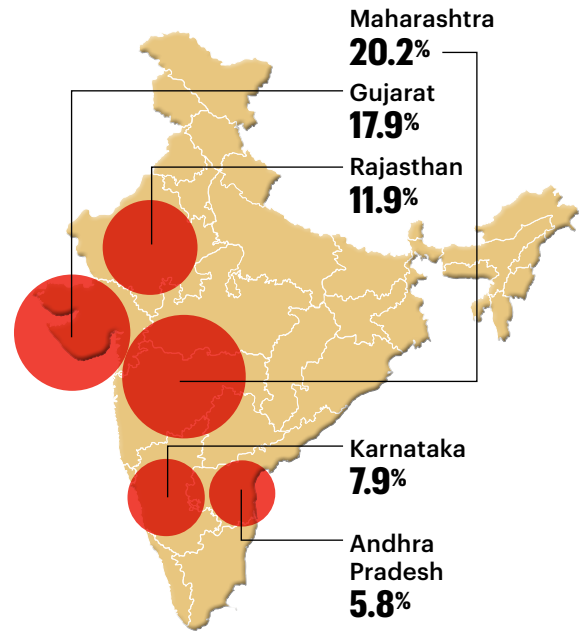
**Stalling rate of private investment (%)**




## THREE STATES TAKE THE LEAD

Maharashtra, Gujarat and Rajasthan emerged as the top destinations for private capex in FY26; the top five states accounted for 63% of private capex announcements

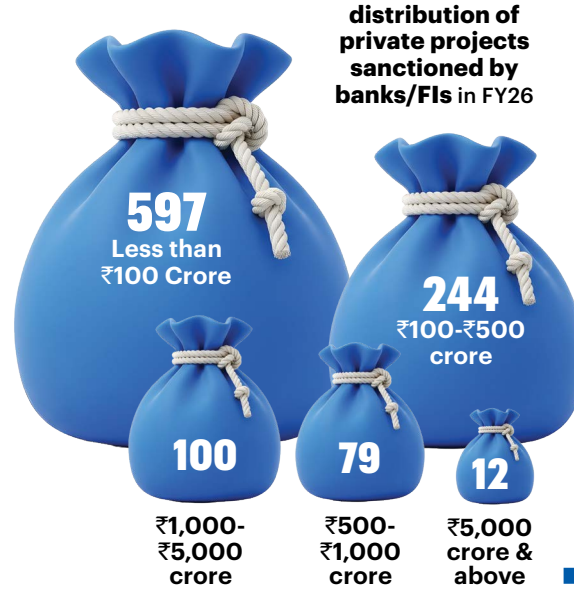
**States having highest share (%) of private capex** (FY26)




## INVESTMENT BASKET

Small projects (less than ₹100 crore) sanctioned by banks and FIs dominated by number, while larger investments accounted for the bulk of project costs, at ₹4.3 lakh crore in FY26

**Size-wise distribution of private projects sanctioned by banks/FIs** in FY26




**SOURCES:** CII RESEARCH; CMIE CAPEX DATABASE; RBI

# “Health a structural, **multi-decade opportunity**”

**Ravi Lambah, Head, Strategic Initiatives, Temasek International; Head, India, Temasek Global Investments, on Indian healthcare, AI, medtech, and more**

BY **NEETU CHANDRA SHARMA**

PHOTOS BY **MILIND SHELTE**

**Manipal Hospitals' acquisition of Pune-based Sahyadri Hospitals highlights the consolidation and expansion underway in India's healthcare sector. For Temasek, the deal is an example of how healthcare platforms can scale into strategically important geographies through mergers & acquisitions (M&As), while access to capital can support further organic expansion.**

**Temasek has invested in Indian healthcare for two decades and now counts the sector among its top three areas of investment in India. Its healthcare portfolio spans hospitals, diagnostics, pharmaceuticals, medical devices and digital health, with investments including Manipal Hospitals, Cloudnine, and PharmEasy, which has a significant diagnostics business through Thyrocare.**

**The investment firm sees a long runway for growth as rising incomes, changing disease patterns and demand for quality care drive the sector forward. It sees opportunities not only in multi-specialty hospitals but also in single-specialty care, diagnostics, affordable medical devices and digital health.**

**Ravi Lambah, Head, Strategic Initiatives, Temasek International, and Head, India, Temasek Global Investments, spoke with *Business Today* about Manipal's expansion, the next phase of growth in Indian healthcare, opportunities across the ecosystem, the role of AI, and the gaps around capacity, affordability and access that still need to be addressed. Edited excerpts:**

**Q: Temasek has invested across multiple segments of India's healthcare ecosystem. What is your view on the evolution of the sector?**
**A:** We have invested in Indian healthcare for two decades, and our conviction has strengthened as the sector has matured. The opportunity was initially anchored in under-penetration, rising incomes, and growing demand from families seeking better care. Over time, that has grown into a much broader ecosystem opportunity. That is why healthcare is a structural, multi-decade opportunity, and is now one of our top three sectors in India.

**"India has made real progress, but there is always more we can do. Capacity is one area, particularly in expanding the clinical workforce and reaching Tier-II and Tier-III cities"**

**Q: From an investor's perspective, what will define the next phase of growth?**
**A:** The next phase for us is about scaling our healthcare portfolio responsibly. The industry continues to evolve, with ongoing improvements in both access and affordability. We will work with our portfolio companies to scale organically and through M&As, focusing on building a high-quality presence across geographies. Manipal Hospital's acquisition of Sahyadri Hospitals in Pune is an example of scaling in a strategically important geography. The sector has a long runway for growth, and we will continue to engage.

**Q: Where do you see the biggest opportunities for value creation over the next decade?**
**A:** We see opportunities across the healthcare services ecosystem. In multi-specialty chains, where we have deployed significant capital, growth will be driven both organically and through consolidation.

Single-specialty models are scaling rapidly, while diagnostics and medical devices remain significant growth opportunities. We are invested in PharmEasy, which has a significant diagnostics business through Thyrocare. In medical devices, we see opportunities to scale businesses within the "Make in India" framework by accessing international IP for both domestic and global markets. Value creation will also be significantly driven by scaling and deployment of AI across the ecosystem.

**Q: What are the biggest gaps that still need to be addressed to build a stronger and more resilient ecosystem?**
**A:** India has made real progress, but there is always more we can do. Capacity is one area, particularly in expanding the clinical workforce and reaching underserved Tier-II and Tier-III cities. Affordability is another. Insurance penetration is improving, but many families still face significant out-of-pocket costs. We also need more investment in primary and preventive care. There is also a significant opportunity in medical devices and higher-value consumables supply chain.



**Q: The sector is witnessing a new wave of expansion and consolidation. How do you see the competitive landscape evolving?**
**A:** The sector has matured considerably, and we expect more organised platforms to emerge. There is also ample room for well-run single-specialty platforms to grow, especially where focused expertise and standardised care pathways can deliver strong outcomes. Our investment in Cloudnine Hospitals, which caters to the fast-growing mother and child opportunity, is a good example.

**Q: India has established itself as a global force in pharmaceuticals and is expanding its footprint in healthcare services and life sciences. Where do you see the country's next competitive advantage emerging from?**
**A:** India is already well-established in pharmaceuticals. We see the next wave of opportunity in affordable medtech, digital health, and selected areas of biotech and life sciences. India has a combination of scientific talent and manufacturing depth, as well as a large domestic market. That is needed to build products for both domestic and international markets.

**Q: Innovation is reshaping healthcare. How is it influencing investment decisions, and what opportunities does it create for Indian healthcare businesses?**
**A:** We see healthcare as an industry that will be enabled by AI, whether through improved processes, customer experience, diagnostics, or drug discovery. For India, the most compelling opportunity is whether technology can help bring down the cost of good care or extend services into communities where specialists are harder to access. We are encouraging our portfolio companies to integrate AI effectively into their business models. Given the depth of tech talent in the country, I think Indian businesses can adapt well.

**"We will work with our portfolio companies to scale organically and through M&As, focusing on building a high-quality presence"**

**Q: What policy and regulatory reforms would have the greatest impact on attracting long-term capital and accelerating the growth of India's healthcare sector?**

**A:** Policy direction has been encouraging. Ayushman Bharat continues to widen access, health spending has risen steadily, and PLI schemes alongside FDI liberalisation are drawing more long-term capital into the sector. With this foundation in place, the next priority is to sustain momentum on workforce capacity, insurance coverage, and primary care.

**Q: India's healthcare needs are changing rapidly. How should healthcare businesses adapt to these shifts?**

**A:** Delivery models will need to evolve with demand. As chronic conditions become more common, healthcare businesses will need to build more capacity in specialities where patients require ongoing management rather than one-off treatment. We also expect more investment in day-care, outpatient, and diagnostics-led formats. Elder care will also become increasingly important.

**Q: As India moves towards becoming a developed economy by 2047, what kind of healthcare ecosystem should the country aspire to build?**

**"As chronic conditions become more common, healthcare businesses will need to build more capacity in specialities"**

**A:** By 2047, we would hope to see quality healthcare become much more accessible across India, not just in the major metros, but also in Tier-II and -III cities. Care delivered through robotic systems will become the norm. That means more capacity, stronger clinical talent, better diagnostics, and a stronger primary and preventive care layer. We would also hope to see India strengthen its global position across the spectrum. BT

*@neetu_csharma*

# CAUTIOUS CONFIDENCE

INDIA INC'S CONFIDENCE IMPROVES AHEAD OF THE FESTIVE SEASON, BUT THE WEST ASIA CONFLICT, HIGH OIL PRICES, AND UNCERTAIN CONSUMER DEMAND KEEP BUSINESSES ON GUARD **BY SURABHI**

**MIXED SIGNALS**

The *BT*-C Fore BCI for Q3FY27 reveals that the conflict in West Asia and concerns over poor monsoon have kept sentiment muted

West Asia war remains the top concern in Q3FY27; most have slowed capex plans

Experts concerned over inflationary pressures from surging crude oil prices, below normal rains

Business confidence in Q3FY27 has improved compared to Q2FY27 but remains below the 50-mark

Companies hopeful that festive season will bring some improvement in demand but hiring plans are conservative



**BUSINESSES ACROSS SECTORS** and sizes are more optimistic about the economy for the upcoming October-December quarter, but the improvement in sentiment comes with a dose of caution, and little sign that companies are ready to loosen their purse strings.

The *BT*-C Fore Business Confidence Index (BCI) rose to 48.2 on a scale of 100 for the third quarter of FY27 from 44.6 in the previous quarter, driven in part by expectations of higher festive-season demand. However, it remained below the 50-mark for the third consecutive quarter, suggesting subdued corporate sentiment.

Companies are wary about the renewed conflict in West Asia, with crude oil prices trading above $100 a barrel, while deficient rains have raised concerns about rural incomes and inflation. A fresh risk has also emerged from the possibility of 100% US tariffs on countries buying Russian crude, including India.

The caution is evident in what companies plan to do. Only 11% of the 500 chief executive officers and chief financial officers surveyed plan to hire more workers for the festive season, while 14% say they are laying off workers and 52% have no plans to hire. On the investment front, just 6% plan to expand facilities or increase capital investments.

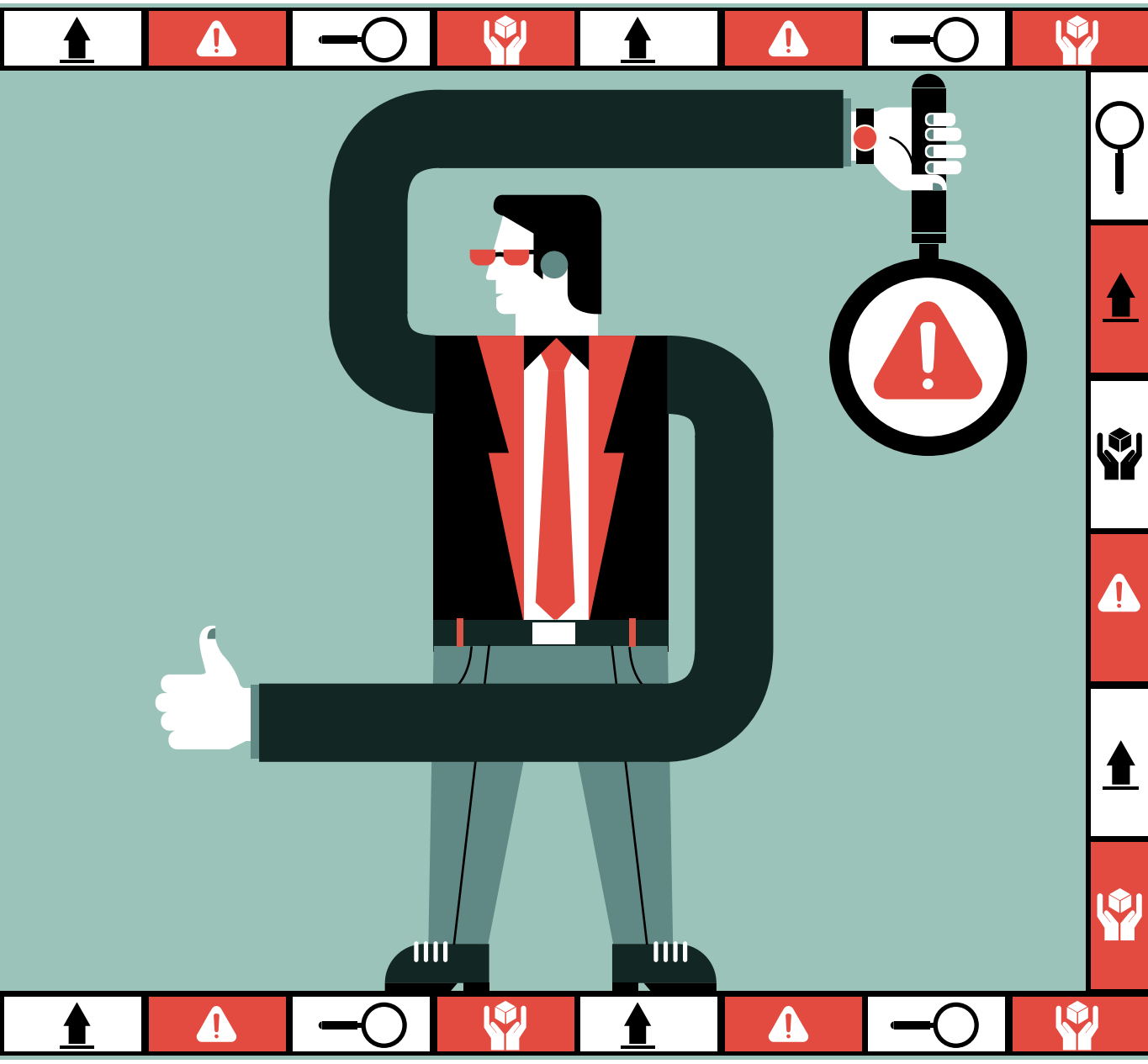


ILLUSTRATION BY **RAJ VERMA**



There is also pressure on prices. While 22% of companies surveyed have already increased prices, another 25% plan to do so. The rest remain undecided, suggesting that companies are still assessing how much of the rising cost burden they can pass on to customers.

The survey was conducted between September 9 and 18, as the festive season was getting under way with Ganesh Chaturthi. That makes the subdued confidence notable: experts say the third quarter typically brings greater optimism as companies anticipate higher sales and profitability.

## OIL SQUEEZE

Even as the West Asia conflict's fresh escalation led to a surge in global crude oil prices—Brent touched a high of $109.97 per barrel on September 11 and remained largely above $100 per barrel— the average price of the Indian crude oil basket also shot up as supply constraints and higher transport costs impacted prices. The Indian basket, which had averaged $82.04 a barrel in July— the lowest this fiscal—averaged $116.09 in September, an increase of 41.5%. It had risen further to $117.58 per barrel by October 6.

"It takes about 45-60 days in terms of shipping time, leading to higher transportation costs. It has been difficult to find tankers and

94% of oil imports are through foreign vessels. Insurance costs have also risen, all of which have added to the higher price of India's crude oil basket," says energy expert Narendra Taneja, while noting that India has diversified crude oil imports beyond the Gulf countries.

The impact is visible in the BCI survey: 54% of businesses identified the West Asia war as their top concern going into the third quarter, and nearly two thirds remained impacted by it to varying degrees. While 27% said they remain somewhat impacted by the conflict in terms of exports and imports, 45% reported being highly impacted with loss of orders and higher costs.

### GROWTH OUTLOOK

Despite the caution, businesses remain relatively optimistic about economic growth this fiscal year, with 41% of those surveyed saying that the economy will do better than anticipated in FY27 following the 7.8% GDP growth in the first quarter.

Agencies like the World Bank, Asian Development Bank, and the OECD have scaled up their growth forecasts for India for FY27 to 7% and even higher in recent weeks, but it remains below the 7.8% real GDP growth in FY26.

Madan Sabnavis, Chief Economist at Bank of Baroda, also expects the economy to grow by 7-7.2% this fiscal and underlines that the economy is much bigger in size compared to a decade ago and better placed to absorb shocks. However, while Q1 and Q2 growth are seen to be on the higher side partly due to the base effect, he says growth may slow down in Q3FY27.

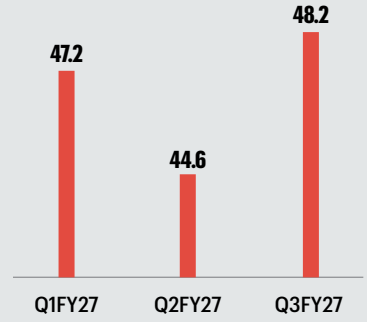
According to Sabnavis, the third quarter is a critical time as it is marked by several festivals and is also the post-harvest time that determines rural income and consumption. "With large parts of the country registering poor rains, there is apprehension about crops and income that will impact rural spending," he says.

Reserve Bank of India Governor Sanjay Malhotra, in his monetary policy



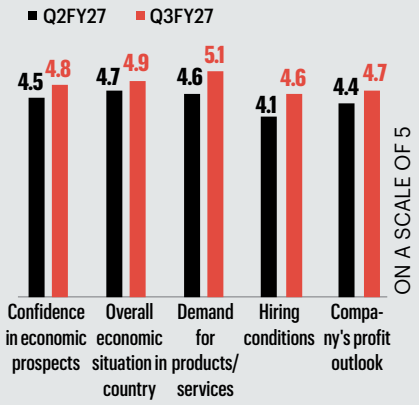
## A SLIGHT UPTICK

• Business confidence in Q3 rises sequentially but still in contractionary zone



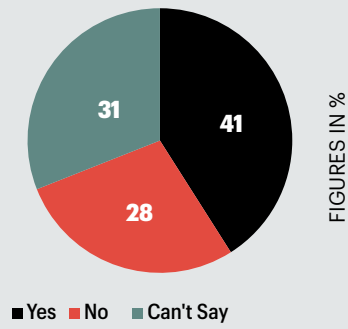

## FESTIVE CHEER?

• Season unlikely to lift mood much

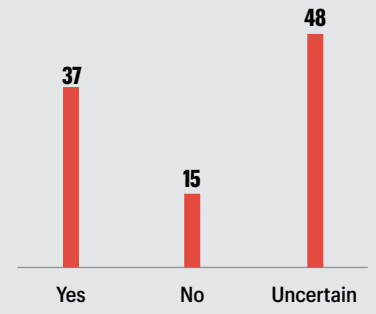



## INDIA INC SEES MANY CHALLENGES AHEAD

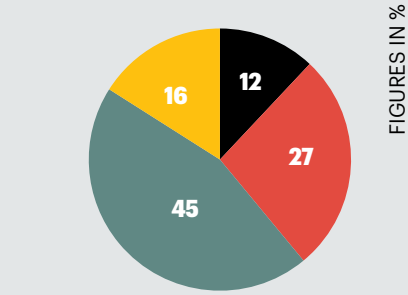
• With Q1 growth at 7.8%, do you expect the economy to perform better than anticipated in FY27?



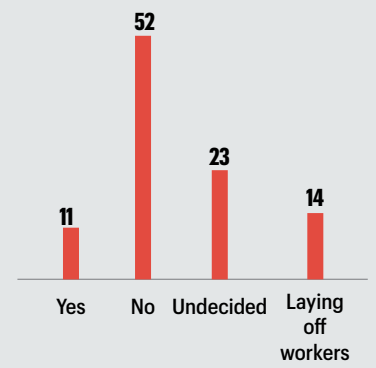

• With the festive season now starting, do you expect higher consumer demand?




• Six months since the war in West Asia began, how is your business impacted?

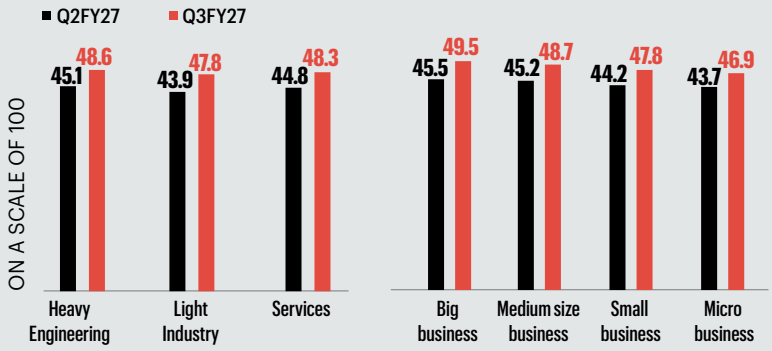



• Will you hire more workers to meet festive season demand?

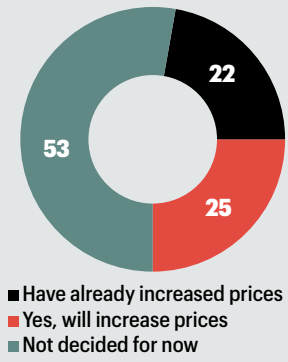


## STILL DOWNBEAT

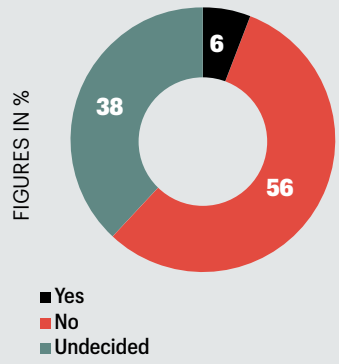
• Confidence improves across sectors, size of industries, but remains belowe 50



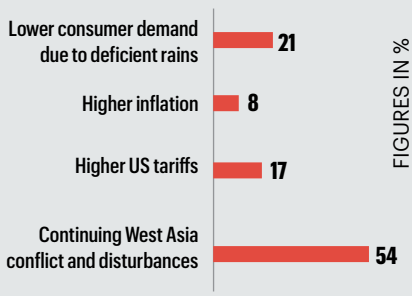

**Are you looking at a price hike going forward?**



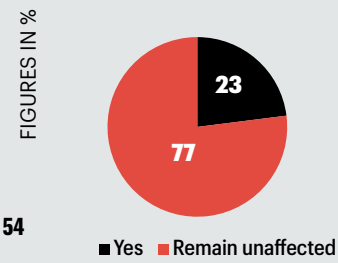

**Are you planning expansion in facilities and capital investments?**




**What is your top concern for the third quarter (October- December) of the fiscal year?**




**Have the new free trade agreements, including the one with the UK, helped boost export orders? (This question was addressed to only those who are engaged in exports)**




**SOURCE** THE *BT*-C FORE BUSINESS CONFIDENCE SURVEY CONDUCTED BETWEEN SEPTEMBER 9 AND 18, 2026 OF 500 CEOs OR CFOs REPRESENTING VARIOUS INDUSTRIES AND COMPANY SIZES ACROSS 12 CITIES—DELHI, MUMBAI, CHENNAI, HYDERABAD, BENGALURU, KOLKATA, CHANDIGARH, LUCKNOW, NAGPUR, KOCHI, VIZAG, AND BHUBANESHWAR

statement on October 7 also underlined the resilient private consumption and strong investment activity but said global economic uncertainty and strong El Nino conditions pose concerns. However, the MPC upped the growth forecast to 7.1% in FY27 from its earlier forecast of 6.7%.

### DEMAND UNCERTAIN

Consumer demand is also keeping some businesses wary. While 37% said they expect higher demand, 21% identified lower consumer demand due to deficient rains as their chief concern for the third quarter.

Free trade agreements, meanwhile, appear to have had a limited impact on exporters' sales. Among businesses engaged in exports, only 23% said the trade deals have helped boost their sales.

Inflation, meanwhile, could add to the pressure on consumption demand. Retail inflation has been on the upward curve in recent months, rising to a 20-month high of 4.82% in August from 2.74% in January. Malhotra also noted that it is clear that inflation and its outlook are not benign as they were last year. The RBI has pegged retail inflation at 5.2% for FY27 and the MPC unanimously chose to hike the repo rate by 25 bps.

Sabnavis says there is expectation of higher inflation from deficient rains, which could impact discretionary spending. If oil prices continue to remain above $100 per barrel, the government could consider increasing retail prices, but it is unlikely to do so immediately.

Taneja believes any such decision may be taken only after mid-November. "There is talk of another US-Iran ceasefire, which will help to immediately bring down prices in the international markets," he says, adding that the government would also like to shield customers in the festive season, at least until Diwali.

A sustained easing of tensions in West Asia and greater clarity on rural incomes could determine whether India Inc's cautious stance gives way to stronger hiring and investment. **BT**

*@Surabhi_prasad*

# "Private sector capital is starting to move"

**Neelkanth Mishra, India's Executive Director at the World Bank, on his outlook for the Indian economy**

BY **SURABHI**

**THE PRIVATE CAPEX** cycle in India is beginning to move, says Neelkanth Mishra, India's Executive Director at the World Bank, as investments in sectors like steel, cement, power, and data centres gain momentum. He says 7%-plus GDP growth is achievable but also warns of headwinds from high crude oil prices and cost of capital. Edited excerpts:



▶▶▶

***With the West Asia war and a poor monsoon, do you think India's current growth trajectory is sustainable?***

Yes, I think it is sustainable. The real estate cycle has turned, and the investment cycle is turning. We are out of the deleveraging that was forced by the previous excesses in lending and the IBC (Insolvency and Bankruptcy Code). Our trend growth is well above 7%, and there is no reason why we should not be able to manage it.

There are two significant growth headwinds. First are energy prices. It does not look like the war in West Asia will end soon, and it may escalate further before de-escalating, which means energy prices will be higher for longer. The government is providing some fiscal support to cushion the impact on the economy, but depending on where oil prices average over the next 12 months, there will be some headwind. At $100-$105 dollars per barrel, it is a 2% of GDP headwind to the economy. Some of this is being offset by the government's fiscal interventions, which means GDP growth can be impacted by 0.5-1%.

**"THE REAL ESTATE CYCLE HAS TURNED, AND THE INVESTMENT CYCLE IS TURNING. WE ARE OUT OF THE PREVIOUS DELEVERAGING PERIOD"**

The second headwind is from the cost of capital. Earlier, one could finance 2-3% of GDP worth of current account deficit (CAD), but today even 1% is harder to finance, given rising yields and the global growth slowdown. But 7%-plus growth rate is still doable.

▶▶▶

***What is your take on inflation?***

Oil prices are averaging $100–105 instead of $70, and energy constitutes a significant part of the consumption basket, both directly and indirectly. It would therefore, be unwise to assume that inflation will not rise. Prices will have to increase.

Unless you expect crude oil prices to rise from $70 to $105 and then to $150, it is unlikely that this alone would warrant rate hikes. It would be viewed as a step-change adjustment in the process. The question is whether this will significantly affect consumption. That is how the growth headwind manifests itself: consumers are paying considerably more for the same volume of goods they previously purchased.

PHOTO BY **HARDIK CHHABRA**

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***Equity markets and foreign investors do not seem to reflect the optimism surrounding the fundamentals of the Indian economy. Why is that?***

The stock market and the economy are related only broadly. There is no consistent correlation. The headline indices are not behaving that well, but that is not a reason to disbelieve economic growth.

The FIIs (foreign institutional investors) are a different issue. If you are an asset allocator and US 10-year yields are 5.3%, it is given that there will always be an equity risk premium. It is hard to expect that FPIs (foreign portfolio investors) will be warm towards India because India is not an asset class yet.

If interest rates have gone up in the US, especially in the last two months, it is difficult to see FPIs getting flows, and flows coming to India. That has nothing to do with whether they believe in the India growth story or not.

▶▶▶

***Why has the private capex cycle still not picked up?***

In Q1 FY27, GDP and investment growth were robust. A lot of it is private. Private sector investments may not be commensurate with the capital or retained earnings that they are generating, but they are starting to invest in sectors like steel and cement. Power sector capex has picked up. Many large groups have announced expansion of data centres, although clarity on silicon prices is missing. Private sector capital is starting to move. It is not as bad as it was three years back. But the growth we saw between say 2007 and 2012-13, may not be visible in the next one or two years.

▶▶▶

***What are your priorities as India's Executive Director at the World Bank?***

The World Bank is going through transition, and several policies need to be defined and adapted. The board also has a role in allocation of capital to various geographies and products.

The second area of work is channelling World Bank capital and knowledge to the Indian economy. We have identified urbanisation, tourism, critical minerals, and agricultural exports as the four areas where we need the convening power of the World Bank, or someone who has authority and knowledge of the World Bank, to work with the state governments.

The third area of work is that in Washington D.C., the understanding of the Indian economy is a bit outdated from 15-20 years back. It is important to engage with think tanks and industry bodies on work that India has been doing in sectors like semiconductors, manufacturing, infrastructure, and availability of power. **BT**

*@surabhi_prasad*

PHOTOS BY **HARDIK CHHABRA**

# "No other democracy has grown as sustainably as India"

**Arvind Panagariya, Chairman of the 16th Finance Commission, says the crude oil situation is unlikely to get any worse**

BY **SURABHI**

**CONFIDENT ABOUT THE** fundamentals of the Indian economy, Arvind Panagariya, Chairman, 16th Finance Commission and former Vice Chairman, NITI Aayog, remains optimistic about the outlook and hopes that the economy would be able to sail through the current crisis emanating from the war in West Asia.

Panagariya, who is now a professor at Columbia University also expects benefitting from the trade pacts with the UK and the EU. He also says that the lack of a trade deal with the US will not lead to any adverse impact on India's growth. Edited excerpts:

▶▶▶

***The Indian economy has been performing well, with 7.8% growth in Q1FY27 despite the West Asia war. Do you think this can be sustained?***

We have sustained this growth for quite a while despite several shocks. There were at least three to four shocks since 2003-04, including the global financial crisis shock. Then in 2011-12 and 2012-13, when we were amongst the 'Fragile Five' economies, followed by the NPA shock and the Covid-19 pandemic.

In spite of all this, India's GDP growth rate over this almost 20–21-year period was reasonably close to 7%. My calculations in real dollar terms

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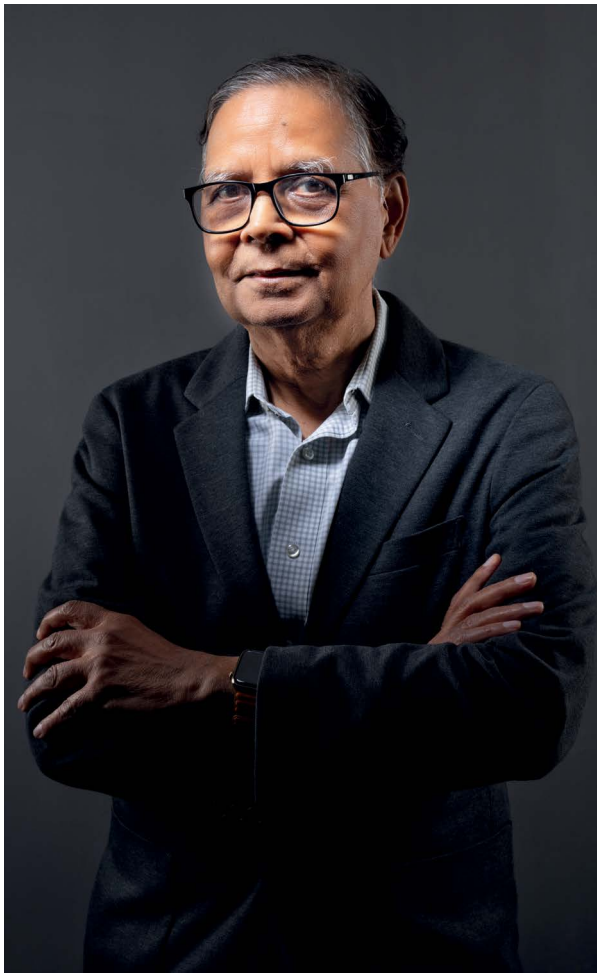


**"THERE IS SOMETHING FUNDAMENTAL WHICH IS ROBUST ABOUT THE ECONOMY. IT IS A MATTER OF RIGHT POLICIES WHICH BEGAN AFTER 1991"**

places the growth rate well above 7%.

There is something fundamental which is robust about the economy. It is a matter of right policies which began after 1991.

We have continued on the growth path and progressively also tackled some of the reforms that, at one point, did not seem possible—such as the Insolvency and Bankruptcy Code, the goods and services tax, and the labour laws. If we continueon this path, we can sustain this growth. Our reform process also has been slower because of the democratic pushbacks and so our growth has not been as rapid as that of countries like South Korea, Hong Kong, Singapore, Taiwan, China.

But if one only looks at the democracies, we have beaten everyone.

I don't think there is any other democracy which has grown faster than India for sustained periods like this.

▶▶▶

***Crude oil with prices over $100 per barrel remains a be big shock for India. How do you see it panning out?***

*Ex ante*, I would have also predicted that it might shave off a percentage point of growth. But now that we have seen it, the economy has handled it quite well, and we have been able to maintain 7% plus growth, despite oil price staying high. That makes me a little more optimistic. Probably now, as far as oil is concerned, things will not get any worse. Currently, it looks like the West Asia war is not going to intensify in any major way.

▶▶▶

***How do you see the trade deals India has signed helping the export sector?***

It is still a little bit early, although some people say that impact is beginning to show.

In my view, the two big free trade agreements India has signed are the ones with the United Kingdom and the European Union (EU). The deal with the EU is disproportionately large but the agreement is yet to be signed. I am very hopeful that it will have a very significant impact.

▶▶▶

***Have the recent concerns around the balance of payment and financing current account deficit been addressed effectively?***

If you look at the current account deficit, it's not large. It should not be a problem.

▶▶▶

***US Trade Representative Jamieson Grier has said the India-US bilateral trade pact is not imminent. Does that put India at a disadvantage?***

It's status quo. It's an advantage which would have accrued but will not accrue for now. That's all. A lack of a deal doesn't make things worse, but the deal would make it better. I see no adverse impact from it in terms of economic growth. BT

@surabhi_prasad

# From 9th-Grade Dropout to ₹2000-Crore Empire

## The Rise of Entrepreneur Amit Majithia

**Amit Majithia**
Founder Of Tashan Club Chains And Tashan Bolly Studio Music Lable.

From 9th-Grade Dropout to ₹2000-Crore Empire: The Rise of Entrepreneur Amit Majithia

True entrepreneurship is defined by grit, vision, and real-world impact. Amit Majithia embodies this ethos—transitioning from a 9th-grade dropout to the architect of a ₹2000-crore brand ecosystem. As the founder of Tashan Club Chain, Tashan Bolly Studio, CBTFMyTube.com, and the self-funded CBTF Foundation, Majithia has created a footprint across hospitality, digital entertainment, media production, and ground-level social reform.

**BUILDING A ₹2000-CRORE MULTIVERSE**

Without traditional academic degrees, Majithia navigated the commercial landscape using market foresight and decisive action. His portfolio spans key growth industries:

- Hospitality & Nightlife: Leading the Tashan Club Chain, he built a dominant presence in premium lifestyle and nightlife venues across major international hubs, including Dubai.
- Music & Media Production: Through Tashan Bolly Studio and associated entertainment channels, Majithia established a dedicated platform producing mainstream music, Bollywood content, and talent-driven projects.
- Digital Video Ecosystem: He spearheaded CBTFMyTube.com, a specialized video-sharing platform tailored for cricket statistics, match analyses, news, and digital media stream hosting.

Today, his commercial reach is backed by an online community of over 31 million followers across social media platforms, cementing his role as a prominent content creator and brand builder.

**REBUILDING LIVES: THE ₹1.66-CRORE ASSAM FLOOD RELIEF PROJECT**

Beyond commercial ventures, Majithia's defining work lies in direct community action. When severe flooding devastated rural communities in Assam, leaving families displaced and helpless, he launched a structured, long-term rehabilitation effort.

Rather than offering temporary aid packages, Majithia mobilized ₹1.66 crores with a network of 20 close friends to execute a complete village adoption initiative:

- 124 Complete Homes: Constructed 124 durable houses from the ground up, equipped with essential household items and furniture to help families restart their lives.
- Sanitation & Water Infrastructure: Installed private handpumps and sanitary toilet facilities for every individual household to ensure long-term health and safety.
- 90-Day Execution: The entire village transformation was completed within a strict 90-day window, setting a operational benchmark for disaster recovery projects.

**CBTF FOUNDATION: A SELF-FUNDED COMMITMENT**

At the core of his philanthropic endeavours is the CBTF Foundation (cbtffoundation.com), operational since 2018. Driven by a strict no-public-donation policy, the NGO is 100% self-funded directly through Majithia's business revenue.

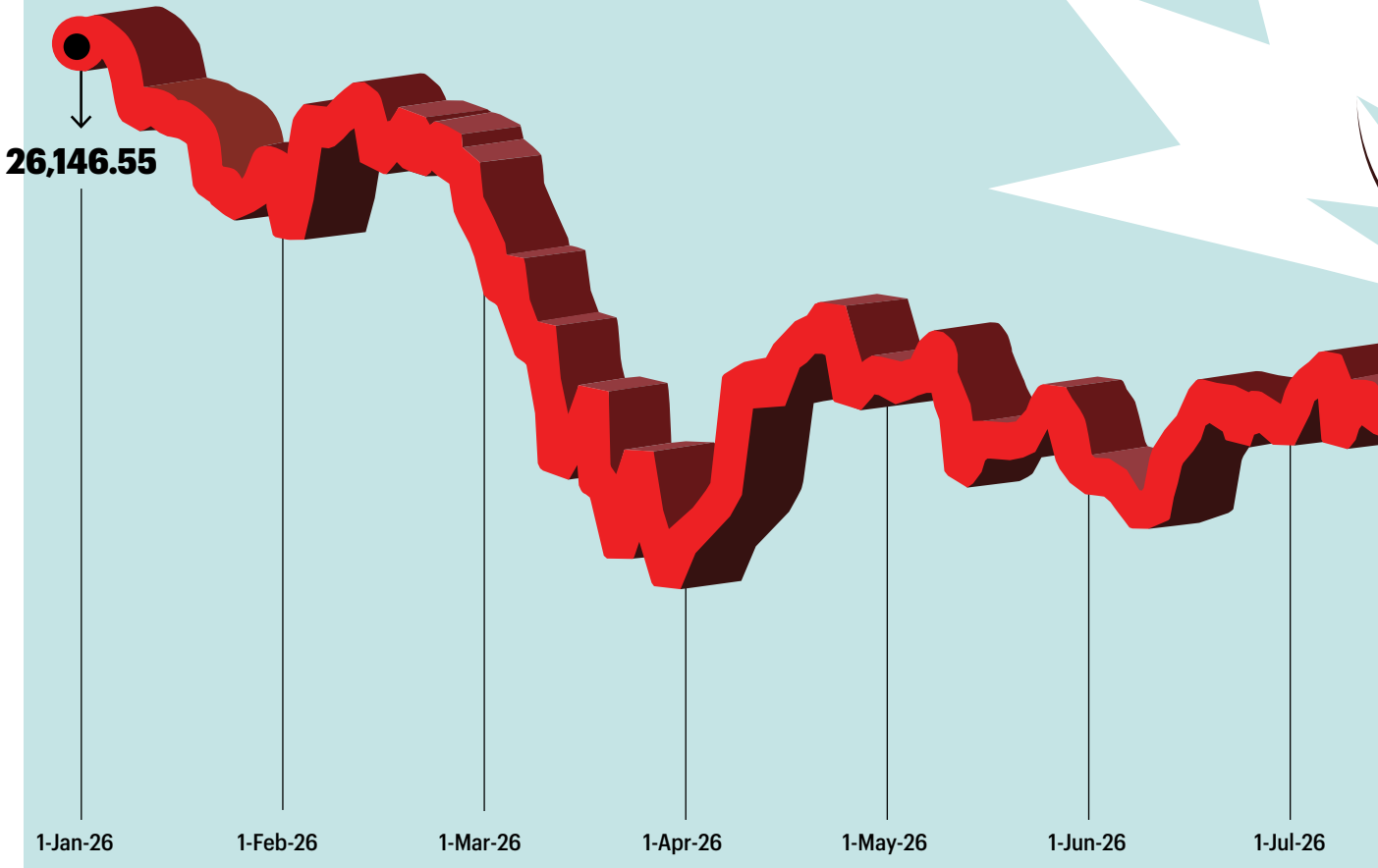
| Key Aspect | Details |
|---|---|
| Funding Model | 100% Self-Funded (Zero public donations accepted) |
| Annual Disbursal | ₹5+ Crores allocated yearly to direct welfare programs |
| Core Focus | Disaster recovery, child education, and healthcare infrastructure |

Amit Majithia continues to demonstrate that real entrepreneurial success is measured not just by brand valuation, but by the tangible positive impact created in people's lives.

# AFTER THE FALL

**The Nifty's steep correction has shaken investors, but falling valuations and resilient economic growth are prompting fund managers to see opportunity amid the turmoil**

BY **SHAILENDRA BHATNAGAR**




## MARKET CALL: STAY PUT

- The Nifty's forward P/E has fallen to around 17–18x from 23–24x in 2024, bringing several blue-chip stocks below their 10-year median valuations
- More than ₹2 lakh crore was raised in each of FY25 and FY26 through IPOs, diverting some retail money towards new issues while giving promoters and private-equity investors a route to cash out
- Fund managers see opportunities in a stock-picker's market, but argue against indiscriminate buying, favouring a diversified mix of equities, gold and corporate bonds for long-term investors

ILLUSTRATION BY **RAJ VERMA**

▶ **Large caps have borne the brunt of the sell-off, while small- and mid-cap indices gained 25–30% between April and September, backed by stronger earnings and margins**

▶ **Rising crude prices and US bond yields, along with heavy foreign selling, have weighed on Indian equities even as the economy continues to grow at 7.8%**

# A

**ABHISHEK SARASWAT HAS** been a regular investor in Indian equity funds for the past five years. An engineer by qualification, the Noida-based 33-year-old works with a global tech multinational corporation and has deployed about 15% of his income in monthly systematic investment plans (SIPs) in several equity schemes. "The first two years were super good," Saraswat says. "But the last 24 months have been painful—no returns."

Saraswat's sentiment echoes that of thousands of retail investors who have been left on the fence as Indian equities took a deep dive after years of strong gains. Posts on social media, conversations in dealing rooms, drawing rooms and boardrooms, around coffee machines in offices and on the street, continue to focus on a bloodbath that shows no signs of letting up.

Steadily going downhill, with only a few spurts of growth, the Nifty 50 has fallen for eight consecutive weeks—its longest losing streak in the past two decades—as sustained selling by foreign investors, rocketing US bond yields, rising interest rates and surging crude oil prices have overshadowed an economy galloping at 7.8% and made Dalal Street one of the worst-performing major markets globally (*see charts*).

Compared to its previous peak of 26,277 in September 2024, the Nifty 50 has fallen about 15% to date, is down about 20% in dollar terms so far this year, and is now threatening to test two-year lows.

For investors like Saraswat, the question is obvious: Is this the time to stay away from equities—or the time to buy? Market veterans argue it is the latter.

"I believe we are somewhere in the endgame here," Dinshaw Irani, CEO of Helios Capital, told *Business Today*. "I think it is time when one should be braver to start averaging as such, and that's what maximizes the returns every time. So, I love it."

That may sound counterintuitive when the headlines are dominated by a falling market, an expanding conflict in the Middle East, and a currency that has been hit hard —the rupee is down 6.6% this year and is the second-worst performing Asian unit so far in 2026 (*see charts*).

But for long-term investors, the sell-off has done something that the relentless rally of the past few years did not: it has brought valuations back towards more reasonable levels.

The Nifty is now trading at around 17-18 times one-year forward earnings, compared with 23-24 times in 2024. Consequently, several blue-chip stocks, including HDFC Bank and Tata Consultancy Services, have been de-rated and are trading substantially below their 10-year median valuation multiples on price-to-earnings and price-to-book ratios.

To be sure, the risks of the market falling further are very real, but the low valuations are prompting fund managers to say this is precisely why investors need to distinguish between what is happening to the market and what is happening to the Indian economy.

"The Nifty is not India," says Ashish Chaturmota, Managing Director for the PMS business at JM Financial. "The market is now segmented into two halves: large caps were overvalued and showed just 6-8% earnings growth. Therefore, they got sold. On the other hand, small-caps and mid-caps had lots of strong themes running, plus sustainable earnings growth in excess of 20%."

While large caps have been battered, small- and mid-cap indices rallied 25-30% between April and September, helped by strong earnings growth during the first quarter, expanding margins and relatively low equity floats. Fund managers rode that rally, even as foreign investors focused their selling on large caps.

According to Irani, who runs top-performing mid-cap and small-cap equity funds, those standing on the sidelines are missing the broader picture. He says Dalal Street's fall is mainly because of two factors—rising oil prices and surging US bond yields—on which investors, corporates and the government have little or no control.

"These are imported problems. Focus instead on GDP expansion and strong credit growth," he says. "This is a stock picker's market, and there are plenty of good stories."

The problem is that investors have been conditioned by a very different market. For two years, many have continued investing through SIPs even as returns have increasingly lagged fixed deposits.

At the same time, Dalal Street has seen an explosion of primary-market fundraising over the past two years —₹2,09,344 crore in FY25 (highest in three decades) and ₹2,04,713 in FY26, according to data from Prime Database. In the current fiscal, ₹1,49,572 crore has been raised till September 30, including the bumper listing of the National Stock Exchange, with issue size of ₹22,568.9 crore (*see charts*).

Retail investors, attracted by spectacular listing gains in several of these IPOs, have increasingly diverted money from existing equity holdings to participate in new issues.

The IPO rush has also allowed private-equity funds, promoters and pre-IPO investors to monetise holdings

## EQUITY FALL

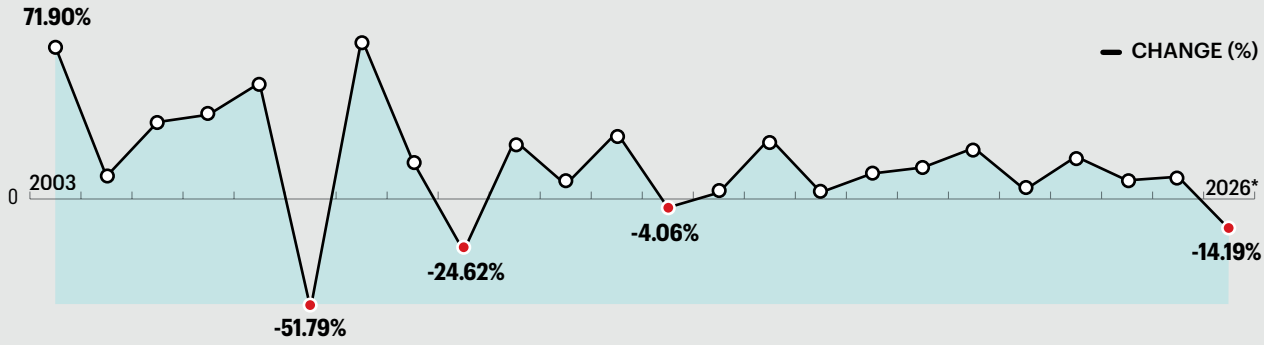
• The BSE Sensex is among the worst performing indices in 2026

| RANK | INDEX | RETURNS |
|---|---|---|
| 1 | Kospi Index | 74.91% |
| 2 | Ngx All Shr | 74.39% |
| 3 | Taiwan Taiex Index | 63.83% |
| 4 | Gse Composite Index | 44.00% |
| 5 | Msci Colcap Index | 39.95% |
| 6 | Nikkei 225 | 35.29% |
| 7 | Tunis se Tunindex | 32.77% |
| 8 | Nairobi All Share | 30.98% |
| 9 | Budapest Stock Exch Indx | 30.67% |
| 10 | Msx30 Index | 29.27% |
| 11 | Hang Seng China ENT Indx | -8.52% |
| 12 | Omx Copenhagen 20 Index | -10.94% |
| 13 | Blom Stock Index | -11.58% |
| 14 | S&P Merval TR Ars | -11.64% |
| 15 | Masi Free Float Index | -12.00% |
| 16 | Psei - Philippine Se Idx | -12.97% |
| 17 | Sri Lanka Colombo All Share Index | -13.66% |
| 18 | QE Index | -14.72% |
| **19** | **Sensex** | **-21.50%** |
| 20 | Jakarta composite index | -35.29% |

**SOURCE** BLOOMBERG

## THE BOUNCE-BACK FACTOR

- Historically, the Nifty 50 has tended to resume high growth after a tepid year




**FIGURES** ARE NIFTY50 CLOSING ON 31 DECEMBER OF RESPECTIVE YEARS ***DATA** OF 2026 REPRESENTS CLOSING VALUE OF 1 OCTOBER **SOURCE** BLOOMBERG

## THE CRASH

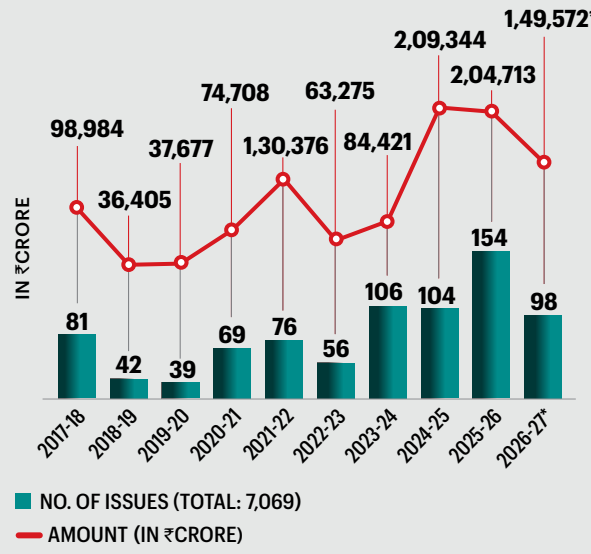
- The Indian rupee is the second-worst performing currency among Asian peers this year

| CURRENCY | SPOT RETURN |
|---|---|
| South Korean Won | 5.66% |
| Chinese Renminbi | 4.21% |
| Offshore Chinese Renminbi | 3.83% |
| Singapore Dollar | 0.39% |
| Malaysian Ringgit | -0.62% |
| Hong Kong Dollar | -0.82% |
| Japanese Yen | -1.03% |
| Taiwanese Dollar | -1.37% |
| Philippine Peso | -6.33% |
| Thai Baht | -6.44% |
| **Indian Rupee** | **-6.61%** |
| Indonesian Rupiah | -7.02% |

**NOTE** FIGURES ARE SPOT RETURNS OF ASIAN CURRENCIES VS US DOLLAR 2026 YTD **SOURCE** BLOOMBERG

## PRIMARY HARVEST

- The previous two fiscal years have seen the highest money raised from public offerings




**NOTE** *AS ON SEPTEMBER 30, 2026 **SOURCE** PRIME DATABASE

at valuations that would have been difficult to achieve in the secondary market.

The local bottling unit of Coca-Cola is the latest example, preparing to raise about $1 billion through a sale of shares by existing investors as bankers tap into strong demand for quality companies.

The result is a market where the headline Nifty number masks a much more complicated picture.

Foreign investors have pulled out about ₹4 lakh crore so far this year, with selling concentrated in large caps, as money moved towards the AI-led technology trade in markets such as South Korea, Taiwan and Japan, where major indices surged.

In this tumultuous backdrop, both Irani and Chaturmota believe the India story remains relevant and that investors need to ride the correction out.

Their message is not to buy everything simply because prices have fallen. The message is that a judicious mix of small-cap and mid-cap funds with a sprinkling of gold and corporate bonds in the overall mutual fund basket will help investors achieve long-term wealth creation goals. BT

*@shail_bhatnagar*

# INDIA'S S

## INDIA'S PRIVATE SPACE COMPANIES HAVE B
## ACHIEVE SCALE, AND COMPETE IN A

**ON AUGUST 21,** two days before National Space Day, the founders of 20 space start-ups, including Skyroot, Agnikul, Pixxel, Dhruva Space, Bellatrix, Digantara and GalaxEye, gathered at Seva Teerth in New Delhi to meet Prime Minister Narendra Modi. The meeting offered a snapshot of how far India's private space industry had come: from an idea to orbit.

It began as a working session, with actual hardware on the table. Dhruva Space showcased its P-30 platform and ground station antenna; Digantara displayed its second-generation optical payload. Around the table, the conversation turned to scale: from one rocket a month to one a week, and eventually several launches a day.

"His (PM Modi's) message to all of us was to make India the 'go-to' nation for solving space technology problems that anyone is facing anywhere in the world," Srinath Ravichandran, co-founder and CEO of Agnikul Cosmos, wrote on LinkedIn after the meeting. Modi's own message on X was that India "can offer the world a strong, innovative and commercially viable model in the space sector".

ILLUSTRATIONS BY **RAHUL SHARMA/AI**

# PACE BET

**BUILT THE TECHNOLOGY. NOW THEY MUST FIND CUSTOMERS, MARKET DOMINATED BY SPACEX** • BY **PRADIP K. SAHA**

The ambition is clear, but scale poses a difficult question: who will buy what India's space companies are building, and at what price?

The customer base is thin—India's space industry remains heavily dependent on government demand, with Isro still the primary local anchor—leaving most of the country's $8.4-billion space industry dependent on overseas orders. Skyroot, for instance, expects two-thirds of its demand to come from overseas.

However, the export pitch is not so straightforward. Among major spacefaring nations, India's average launch cost per kg is the highest, because Indian rockets are expendable and fly only a handful of times a year.

A recent study by economists Alessio Terzi of the University of Cambridge and Francesco Nicoli of Politecnico di Torino estimated India's 2025 cost of delivering a kg of payload capacity to low-Earth orbit at $13,302—the highest among the six spacefaring regions it studied. The estimate is modelled from public cost data (government reports, academic papers, public documentation)

# IN THE SLOW LANE

- Funding, launches, and policy support have accelerated, but there are few buyers
- Most companies still depend on overseas customers, particularly the US and Europe, while Indian demand remains narrow, fragmented and early-stage

standardised to 2024 dollars. The model uses launch vehicles' rated capacity, not the payload actually flown.

India's cost was more than four times the US's $3,225, and much higher than Europe's $9,897, Russia's $6,682, China's $5,809 and Japan's $5,287.

Even as India's private space industry has made remarkable progress on the hardest engineering problems, the harder test now is commercial.

According to Yashas Karanam, co-founder and COO of Bengaluru-based Bellatrix Aerospace, India has close to zero of the global commercial space market today and about 2% of the overall global market, which could reach 8-10% within a couple of years if this cohort of start-ups reaches commercialisation.

## IN PURSUIT OF A MARKET

The meeting with the prime minister came against the backdrop of the best year yet for India's private space industry: the first unicorn in Skyroot, the first private orbital launch in Vikram-1, record fundraising—private investment in the sector has risen nearly three-fold since 2024, to $349 million by September 2026, according to Venture Intelligence—and two new government schemes.

But behind the milestones sits an uncomfortable reality. Before the sector opened to private participation in 2020, Isro was the only anchor for India's space industry. There was no private market. That has not changed much.

In the early 2010s, Rohan Ganapathy and his mates at Coimbatore's Hindusthan College of Engineering & Technology were experimenting with a water-based electric thruster for small satellites as a student project. The turning point came when the then Isro Chairman A.S. Kiran Kumar saw their test results and asked them to present to Isro. The presentation led to the first order for the thruster. To fulfil that order, Ganapathy and his co-founders, including Karanam, incorporated Bellatrix Aerospace in 2015. However, over the next decade, Bellatrix found that both its suppliers and customers were outside India. Importing parts, integrating and testing them at home, then selling the finished product back to Europe or the US was not sustainable. So it started building propulsion systems in-house.

"Most of our money is coming from export orders right now, and we are in the process of scaling up manufacturing," says Karanam, adding that Bellatrix's goal is to become the world's number one or two propulsion supplier through vertical integration.

Bellatrix designs and tests its systems at its headquarters in Bengaluru and a propulsion lab at the Indian Institute of Science, and has a capacity for 250–300 systems a year. It plans to build a $76-million factory on a five-acre site at the KIADB Aerospace Park near Bengaluru airport. Karanam says doubling capacity again will take around $100 million.

Bengaluru-based Pixxel went through a similar experience. When the hyperspectral earth observation company started in 2019, it set up an entity in the US alongside the one in India.

"That's where the budgets were. We had opportunities in the US and Germany, which were countries that would give you licences if you set up some local presence there," says co-

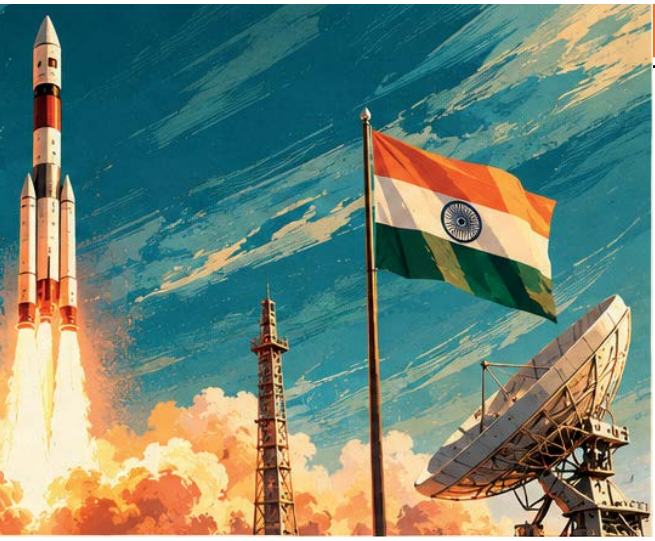
### GOVERNMENT SPACE SPENDING

**$78 BN**
**UNITED STATES**

**$19 BN**
**CHINA**

**$1.4 BN**
**INDIA**

**NOTE** DATA FOR 2025
**SOURCE** SPACE FOUNDATION



- Bellatrix abandoned launch vehicles for propulsion, while Digantara moved from debris tracking to defence, showing how companies are adapting as the market takes shape
- Reusable rockets have dramatically lowered launch costs, while SpaceX's scale in launches, Starlink and government contracts makes it difficult for smaller companies to compete head-on
- Geopolitical tensions are increasing the value of sovereign space capabilities, but Indian companies ultimately need repeat customers, large contracts and global markets to survive

## THE SPACEX SHADOW

**Total PE-VC capital raised by India's private space tech sector, 2019–2026 YTD**

**$1 billion**

**SpaceX Pentagon contracts, 2026 alone**

**$8 billion**

**India, private sector: commercial launches planned in FY27 and FY28**

**Eight**

**Space X launches in 2025**

**165**



founder and CEO Awais Ahmed. "The first thing for any space company to do is get paying customers, so that you can sustain yourself through revenues and not from raising financing."

With funding scarce at the time, Pixxel had to earn its way on revenue, and the customers were abroad. "India has opened up, but (it is) still fairly early. The US and Europe have the numbers, much larger budgets for space, even in defence and intelligence," says Ahmed.

Going global forced a harder question: how does a small Indian player win against bigger, better-funded rivals? Pixxel's answer was to avoid competing on price at all.

"Our thinking was: where is the gap that we can fill where we don't have to compete on price?" says Ahmed. That gap was found in hyperspectral imaging, which captures data across narrow wavelengths of light, including ranges invisible to the human eye.

Where an ordinary satellite can only tell a farm from open ground and read the basic health of a crop, hyperspectral data can show the nutrients in the soil, its moisture levels, a crop's species, and the fertiliser it needs. It can also detect chemical, biological, and physical changes in the earth's surface that conventional optical satellites cannot.

US space agency NASA launched its first hyperspectral imaging satellite back in 2000, but it was decommissioned in 2017. Pixxel moved to fill that gap.

"We started providing a data set that shows things to people they can't see otherwise," Ahmed says. The bet paid off. "We ended up signing a contract with NASA and NRO (US National Reconnaissance Office). And that was the vindication. Build something so unique that people anywhere will pay a premium for it."

Finding customers, however, is only the first step. The harder part is generating enough repeat demand to build a market.

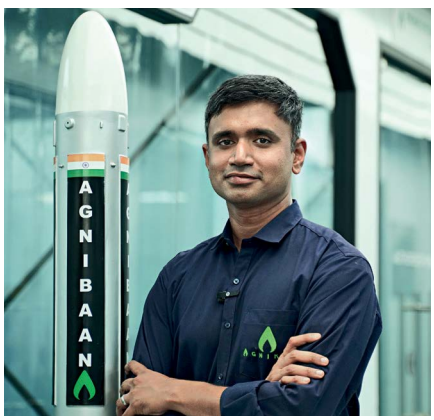
For most Indian companies, the value of space data remains difficult to translate into an immediate business case, say experts. Take agricultural lending. Satellite imagery can show what is happening to a crop, but the commercial prospect hinges on whether a

# THE KEY PLAYERS

**The companies and their founders that are taking Indian space technology global. They aim to compete on value offered rather than price alone**

## PIXXEL

**Founders:** Awais Ahmed, Kshitij Khandelwal
**Founded:** 2019
**Funding:** $208 million
**Area of work:** Hyperspectral earth observation



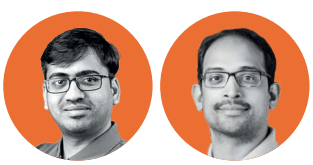

## AGNIKUL COSMOS

**Founders:** Srinath Ravichandran, Moin SPM
**Founded:** 2017
**Funding:** $69 million
**Area of work:** Launch service provider

## BELLATRIX AEROSPACE

**Founders:** Rohan M. Ganapathy, Yashas Karanam
**Founded:** 2015
**Funding:** $34 million
**Area of work:** Satellite propulsion

## SKYROOT AEROSPACE



**Founders** Pawan Kumar Chandana, Naga Bharath Daka
**Founded** 2018
**Funding** $160 million
**Valuation** $1.1 billion
**Area of work** Manufacturing orbital launch vehicles

## DIGANTARA INDUSTRIES

**Founder:** Anirudh Sharma
**Co-founders:** Rahul Rawat, Tanveer Ahmed
**Founded:** 2020
**Funding:** $65 million
**Area of work:** Space domain awareness

## ABYOM SPACETECH & DEFENCE

**Founder:** Jainul Abedin
**Founded:** 2020
**Funding:** $5 million
**Area of work:** Making reusable launch vehicles

## ASTROPHEL AEROSPACE

**Founders:** Immanuel Louis, Suyash Bafna
**Founded:** 2022
**Funding:** $800,000
**Area of work:** Developing launch vehicles

## DHRUVA SPACE









**Founders:** Sanjay Nekkanti, Krishna Teja Penamakuru, Abhay Egoor and Chaitanya Dora Surapureddy
**Founded:** 2012
**Funding:** $28 million
**Area of work:** Full-stack satellite platforms, ground stations, and launch-integration services

## EON SPACE LABS

**Founders:** Sanjay Kumar, Punit Badeka, Manoj Kumar
**Founded:** 2022
**Funding:** $1.7 million
**Area of work:** Building optical and imaging systems, electro-optical payloads

## GALAXEYE SPACE

**Founders:** Suyash Singh, Denil Chawda, Kishan Thakkar, Pranit Mehta, Rakshit Bhatt
**Founded:** 2021
**Funding:** $24 million
**Area of work:** Multi-sensor satellite imaging

## ETHEREALX

**Founders:** Manu J. Nair, Shubhayu Sardar, Prashanth Sharma
**Founded:** 2022
**Funding:** $28 million
**Area of work:** Reusable launch vehicles

lender can act on the information—expand lending, price risk better, cut defaults—by enough to justify paying for the data.

The same problem appears in infrastructure: a satellite can spot changes across thousands of kilometres of roads, pipelines or construction sites, but turning that insight into a recurring contract requires proving that acting on the information saves the customer more money than the cost of the data.

The ecosystem, as a result, is stuck in a chicken-and-egg loop. Customers are reluctant to build new workflows around satellite intelligence before its value is proven; but space companies need customers to generate the revenue and data needed to prove that value. Until that cycle is broken, government and foreign customers will remain the industry's deepest pockets.

To be sure, companies are waking up to the value of space data. Reliance Jio, for example, is exploring a sovereign satellite constellation; Airtel is integrating satellite connectivity into its telecom business; and banks and insurers are beginning to use satellite intelligence for agricultural lending and risk. However, these markets are still narrow, concentrated and emerging.



This creates a second bottleneck: growth capital. Start-ups need larger cheques as they move from prototypes to production, yet the funding pattern in Indian spacetech has been dominated by early-stage capital. Only two commercial space companies—Skyroot Aerospace and Pixxel—have been able to raise more than $100 million. Experts say capital is difficult to raise without visibility on future revenues. And revenue visibility, in turn, depends on having anchor customers.

Anirudh Sharma, founder and CEO of domain awareness startup Digantara, points to SpaceX, or Space Exploration Technologies Corp., the aerospace company founded by Elon Musk.

"Even before SpaceX launched the Falcon, they had contracts from NASA, from defence. We don't yet have that established mechanism to win contracts at a developmental stage," he says. "Today, all the development needs to be funded internally."

The government has become an "anchor customer", he says, but it is not yet consistently acting as a developmental partner that commits to buying a technology before it is proven. When a government commits to buying 30 or 40 systems once the first one works, a company can build its factory, hire engineers and raise capital against plausible future revenue. However, if there is commitment for only one system, the economics don't work. "We don't need grants at this point in time," Sharma says. "We need orders, contracts to develop capacity for the country."

Skyroot is using Siemens' Polarion to unify its software and systems development processes

In some sectors, such as defence, it is already underway. In 2024, the Cabinet Committee on Security headed by PM Modi approved ₹26,968 crore for Phase III of the space-based surveillance mission, aimed at improving land and maritime domain awareness for civilian and military applications. Under the programme, a constellation of 52 satellites is scheduled to be deployed by 2029, of which 31 are to be built by private space companies.

Then, in January this year, IN-SPACe, the Indian National Space Promotion and Authorisation Centre, an interface between the government and private space companies, signed an agreement with Allied Orbits,

IMAGE COURTESY **SKYROOT AEROSPACE**



**INDIA'S SPACE ECONOMY**

**$8.4 BN**
**IN 2022**

**$44 BN**
**BY 2033**

**GLOBAL SPACE ECONOMY**

**$630 BN**
**IN 2023**

**$1.8 TRN**
**BY 2035**

**NOTE** INDIA ESTIMATES ARE IN-SPACE PROJECTIONS
**SOURCE** WORLD ECONOMIC FORUM-MCKINSEY REPORT

a consortium led by Pixxel and including Dhruva Space, PierSight and SatSure, for the country's first indigenous, fully commercial Earth-observation constellation under the public-private partnership framework. The consortium will build, own and operate 12 satellites, investing more than ₹1,200 crore, with the network scheduled to be operational by 2029.

These are still early days, but the direction is clear. SpaceX has already shown how government contracts can help a space company build the capabilities and scale to become a commercial business. It is SpaceX that these founders most want to emulate, which, ironically, is also their biggest competitor globally.

## THE (SPACE)X FACTOR

When Bellatrix's founders were starting out, they considered building launch vehicles themselves. Then SpaceX demonstrated reusability—the first Falcon 9 booster landing in 2015, the first reflight in 2017—and the economics of the industry shifted overnight.

Karanam watched a wave of companies raise funds to build small rockets and eventually decided to focus on propulsion instead. "That's when we realised, since SpaceX is building reusable launch vehicles, companies which have raised like $50 million each, most of them are going out of business," says Karanam. "So that's when we thought, let us not do both launch vehicles and propulsion, let us stick to just propulsion."

**"We have got a lot of good traction from the global market. Most of our money is coming from export orders right now, and we are in the process of scaling up manufacturing"**

**YASHAS KARANAM**
CO-FOUNDER & COO, BELLATRIX AEROSPACE

In the era before reusability, a single launch could cost tens of millions of dollars. Falcon 9 pushed costs towards roughly $2,500–3,000 per kg, says Digantara's Sharma. "If that wouldn't have happened, I don't think we would have had any company launching satellites today," Sharma says.

And SpaceX is just getting started. It listed in June in the largest IPO in history, raising $75 billion and entering the public markets at a valuation of roughly $1.8 trillion. Last year, SpaceX reported $18.7 billion in total revenue, of which $11.4 billion came from its satellite internet constellation Starlink. In comparison, India's entire space economy is valued at around $8.4 billion, expected to grow to $44 billion by 2033.

There is good reason for the difference. In 2025, SpaceX launched 165 times, or one Falcon 9 flight every 2.2 days through the year, data from Space.com showed. That was nearly twice China's national output and accounted for more than 80% of all US launches. SpaceX now holds more than 80% of the global commercial launch market by mass to orbit.

The market could see another reset from Starship, SpaceX's fully reusable super-heavy rocket. On September 28, on its 14th test flight, it reached orbit for the first time, despite an engine failure that cut the mission short. It deployed 26 next-generation Starlink satellites, which SpaceX said added as much network capacity as about 20 Falcon 9 flights.

SpaceX has not disclosed what the flight cost, so the economics remain unproven. But in its IPO filing, SpaceX said the first Falcon 9 cut launch costs to about $2,700 per kg in 2010, against a historical average of $18,500. With Starship, it aims to cut that average by 99%. That would make the economics of small launch vehicles difficult to defend.

That's not all. By August, SpaceX had secured more than $8 billion in Pentagon contracts in 2026 alone. The response from Indian launch companies is to find another edge. Chennai-based Agnikul, for example, is attempting to bring reusability to the small-launch segment, while Bengaluru-based EtherealX is building a fully reusable medium-lift rocket aimed at the Falcon 9 class. For others, the answer is to build capabilities that India cannot afford to source elsewhere. "We do not believe the right strategy is to build a smaller version of SpaceX," says Immanuel Louis, Co-founder and COO of Pune-based Astrophel Aerospace. "SpaceX solved a particular problem at a global scale, but India still requires sovereign, locally manufactured and responsive space infrastructure."

Pixxel's Ahmed agrees: "Let's say, I want to get a detector from a foreign country, and for whatever reason geopolitically, the relationships go down. You don't want to be waiting for the detector," he says.

## WAR AND SPACE

In 2019, when Digantara was still in incubation, it presented a concept for using LiDAR to track space objects at the International Astronautical Congress in Washington, D.C. LiDAR, or light detection and ranging, uses pulses of laser light to measure the distance and movement of objects.

At the time, space surveillance was largely controlled by governments, and private participation was minimal. Buoyed by people's interest in the project, a clear gap in the market, and a ₹25 lakh grant from the Indian Institute of Science, Digantara finally launched as a business in 2020 as a space domain awareness company. In simpler words, it was a space traffic management company, tracking debris and objects that could damage satellites. There were roughly 2,500 satellites in orbit. Their pitch deck talked about navigational safety and protecting satellite operators from collisions.

Then the world changed. As the conflicts in Ukraine and then Iran unfolded, Sharma watched the global space industry pivot.

"All of these conflicts led us to evolving as a defence company. That's when we figured out our go-to-market," says Sharma. "You've seen companies in earth observation that initially started as commercial observation companies, but today they're heavily into defence, and all the data they generate from orbit goes to intelligence agencies — because they have more willingness to pay, and there is a pathway. The demand is already created." Digantara pivoted from space situational awareness to a space surveillance and intelligence company.

"If you map out the economic activity in the space industry, 70% of revenue is coming from government and defence for all companies, regardless of what they do, and 30% is coming from commercial customers and other businesses," says Sharma. "Unless you do something which can directly impact an individual's life—like Starlink providing internet to consumers—you're doing B-to-G or B-to-B."

Even before Starlink and the Falcon 9, NASA had selected SpaceX for its Commercial Orbital Transportation Services programme, providing up to $396 million to develop the Dragon spacecraft and Falcon 9. NASA later awarded SpaceX contracts worth $1.6 billion for 12 commercial resupply missions to the International Space Station.

Then Starlink became SpaceX's own captive customer, generating a steady need to put satellites into orbit. That demand allowed SpaceX to fly frequently, build launch cadence and spread costs over a much larger number of missions, helping other companies to send their payloads into orbit, eventually bringing down the cost of launch.

India doesn't have an equivalent captive customer. Its launch companies need demand to achieve the scale that would make their rockets economical, but the market is too small to provide anything like Starlink's stream of satellite orders. That leaves them with two broad routes to commercial scale: find a market abroad or help create enough demand at home.

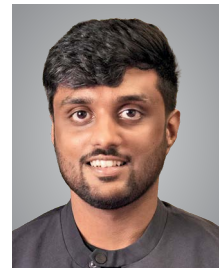
The first means competing globally. "But a smaller Indian company cannot simply offer the same product as a better-funded American or European rival and expect customers to switch," says Radha Krishna Kavuluru, a former scientist at Isro and a spacetech entrepreneur. "There are essentially two ways to win a global customer: be dramatically cheaper, or offer something sufficiently different that price is no longer the deciding factor."

Finland's ICEYE illustrates the second route. By putting synthetic-aperture radar into small satellites, it created a capability that had traditionally required much larger spacecraft. The differentiation gave customers a reason to buy its data, rather than simply choose a cheaper version of a product.

The alternative is to build the market at home. There are already pieces in the pipeline: Reliance Jio's plan for a sovereign satellite constellation and government's space surveillance mission are some examples. The largest is satellite broadband, where global players, including SpaceX's Starlink, are waiting to launch in India.

But companies need visibility beyond the programmes and tenders already on the table. "If the government knows it will need particular capabilities several years from now, publishing that direction could allow companies to develop the technologies and manufacturing capacity before orders arrive," says Kavuluru.

Companies need enough predictable demand to invest, build capacity and keep selling. The question is not simply whether India can produce another Skyroot, Pixxel or Digantara. It is whether the country can create the conditions under which these companies have enough demand to keep building, flying and selling. SpaceX had NASA first and Starlink later. India will need to find its own Starlink moment. BT



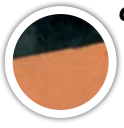
**"Even before SpaceX launched the Falcon, they had contracts from NASA, defence. We don't yet have that established mechanism to win contracts at a developmental stage"**

**ANIRUDH SHARMA**
FOUNDER & CEO, DIGANTARA

# BEYOND ISRO'S BLUEPRINT

## DECADES OF STATE-LED SPACE EXPLORATION BUILT THE FOUNDATION FOR A PRIVATE SPACETECH INDUSTRY THAT MUST NOW BUILD ON IT TO CREATE TECHNOLOGY AND BUSINESSES OF ITS OWN

BY **PRADIP K. SAHA**



***CONTROL TO ALL STATIONS***

*There is a hold observed at T-minus five minutes.*

*All stations are requested to be on standby mode."*

The countdown had stopped.

On July 18, Vikram-1, India's first private orbital-class rocket, built by Skyroot Aerospace, was scheduled to lift off at 11:30 am from Sriharikota's First Launch Pad. Hundreds of visitors, including children, had gathered to watch the launch. Thousands more were following the broadcast across the country.

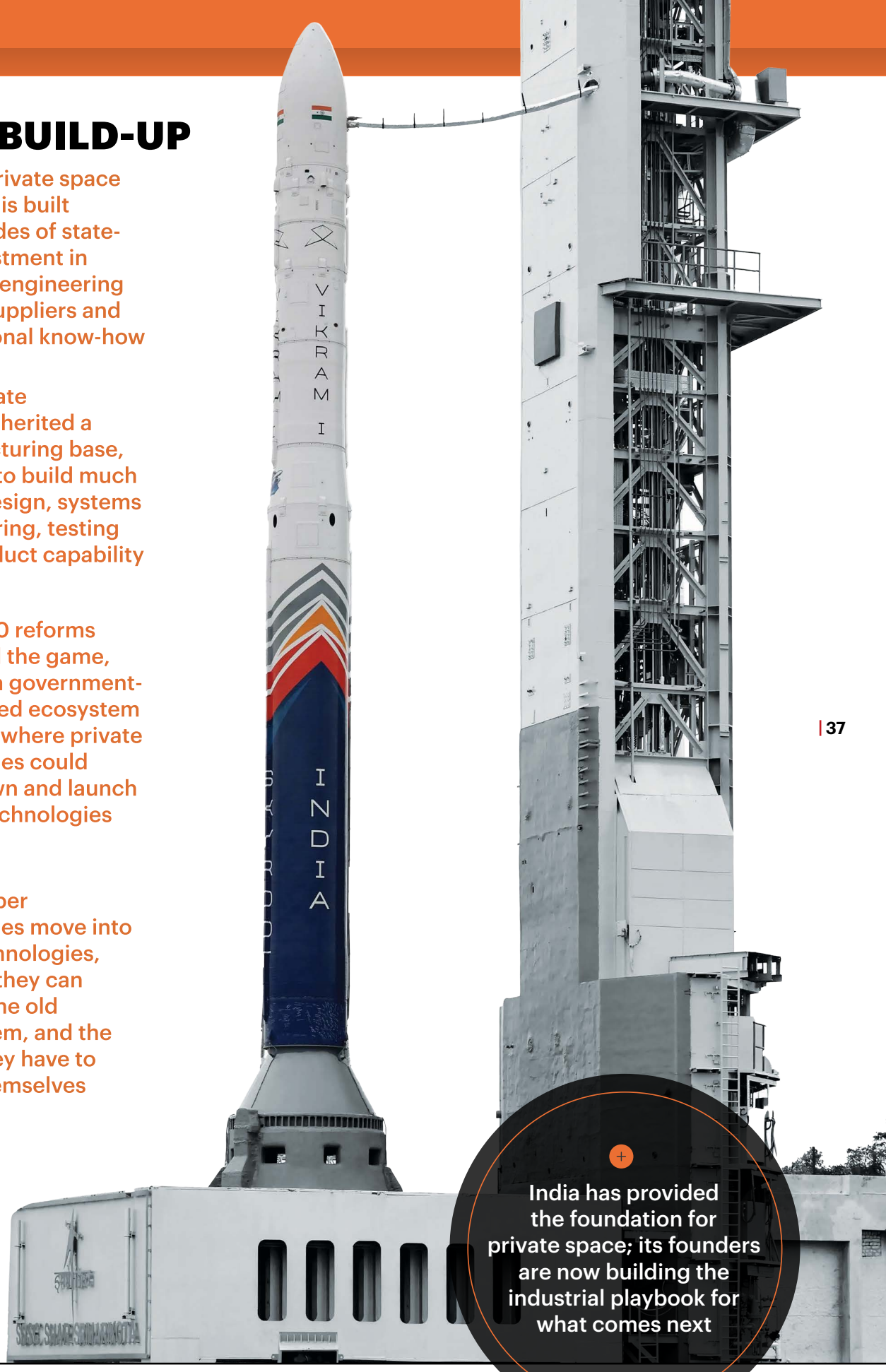
The sudden pause in the countdown shook up the mood inside the launch control centre at Skyroot Aerospace's Infinity Campus in Hyderabad. Engineers rose from their consoles. Phones turned busy. People scrambled between workstations, checking screens, whispering to one another. The tension was palpable.

Beyond Skyroot's campus, the hold message sent goosebumps through India's entire space community. Watching the broadcast in Chennai, Srinath Ravichandran, Co-founder and CEO of Agnikul Cosmos, felt a familiar knot in his stomach.

"We had to abort two launches. One of them was T-129 seconds and the other was T-11 seconds. So, when they had that delay because of a hold, it reminded me of our own launch and our own countdown as well," says Ravichandran, who co-founded Agnikul in 2017 after leaving a career on Wall Street. "It means a

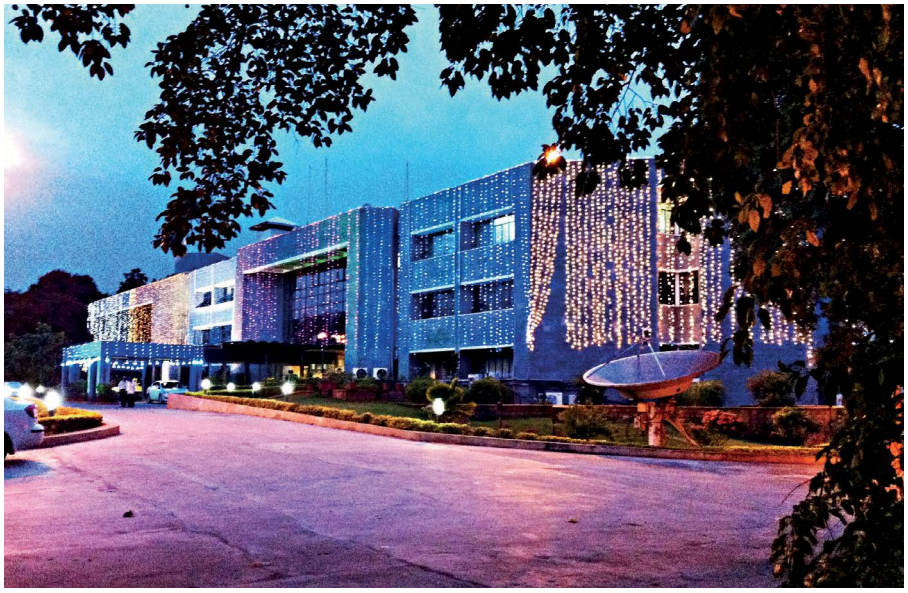
# THE BUILD-UP

- India's private space industry is built on decades of state-led investment in rockets, engineering talent, suppliers and institutional know-how
- The private sector inherited a manufacturing base, but had to build much of the design, systems engineering, testing and product capability
- The 2020 reforms changed the game, turning a government-dominated ecosystem into one where private companies could build, own and launch space technologies
- The deeper companies move into new technologies, the less they can rely on the old ecosystem, and the more they have to build themselves




**India has provided the foundation for private space; its founders are now building the industrial playbook for what comes next**

# THE SPACE STACK



## FOUNDATION

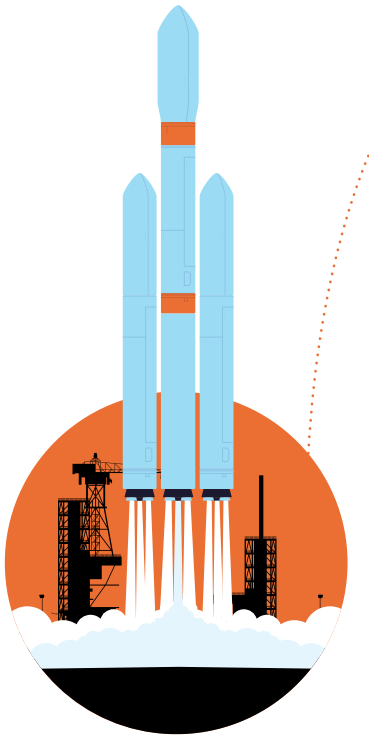
**ISRO'S INDUSTRIAL BASE**

▸ **Precision manufacturing | Cryogenic engineering | Composites & materials | Electronics & avionics | Testing | Launch infrastructure Talent & experience**

## 1. LAUNCH

▸ **Gets payloads into orbit**

▸ **Skyroot | Agnikul Abyom, etc.**





lot to actually be going through a hold when the world is watching."

For Skyroot, this was the culmination of years of work. The stakes were high. Twice over the previous 14 months, Isro's own PSLV, the agency's most reliable rocket, had failed from this same pad: first in May 2025, when PSLV-C61 lost its payload to a Stage III anomaly. This January, PSLV-C62 suffered another Stage III anomaly.

Vikram-1 had to work. Everything had gone to plan. Until five minutes to lift-off.

A software glitch was detected in the automatic launch sequence. The issue was corrected and the launch sequence restarted from T-20.

The countdown resumed. This time, the systems worked perfectly. At 12:05 pm, Vikram-1 lifted off.

"Lift off normal," announced mission control. The spectators at Sriharikota burst into spontaneous applause, but the faces inside the control centre stayed tense.

T+30 seconds. T+1 minute. As the rocket reached a 22 km downrange, the control room went up in cheer. Stage separation followed, then second-stage engine ignition.

Vikram-1 was flying.

"It still feels like a dream to me," Skyroot's Co-

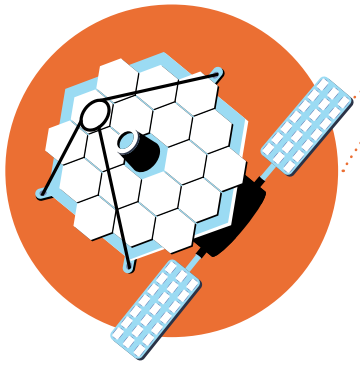
PHOTO BY **GETTY IMAGES**

## 2. SATELLITES

- Earth observation
- Scientific missions

## 3. SUBSYSTEMS

- Propulsion | Avionics
- Power | Bellatrix | Manastu
- XDLINX, etc.



## 4. GROUND INFRASTRUCTURE

- Ground stations | Mission operations | Tracking | Launch integration | Data downlink
- Dhruva | Astrome, etc.

## 5. APPLICATIONS

- Agriculture | Defence

**IN MAY 2020, FM NIRMALA SITHARAMAN ANNOUNCED STIMULUS MEASURES FOR THE CREATION OF IN-SPACE**

ILLUSTRATIONS BY **RAJ VERMA**



founder and CEO Pawan Kumar Chandana told a television channel after the launch. "The next phase is to reliably reach orbit... and then get to commercialisation." Chandana didn't respond to *BT*'s emails and messages requesting comments for this story.

In Bengaluru, Yashas Karanam, Co-founder and COO of Bellatrix Aerospace, was watching the launch at his office, with a team member running live commentary over the feed.

"We were all super excited, wishing that it would be a success, because it sends out a very strong signal to the global ecosystem that India means serious business," says Karanam. "Having a launch vehicle succeed on the first attempt, at full scale... a lot of countries have actually failed to do that."

When Vikram-1 reached low-Earth orbit on its maiden flight, it became the first privately developed Indian orbital launch vehicle to fly from Indian soil. It also made India the third country, after the US and China, with private orbital launch capability. Germany joined them on September 5, when Isar Aerospace put its Spectrum rocket into orbit from Andøya Spaceport in Arctic Norway. It was proof that India's private space industry, which was in the making for decades but opened up for private participa-

**"The private sector has always been involved. The only difference now, which is a big difference, is that private people can be mission owners"**

**SRINATH RAVICHANDRAN**
CO-FOUNDER AND CEO, AGNIKUL COSMOS

tion only a few years ago, could convert Isro's know-how and its inherited base of suppliers into hardware of its own.



## THE ISRO INHERITANCE

Much of what private companies are building today rests on six decades of work by India's state-owned space agency, the Indian Space Research Organisation, or Isro. Before 2020, Isro was India's only face in space. It designed the rockets and also built various satellites. Since 1979, the agency has done more than 100 launches and put satellites into orbit, and even sent missions to the Moon (Chandrayaan) and Mars (Mangalyaan).

Formed in 1969, Isro succeeded the Indian National Committee for Space Research, which India's first prime minister, Jawaharlal Nehru, had established under the Department of Atomic Energy in 1962. Born out of the need for communications, weather forecasting, remote sensing and development of a young, and impoverished, republic, it prioritised reliability, frugal engineering and self-reliance.

Over the next six decades, Isro built a base of more than 500 companies—public and private—across India, with dense clusters around Bengaluru, Thiruvananthapuram, Ahmedabad, Hyderabad and Sriharikota.

By Isro's own estimates, around 80-85% of launch-vehicle inputs come from this network of roughly 500 small to large industries. These suppliers were, and largely still are, geared towards build-to-spec fabrication.

Manish Gupta, general partner at growX Ventures and an early investor in Bellatrix, calls it built-to-print. "Isro developed the designs and owned the technology, it sourced or made the components; vendors received the finalised drawings," says Gupta. "They had to build it precisely to a drawing or specification handed to them and did a very good job of it." That robust supplier base, and its capabilities of building to print, are genuine strengths. But they represent only one part of a space innovation ecosystem. Mission design, system configuration, components development, sensor integration, and the firmware and software that give most subsystems their intelligence were not part of that supplier base's capability.

The question every spacetech founder in India had to answer was how much of the existing aerospace and defence manufacturing base they could actually use. For heavy manufacturing, the answer was: most of it. Sanjay Nekkanti, Co-founder and CEO of Hyderabad-based Dhruva Space, started his company in 2012 on

## TAKING OFF

**PE-VC investment in Indian spacetech rose from around $10 million in 2020 to around $350 million in 2026**

**SOURCE** VENTURE INTELLIGENCE

top of that base. "If you come to my office, you won't see CNC machines or PCB printing lines. There are already so many of them in this country," says Nekkanti. "The same manufacturing guys who build for aerospace also build for defence, industrial, everything. All the ingredients required for a thriving space ecosystem already existed."

But the base only helped if one knew what to ask it for. "For somebody who's been there, done that before, it's relatively fast," says Nekkanti. "When you don't know what you want, that's the problem."

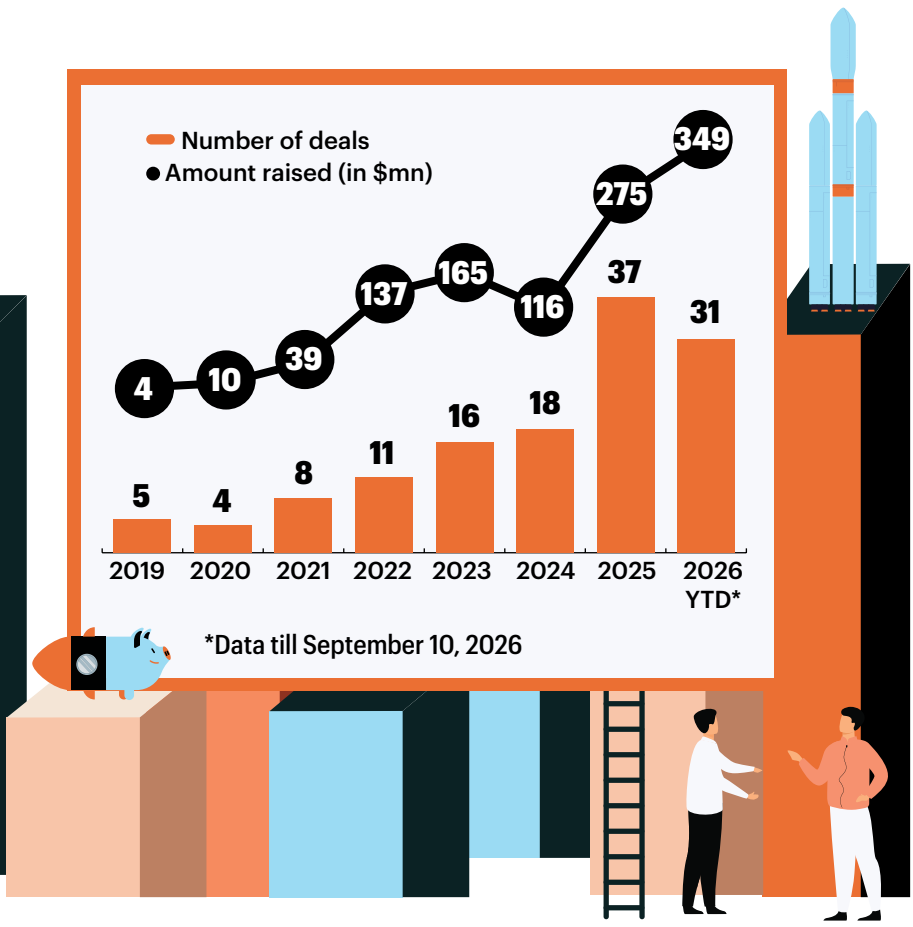
Beyond the manufacturing base, there were people. Several Isro engineers, including Skyroot Co-founders Chandana and Naga Bharath Daka, became entrepreneurs. Many other retired engineers became advisors to these new-age companies. Some scientists even joined them full-time. Being coached by the scientists who had worked under A.P.J. Abdul Kalam, when Isro was young and short of resources, was invaluable, according to Agnikul's Ravichandran.

"That generation of people were inspired. Isro was like a start-up then," says Ravichandran. "That scrappiness, in terms of getting together whatever it takes to build something, fire the engine—that's what I learned."

Isro's name did the rest. When Bengaluru-based Pixxel began selling hyperspectral data to customers in the US, for example, Isro's legacy did some of the early persuading.

"Isro thankfully had set that foundation and set up that reputation," says Awais Ahmed, Co-founder and CEO of Pixxel. "As a private company, we were still unproven, but once they saw that we were coming from India and that we had retired Isro scientists as advisors and some Isro scientists who had joined us full-time, we were taken quite seriously." The deepest inheritance is the know-how. Reaching orbit is a capability only a handful of countries hold, and India already had it. "We know what it means to build stuff that works in space. We know how to go to orbit," says Ravichandran. India was not starting at the beginning.

But the companies still could not own what they made. That changed in 2020, when private players finally got a policy push.




## THE COMING-OF-AGE MOMENT

In May 2020, in the middle of the Covid-19 pandemic, Finance Minister Nirmala Sitharaman announced a series of stimulus measures as part of the government's Aatmanirbhar Bharat package. Among them was the creation of the Indian National Space Promotion and Authorisation Centre (IN-SPACe) to let private firms use the assets and institutional knowledge of Isro and the Department of Space, and participate in space activities.

Six weeks later, the Union cabinet, chaired by Prime Minister Narendra Modi, approved "far-

reaching reforms in the space sector", explicitly aimed at enabling private-sector participation across the full range of space activities.

The reforms created a route for private companies to design, build, own and operate rockets, satellites, ground systems and launch infrastructure, subject to authorisation.

"It was a big relief," recalls Ahmed, who co-founded Pixxel with Kshitij Khandelwal at BITS Pilani in 2019. "We could all build a satellite. But before this announcement came, launching a satellite for a private company was basically illegal."

Isro and IN-SPACe didn't respond to email requests for comments.

The 2023 Indian Space Policy formalised the new architecture. An Indian company could now design its own rocket or satellite, own the technology, and take it to market. This was followed by the easing of foreign direct investment (FDI) rules in 2024, making way for up to 100% FDI into Indian spacetech.

These changes released pent-up ambition. Government data showed the number of active space start-ups jumped to around 440 by 2026 from a few dozen in 2020, as they spread across almost every layer of the space economy.



Skyroot, Agnikul, and Pune's Astrophel Aerospace began building rockets, the vehicles that carry satellites into orbit. Hyderabad-based Abyom SpaceTech & Defence started building reusable rockets, advanced space infrastructure and dual-use technologies. Bellatrix went after propulsion, the systems that move spacecraft once they get there. Dhruva Space built satellite platforms and the ground infrastructure needed to receive and process what comes back.

Pixxel built hyperspectral imaging sensors and EON Space Labs, also from Hyderabad, made electro-optical payloads—the cameras and instruments that collect information from space; Bengaluru-based GalaxEye Space combined optical and radar sensing into one. Digantara started with space domain awareness, tracking satellites, debris, and other objects already crowding the orbit.

Together, they form a chain across the space economy, with each layer enabling the next. "Space is the next frontier," says Jainul Abedin, Founder & CEO of Abyom. "Whether we realise it or not, our everyday lives are becoming increasingly dependent on space. A farmer doesn't necessarily need to know that a satellite is behind the weather data he uses before sowing, the crop insurance he accesses, or the digital maps that help him navigate. But the infrastructure enabling many of these services increasingly begins in space."

Today, India operates just over 50 satellites, spread across communication, navigation, earth observa-

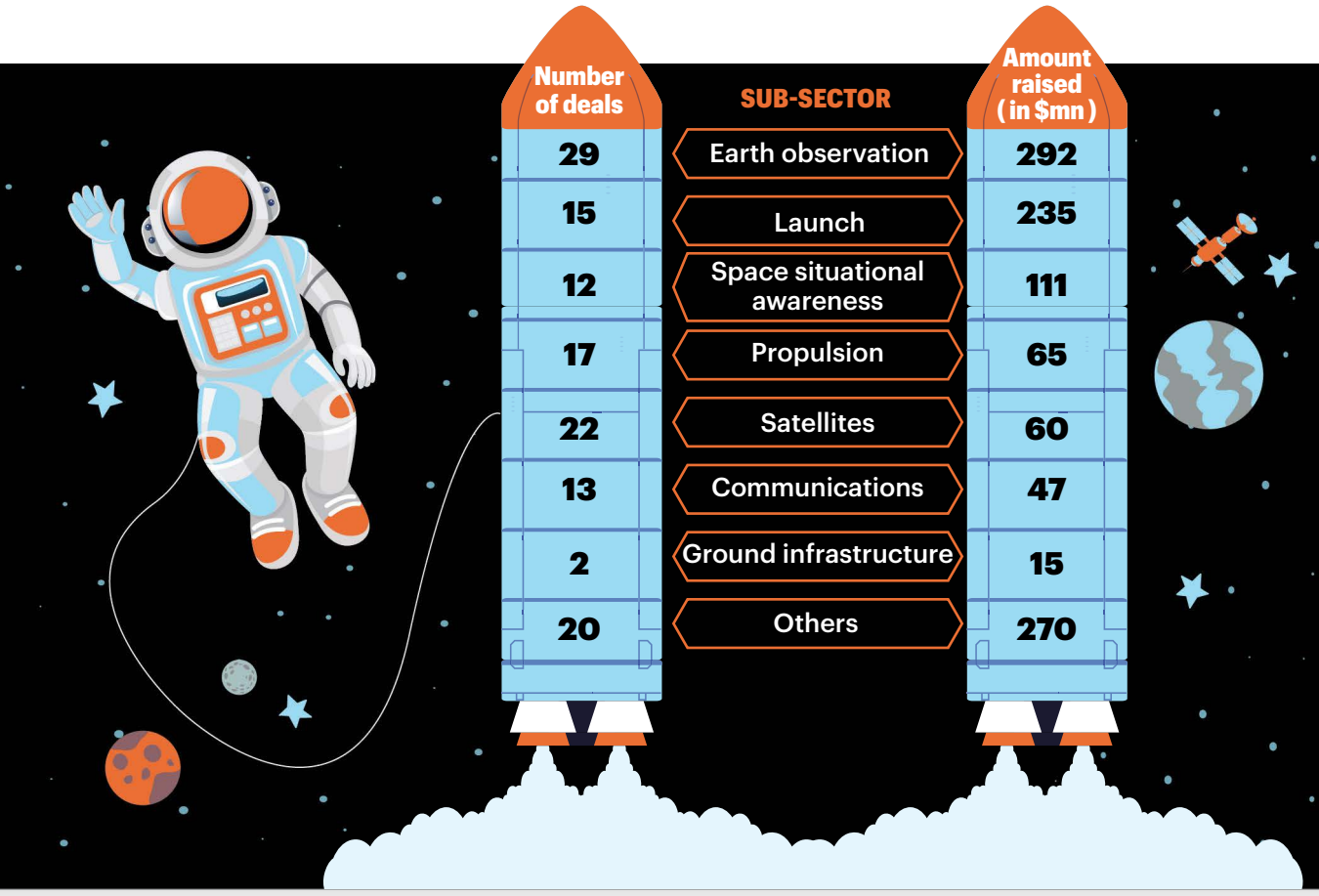
**"As a private company, we were still unproven, but once they saw that we were coming from India, and had retired Isro scientists as advisors, we were taken seriously"**

**AWAIS AHMED**
CO-FOUNDER AND CEO, PIXXEL

## WHERE THE MONEY IS GOING

**PE-VC investments in spacetech companies by segment**

**NOTE** FIGURES FROM 2019-2026 YTD
**SOURCE** VENTURE INTELLIGENCE






tion, and strategic applications. By comparison, the US has more than 11,000 active spacecraft—over 10,000 of them in SpaceX's Starlink constellation—while China's fleet has crossed the 1,000-satellite mark and is expanding rapidly with new navigation, earth observation and broadband constellations.

Private equity and venture capital investment in Indian spacetech companies rose from roughly $10 million across four deals in 2020 to about $275 million across 37 deals in 2025, according to funding data from Venture Intelligence. It has reached $349 million across 31 deals this year as of 10 September. IN-SPACe has set a goal of growing India's space economy from about $8.4 billion now to $44 billion by 2033, raising the country's share of the global market from 2% to 8%.

To put things in perspective, the global space economy was worth about $630 billion in 2023 and could reach $1.8 trillion by 2035, according to a World Economic Forum-McKinsey report.

## PHARMA TO PAYLOADS

On February 3, 2026, a small Indian satellite tracked the International Space Station across the sky and photographed it twice, capturing the orbiting station in 15 angles while both craft moved at roughly 7km per second—fast enough for one to travel from Delhi to New York and back in under an hour. The industry called it in-orbit snooping, a crucial development in modern space security.

The satellite—an 80kg Earth-observation craft called ABA First Runner, or AFR—was launched on June 13, 2023, aboard SpaceX's Falcon 9, as part of the Transporter-8 mission. It was made by Azista Space, in collaboration with Berlin Space Technologies.

When Azista Industries was founded in Hyderabad in 2014, it made organic food products and pharmaceuticals. From there, it started climbing up the value chain, moving from consumer goods into high-reliability, high-margin manufacturing. Its aerospace business emerged in 2016, initially around space and composites. In 2017, it got its

**"Initially, the industry was largely focused on the GEO market. We recognised the shift toward LEO early... None of the existing companies had strong active R&D teams. They were all manufacturers"**

**YASHAS KARANAM**
CO-FOUNDER AND COO, BELLATRIX AEROSPACE

first real exposure to space technology as an Isro subcontractor.

"That experience was invaluable," says Azista Space Managing Director Srinivas Reddy. "It taught us the quality systems, documentation discipline and process rigour that space demands, none of which you can learn from commercial electronics work."

For the next few years, Azista Space built hardware to Isro's drawings, to the agency's standards, for its national missions. It had two options: keep building to Isro's specifications and remain a contractor, or venture out on its own. About three years in, it made a call: it would develop its own products for the global market.

Space hardware demands a different kind of discipline. It has to survive an environment that cannot be fully replicated on the ground and not repaired once launched—vacuum, extreme thermal cycling, radiation and severe launch vibration. It has to work the first time, every time.

"That changes everything upstream. We had to build quality systems and traceability where every component, process and test is documented and auditable end to end," says Reddy. "We had to adopt space-grade parts selection and screening, environmental qualification and testing (thermal-vacuum, vibration, EMI/EMC), cleanroom assembly practices, and rigorous failure analysis and review discipline. Reliability and repeatability, not just a working prototype, become the real deliverable."

The policy change in 2020 became the catalyst it needed.

## A MIXED INHERITANCE

"The private sector has always been involved. The only difference now, which is a big difference, is that private people can be mission owners," says Ravichandran. "We all worked as contractors. We were building something, but the mission was owned by India."

Isro's vendor base could make almost any part to a specification, but it was thin where the requirement was to design, qualify and repeatedly manufacture a specialised part of its own: aerospace-grade valves, cryogenic components, radiation-hardened electronics, optical glass and coatings. So companies had to build these in-house or import.

Immanuel Louis, Co-founder and COO of Astrophel, maps the gap precisely. "India has strong precision-manufacturing capabilities, but gaps remain in aerospace qualification, traceability and low-volume specialised supply chains," says Louis. For example, most Indian companies still import high-pressure and cryogenic valves often from Germany, so Astrophel now builds its own. Roughly 47% of the company's bill of materials by value still comes from outside India.

The harder lesson, Louis says, came after the engineering. "Industrialising the technology is often harder than developing it," he says. "Aerospace is not only a technology problem in India. It is a manufacturing, supply-chain and execution problem." The machining India could do. The qualified, specialised supply, it largely could not.

That gap forced companies to build capabilities in-house. Propulsion was perhaps the clearest case, because there was almost no domestic manufacturing base. Bellatrix, for example, set out to build engines for a new generation of small satellites and found the technology had to be reinvented for smaller scale.

"Initially, the industry was largely focused on the GEO market. We recognised the shift toward LEO early and had to re-invent the wheel, thereby developing solutions specifically for lower-orbit satellites. This meant inventing entirely new products for the LEO regime," says Karanam. "At the time, none of the existing players had strong, active R&D teams—they were primarily manufacturers." Both the company's customers and suppliers were based overseas, and, as Karanam puts it, "there were absolutely no suppliers in the country".

GEO, or Geostationary Earth Orbit, and LEO, or Low Earth Orbit, are the two main types of orbits used for satellites. LEO is closer to earth and satellites in it move quickly, typically at altitudes of 160 to 2,000 km. GEO is much farther away, at about 35,786 km above the equator, where satellites orbit at the same rate as

**Flying high** The launch of Vikram-1, India's first privately developed orbital-class rocket, built by Hyderabad-based firm Skyroot Aerospace

PHOTO BY **ANI**

earth's rotation and, therefore, appear stationary from the ground.

Abyom is another example. A reignitable, throttleable engine didn't exist in India, and it couldn't be reverse-engineered. "Rocket propulsion is fundamentally a deep-tech problem. You cannot simply take an existing rocket engine apart and reverse-engineer it like you might with a conventional automobile engine," says Abedin. "Every element—from combustion and injection to cooling, materials and controls—has to work together under extreme conditions. We have had to build a significant part of this capability ourselves, through continuous testing, iteration and learning."

The gap was widest where the work was design-intensive. Lenses and sensors, for example, need a specialised supply chain: makers of optical glass, of the fine coatings that go on lenses, of ultra-precise polishing and measurement. Almost none of it exists in India. So EON had to build the know-how itself. "We're no longer just assembling imported components," says Co-founder and COO Punit Badeka. "We're designing and engineering sophisticated electro-optical payloads in India."

How much a company inherited then depended on which part of space it had chosen. The closer their work sat to hardware Isro had already built, the more there was to draw on; the further it moved into new territory, the less.

Digantara, a space domain awareness and surveillance start-up, sits at the far end. "We had to build every bit of infrastructure from scratch," says the Bengaluru-based company's founder and CEO Anirudh Sharma. "Isro had built satellites, rockets and missions, but it had never tracked satellites from space." There was no vendor to call, no scientist to borrow, no expertise anywhere in the country to hire.

Between the two ends sat most of the industry, drawing on what existed and building what did not. The supplier base gave the new companies manufacturing capacity. Isro gave them people, testing infrastructure, engineering experience and credibility. The policy reforms gave them permission to become mission owners. But the technology they now had to own, and the supply chains they needed to make it at scale, had to be built by the companies themselves.

Vikram-1 showed that India's private sector could assemble an orbital vehicle and put it into space. The next test is whether India's spacetech companies can compete globally. BT

NSE COMMODITY
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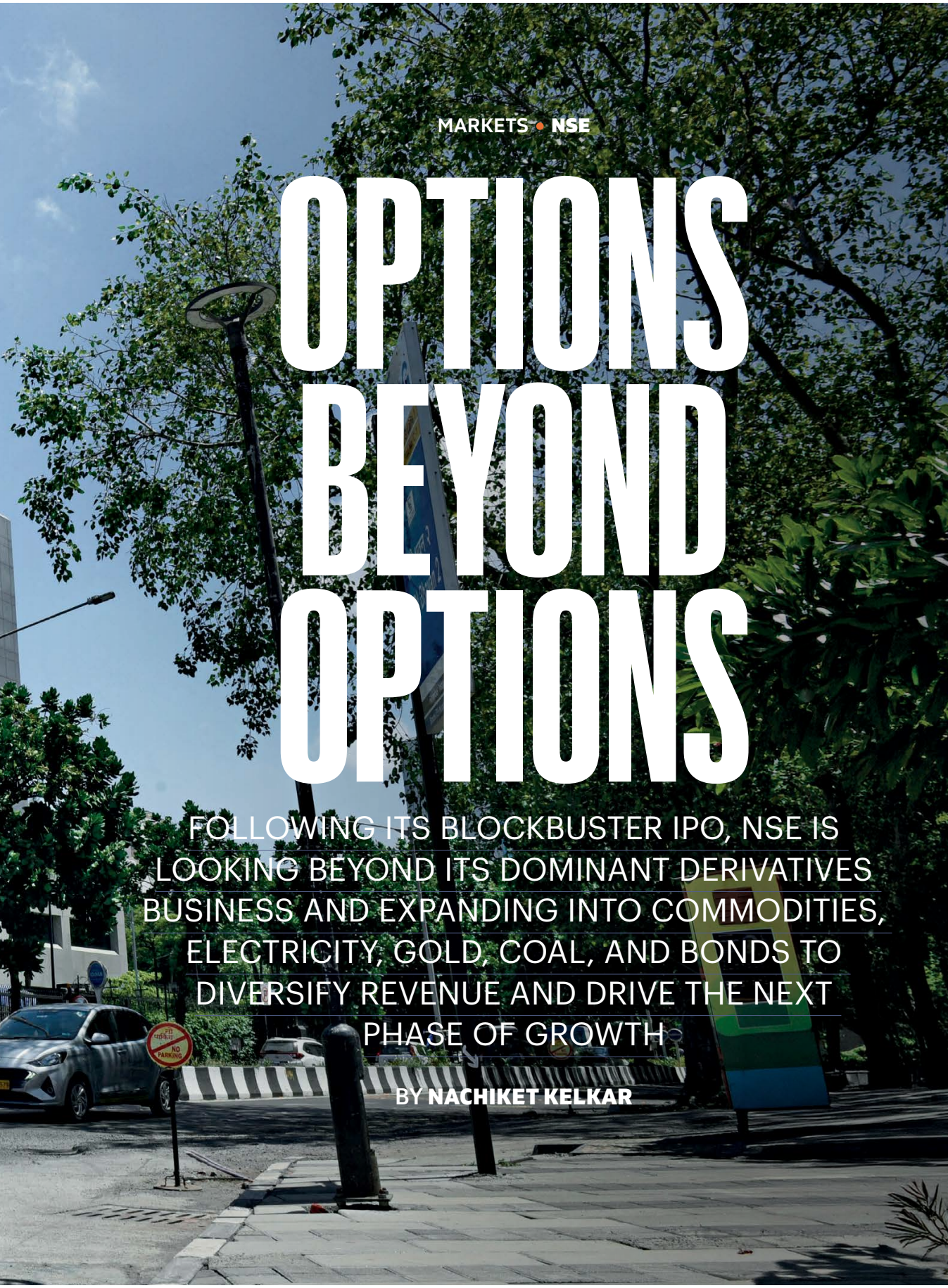
PHOTO BY **GETTY IMAGES**



MARKETS • NSE

# OPTIONS BEYOND OPTIONS

FOLLOWING ITS BLOCKBUSTER IPO, NSE IS LOOKING BEYOND ITS DOMINANT DERIVATIVES BUSINESS AND EXPANDING INTO COMMODITIES, ELECTRICITY, GOLD, COAL, AND BONDS TO DIVERSIFY REVENUE AND DRIVE THE NEXT PHASE OF GROWTH

BY **NACHIKET KELKAR**

**AFTER ABOUT A** decade-long delay, the National Stock Exchange of India (NSE) went public on September 24, in a ₹22,562-crore initial public offering (IPO). It was the second-largest IPO in India after Hyundai Motor India's ₹27,870-crore offering in 2024. The issue was subscribed 5.7 times. Driving the listing ambitions was MD and CEO Ashishkumar Chauhan, who took over in 2022 when the NSE was dealing with the co-location controversy. Under him, the NSE worked through those issues and revived its listing plans.

Chauhan, who says organisations like the NSE are set up once in the lifetime of a nation, fondly remembers his association with the exchange since its inception. And he sees an even brighter future for NSE after the IPO.

"Over the next five years, we see the NSE evolving from being primarily identified as an equity and derivatives exchange into a genuine one-stop shop with deep liquidity across multiple asset classes," Chauhan says in an interview with *BT*. His relaxed demeanour belies the scale of the ambition behind that vision.



## NEW STRATEGY

The NSE's next phase of growth will be driven by a combination of new products, newer market segments, technology, and deeper participation across the financial ecosystem.

"Commodity and energy markets are certainly an important part of the NSE's growth opportunity. The recent regulatory decision to allow FPIs (foreign portfolio investors) to participate in physically settled non-agricultural commodity derivatives can also help broaden participation in the commodity ecosystem," says Chauhan.

The NSE commenced operations in 1994, ushering in a new era of electronic screen-based trading of securities, while settlement in dematerialised securities followed in 1996. Derivatives trading began in 2000.

The rapid economic growth seen over the past 30 years has led to multi-fold growth for NSE as hundreds of companies tapped the stock market. That, in turn, has brought in large institutional investors as well as millions of retail investors into the ecosystem.

Chauhan points out that the NSE's premium turnover in the WTI (West Texas Intermediate) Crude Oil contract has surged from ₹875 crore in January 2026 to about ₹8,200 crore in September 2026. Similarly, in electricity futures, it now has more than 72% market share.

Analysts from PL Capital say the NSE has built a "holistic ecosystem through key market institutions." NSE Clearing, for instance, is the largest clearing corporation in India in terms of cleared value, they say. Separately, NSE Indices had 425 indices, spanning sectors, themes and strategies, as of March 2026, they said.

NSE IX, the international exchange in Gujarat's GIFT International Financial Services Centre, got the global access provider licence earlier this year, enabling it to offer an avenue for Indians to invest in global stock markets.

Last year, transaction charges driven by trading activity made up a large portion of the NSE's revenue. Officials say this should come down with new revenue lines such as data and connectivity, and indices growing at a fast pace. "If you look at the revenue composition, 80% volumes come from trading activity, 20% from related activities. Moving forward, we need to diversify the revenue stream," says Srinivas Injeti, Chairman, NSE, adding the organisation can also monetise data within what is permissible under the regulatory framework. "So, many other products could also be developed," says Srinivas.

## ROBUST NUMBERS

The NSE's total income rose from ₹16,352 crore in FY24 to ₹18,713 crore in FY26. According to Srinivas,

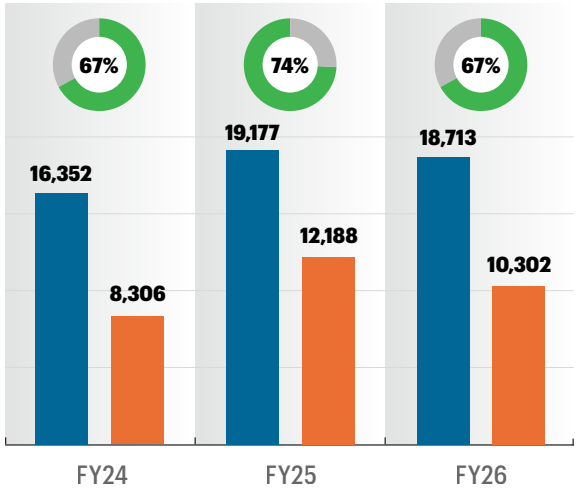
## INCREASED FOCUS ON DIVERSIFICATION

**NSE has been the market leader in segments like equity cash and derivatives, but has been aggressively expanding in newer areas like commodities**

**In March 2026, it launched 10-gram gold futures contracts to broaden retail participation in commodities**

## STEADY EARNINGS

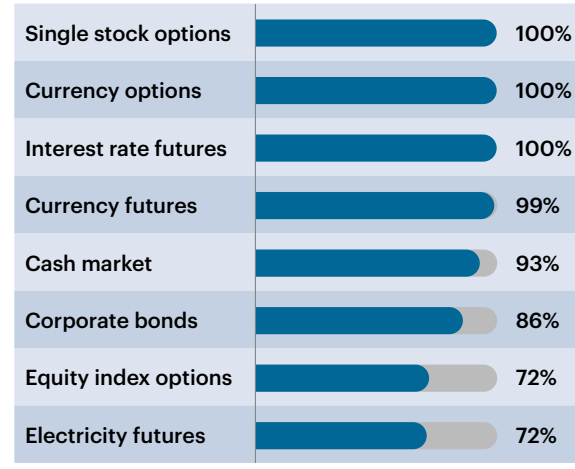
• In the past three financial years, NSE has been able to maintain robust financials despite falling stock markets




TOTAL INCOME (₹CRORE) PROFIT AFTER TAX (₹CRORE)
OPERATING EBITDA MARGIN (%); **SOURCE** NSE PRESENTATION

## DOMINANT PRESENCE

• NSE commands a massive market share across asset classes in FY26. In options, it has complete dominance




**SOURCE** NSE PRESENTATION, RHP



the company's economies of scale mean that many of its new product launches may not require major additional costs, given that it has the technology and manpower in place.

In the cash market, the NSE had a market share of 93% as of March 2026, based on turnover. In equity futures, its share was 99.79%; in currency futures and interest rate futures, it was 100%. In the equity options segment, its market share last year was 74.71%, based on the premium turnover.

"Equity as an asset class remains hugely underpenetrated in India, so the runway is long. As more investors enter the market and more companies tap the capital market, the exchange will benefit. Derivatives will remain at least half the business. But they will diversify, and given that the business is generating a lot of cash, they will not require any major debt to expand," says Avinash Gorakshakar, a Sebi-registered research analyst and founder of Avinash Mentor Research.

The NSE's key strength is not just the market share but the liquidity advantage built over time. "Its unique investor base has grown at 26% CAGR to 132 million. More traders bring more liquidity, which attracts more traders. In simple terms, the NSE offers

**In July 2025, the exchange launched electricity futures contracts, aiming to develop electricity derivatives**

**In April 2026, dated Brent crude oil (Platts) futures contracts were launched**

**NSE wants to expand offerings in index and data services and develop investment products incorporating alternative datasets, including ESG parameters**

# "NSE WILL HAVE DEEP LIQUIDITY ACROSS MULTIPLE ASSET CLASSES"

**NSE's MD & CEO Ashishkumar Chauhan on how the exchange will focus on strengthening the entire market ecosystem rather than depending on a single product. Edited excerpts:**

**What are the new products and segments that NSE has identified for its next growth phase?**
The next phase of growth will be driven by a combination of new products, newer market segments, technology, and deeper participation across the financial ecosystem. We are already expanding some of our traditional businesses.

Of particular interest to us are electricity futures, electronic gold receipts (EGRs), commodity and energy products, bond index futures, among others. We are also looking at opportunities in the bond ecosystem. We received Sebi's (Securities and Exchange Board of India's) approval to launch Corporate Bond Index Futures on September 30, 2026.

EGRs are interesting because they can bring greater transparency and efficiency to the gold market and potentially enable a wider ecosystem of products around gold. Electricity futures have scope for better price discovery in a segment of the power market.

▶▶▶

**How will NSE look to retain its leadership in equity cash and derivatives market in the coming years?**
We do not seek leadership only in terms of market share or volumes. It is about a sustained ability to provide markets that are efficient, fair, transparent, orderly, and low-cost. You would agree that regulation is an important part of the evolution of any market.

Therefore, while the regulators ensure that markets develop in a manner that protects investors and maintains market integrity, we continue to operate within the regulatory framework.

PHOTO BY **MILIND SHELTE**

The derivatives market has gone through a period of significant regulatory change. It has adapted, with activities spread across different instruments, including monthly options, futures, and other segments.

**How do you intend to diversify your revenue streams?**

Diversification is a process. Equity derivatives will remain an important part of NSE's business because they are an important component of the overall capital-market ecosystem. However, the objective is to ensure that our growth is not dependent on a small number of products.

We are, therefore, building multiple revenue streams across commodities, energy, bonds, indices, market data, and technology-led services.

Market data and index-related businesses are already growing, although they are smaller compared with the core transaction business. Over time, we expect areas such as commodities and data monetisation to develop into meaningful, standalone revenue streams.

**"WE ARE BUILDING MULTIPLE REVENUE STREAMS ACROSS ENERGY, BONDS, INDICES"**

There is also potential for growth in the underlying equity and cash markets, as financialisation of savings continues. India's investor base is expanding, and digital adoption is increasing.

**Where do you see NSE in the next five years?**

Over the next five years, we see NSE evolving from being primarily identified as an equity and derivatives exchange into a genuine one-stop shop, with deep liquidity across multiple asset classes. We would like to see deeper and more efficient markets across products, supported by technology, strong risk management, and surveillance infrastructure. BT

*@TheNachiket*

exposure to India's growing participation in financial markets without having to pick which broker or platform will emerge as the winner," says Maahir Mani, research analyst at broking firm Deven Choksey.

These strengths are reflected in the NSE's earnings. In FY26, it reported a net profit of ₹10,302 crore, up from ₹8,306 crore in FY24. In recent years, however, as the government and regulators have moved to curb derivatives activity, the exchange has felt the impact. In FY26, its net profit declined 15.4% from FY25, while total income fell 2.4%. In addition to the curbs in futures and options (F&O), the broader volatility in the stock market due to geopolitical and trade tariff-related tensions also dented investor activity and had a bearing on earnings.

## THE DIVERSITY STORY

Another reason the NSE is diversifying its product basket is the uncertain global envoronment. In 2025, the exchange launched its futures contracts. These are exchange-traded, standardised, and cash-settled contracts that allow market participants to secure pricing for 28/30/31 days of round-the-clock ity delivery, up to three months ahead. These contracts oper-

**SRINIVAS INJETI**
CHAIRPERSON, NSE

**"If you look at the revenue composition, 80% volumes come from trading activity, 20% from related activities. We really need to diversify the revenue streams"**

ate alongside the physical electricity market, providing a complementary financial instrument rather than being part of the physical trading or delivery process.

In March 2026, the exchange introduced 10-gram gold futures contracts, a move that could help investors trade in small quantities and drive retail participation. In May, it announced the launch of electronic gold receipts, essentially gold held in an electronic dematerialised form.

In April, it introduced dated Brent Crude oil futures contracts. These are based on the physical Platts dated Brent crude oil benchmark. Chauhan says the steps taken by Sebi to curb retail activity in the derivatives market were in the interest of investors. From the exchange's perspective, the right thing to do was to diversify.

"We need to find ways to ensure that we not only do not depend on a few products but also expand our products," he says.

Sebi has mandated exchanges to offer weekly index derivatives on only one benchmark index each, instead of weekly expiries. The NSE has been among the most impacted, say analysts.

"The NSE had five weekly expiries—Nifty 50, Bank Nifty, Nifty Financial Services, Nifty Midcap Select, and Nifty Next 50. Following the new guidelines, it retained Nifty as its weekly expiry product, while shifting the remaining indices to a monthly cycle. Consequently, NSE index option premium market share moderated from 87% in November 2024 to 72% in March 2026," say analysts at PL Capital.

**SUDIP BANDYOPADHYAY**
MARKET EXPERT

**"Financialisation of savings will remain a trend. The wealth effect will benefit capital market players over the long term. Newer revenue streams are also emerging for NSE"**

## ENSURING EFFICIENCY

Chauhan says contrary to the exchange's expectation that volumes and revenues will drop by half, last financial year, the dip was only 3%. This, Chauhan says, shows that diversification is working. More is in the pipeline, he adds.

"On a scale of three years, we have been able to reduce dependence on weekly options to 40-42%. We are now in the process of launching a coal spot exchange with some industry leaders. We also will have derivatives on top of that later," says Chauhan.

In April this year, the NSE got the nod from Sebi to invest in the proposed National Coal Exchange of India. The exchange is intended to facilitate electronic spot trading of coal through standardised contracts, enabling transparent price discovery and defined settlement mechanisms for market participants, including producers, consumers and traders.

On September 30, the NSE also received a no-objection certificate from the market regulator for proposed introduction of futures contracts on a corporate bond index. Corporate bond index futures are expected to complement the growing corporate bond market.

"India's capital markets are no longer merely reflecting economic growth. They are actively shaping it by channelling savings into productive investment, supporting entrepreneurship, fostering innovation and enabling efficient capital allocation," says Chauhan.

## ROAD AHEAD

With increased regulations, the NSE is expected to face more challenges. According to market expert Sudip Bandyopadhyay, it is expected to lose market share as competition increases. However, he also notes that the overall market size is increasing too, and that will benefit the NSE over the longer term. "Financialisation of savings is happening and will remain a trend. The wealth effect will benefit capital market players over the long term. The pie is increasing, and newer revenue streams are also emerging for NSE."

As Chauhan sums it up, going ahead, the NSE's growth will be closely linked to the growth of India, and as the economy becomes more formalised, the opportunity for market infrastructure will expand with it. BT

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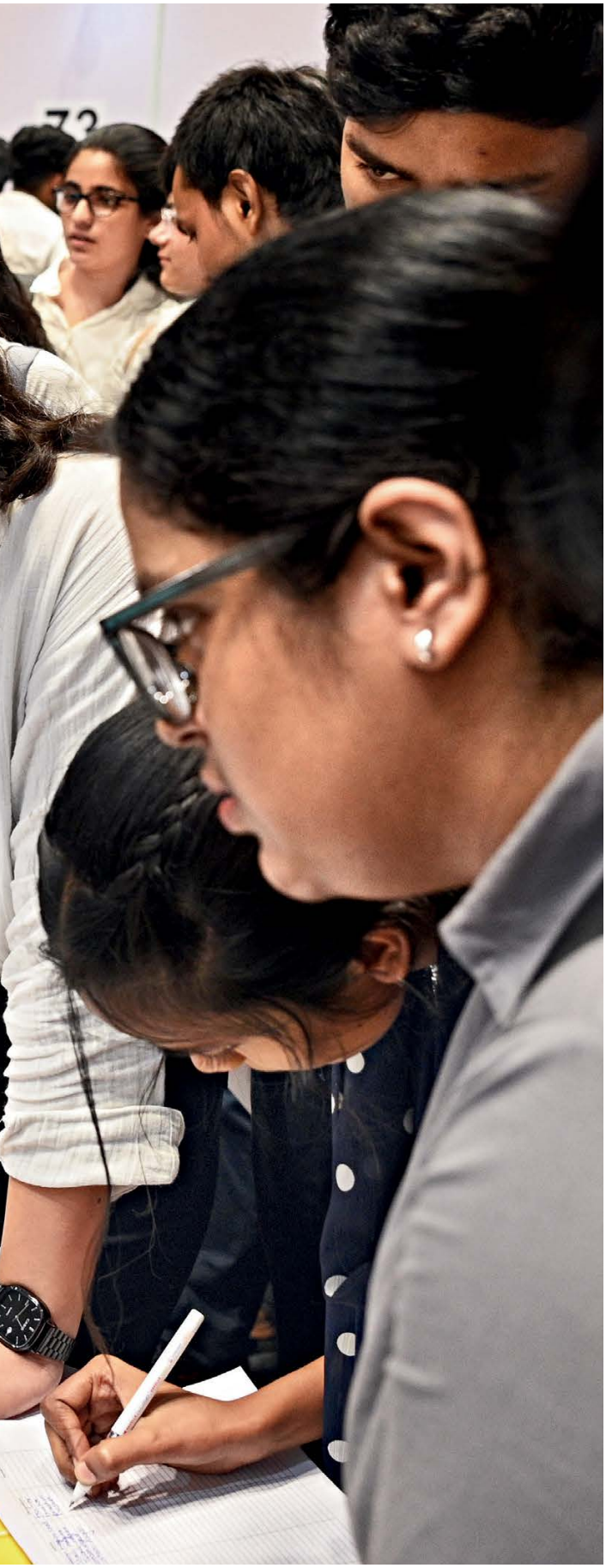


# THE YOUNG AND THE RESTIVE

INDIA IS GROWING FAST AND SO ARE THE ASPIRATIONS OF ITS YOUTH. BUT THERE ARE NOT ENOUGH JOBS FOR THE EDUCATED YOUNG. ENTRY-LEVEL OPPORTUNITIES ARE LIMITED BY INADEQUATE SKILLS, LOW WAGES AND AI, WHILE MILLIONS OF SANCTIONED GOVERNMENT POSTS REMAIN VACANT

BY **SURABHI**

**VIVEK, 23, GRADUATED** with a BA degree two years ago, and has been burning the midnight oil to get a job in the police force. It has been a long, anxiety-riddled wait that was made more harrowing after an earlier exam was cancelled. Today, he works in a packaging factory in Aligarh to supplement the family income, shares a room with three young co-workers, and has enrolled himself in a part-time computer course while he prepares for more exams. "The next exam is scheduled over the next six months," he says. "I hope it takes place and I get recruited."

Vivek is the youngest child—he has two older sisters—of Rupa, a nanny, and Arvind, a driver, in Delhi. While his current job gives him a degree of financial independence, his parents are worried. "We have toiled hard for his education," says Rupa. "He still dreams of getting a police job but it's difficult for him—living away from the family, working and studying at the same time."

Vivek's plight typifies that of millions of young men and women across India. These youngsters, armed with graduate, post-graduate and even technical training degrees, are hunting desperately for work—including appearing for exams to get the coveted government jobs—in a market for which their education did not prepare them.

While regular jobs in the private sector remain difficult to get, government jobs have gotten even tougher with cancelled exams, long recruitment process and fewer appointments. And the pressure on the young to get jobs will only intensify. India's working-age population of (15+ years) is expected to exceed 980 million in the next 10 years, from 616 million at present, data from the Economic Survey 2025-26 showed.

But employment numbers underpin the story. According to the government's Periodic Labour Force Survey (PLFS) report for August 2026, youth unemployment (15 years and above) was at 15.1%, thrice the overall unemployment of 5%. In 2025, these numbers were 9.9% and 3.1%, respectively.

A recent report by CareEdge Ratings noted this can be attributed to the persistent gap between education levels and availability of suitable employment opportunities. "Furthermore, unemployment numbers may overstate gains given rising self-employment and prevalence of disguised unemployment in agriculture. The rate may understate the extent of unemployment in an economy characterised by seasonal, casual, and irregular employment," it said.

And even for those who find work, the problem is often not solved. Indrajit, a BA graduate who currently works as a delivery partner for a food platform, is in his late 20s and actively looking for other work after he was laid off recently from a job at the Delhi airport. Life as a factory worker has little appeal for him due to its low wages. "As a delivery partner, I earn ₹700-800 per day. Many of my friends work as drivers and I am hoping to get a job like that too. I can earn at least ₹30,000 per month then," he says. To cut costs, he has stopped visiting his family, which lives near Lucknow. "I am barely managing in Delhi. Travel adds unnecessary costs," he says.

The International Labour Organization (ILO), in a recent report, said youth unemployment is rising across the world and was 12.4% in 2025 due to slowing global economic growth, diminished job creation, geopolitical tensions and rapid technological change. Job creation is not keeping pace with growth in the youth population in multiple regions.

In India, too, this phenomenon has been evident all through the year 2026, which saw youth pro-

**"Employers are looking for freshers who are upskilled, have good soft skills, can communicate well, and function in a team environment"**

**Sashi Kumar**
Managing Director, Indeed India

PHOTO BY **HARDIK CHHABRA**

## THE GREAT JOB SQUEEZE

1. YOUTH UNEMPLOYMENT IN 2025 WAS THREE TIMES THE OVERALL UNEMPLOYMENT RATE
2. LACK OF ADEQUATE JOBS AND APPROPRIATE WAGES IS TURNING INTO A CHALLENGE
3. PRIVATE SECTOR JOBS NOT ENOUGH TO MEET DEMAND, EMPLOYERS SEEK BETTER SKILLS
4. MILLIONS OF GOVERNMENT POSTS REMAIN VACANT; OUTSOURCING OF JOBS RAMPANT, EXAMS CANCELLED

tests expressing dissatisfaction with the current state of affairs —by industrial workers in the Delhi-NCR belt over low wages and poor working conditions, then by students protesting the cancellation of the NEET exam, and again in states like Jharkhand and Bihar over irregularities in recruitment exams for state administrative services.

Alakh N. Sharma, Director, Institute for Human Development, says jobs and wages are not keeping pace with economic growth. "Unemployment is a problem for the youth, especially in the educated segment in urban areas. In rural areas, the young educated can still report themselves as being employed by working part time in agriculture or a family enterprise, but for the urban, educated youth, there are few such avenues," he says.

Sharma's prognosis is backed by data from the 2025 PLFS report. People with secondary and above education reported the highest unemployment rates among all cohorts at 6.5% in 2025 versus 0.3% for persons who were not literate. Even those with coveted degrees such as engineering and MBA are struggling to find jobs.

### HIGHER EDUCATION ISN'T HELPING

Across the country, engineers and management graduates are finding that their degree may open the door to the job market, but it doesn't guarantee an offer. Students graduating from average MBA colleges say they are increasingly finding themselves caught between a degree and the skills that recruiters demand.

Take Ayush Negi. The 24-year-old completed his MBA from a university in Rishikesh, focusing largely on earning the degree, with little guidance on the practical

skills needed to secure a job. The result, he says, is a widening gap between what students are taught and what the job market demands. Students graduating from average colleges may complete their degrees without a clear understanding of skills required for the roles they are applying for. The problem, he believes, begins much earlier, with limited career counselling and inadequate exposure to changing industry requirements during the academic journey.

His friend and batchmate, Dev Sharma, says recruiters hiring for junior-level roles are looking beyond qualification and seeking candidates with at least the practical and technical skills needed to handle basic operations. "For students who have not been guided towards those skills during college, the gap becomes visible only when they enter the job market," Sharma says.

Both Negi and Sharma are on the lookout for jobs and continue to apply to various companies.

These issues were also highlighted in a recent report by global hiring platform Indeed, which revealed that India's young jobseekers are facing a more competitive and increasingly complex transition from education to employment, with 70% saying securing a first job is harder today than it was three to five years ago.

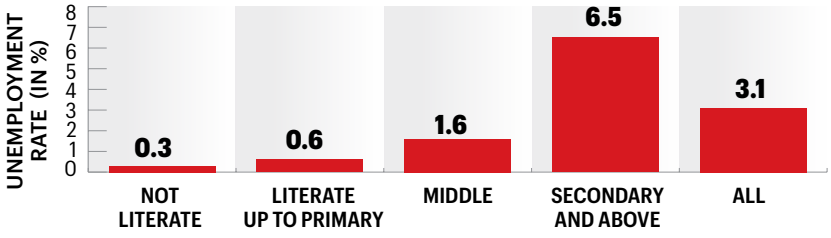
"New data from Indeed highlights how entry-level hiring expectations are evolving amid rising competition, limited career exposure opportunities and growing pressure on young professionals to compromise early in their careers," says Indeed's Fresher Hiring Report, based on a survey of 2,633 job seekers across India.

Sashi Kumar, Managing Director, Indeed India, says July saw an uptick in job creation on the platform after some softening in the previous three months. While there has been a slowdown in jobs in IT, roles for younger people are available in admin and sales support, education and service roles, commercial support, people-oriented functions even in the current reasonably tough job market, he says.

"Employers are looking for freshers who are upskilled, have good soft skills, can communicate well and function in a team environment," he says, adding that there's also a focus on having previous experience.

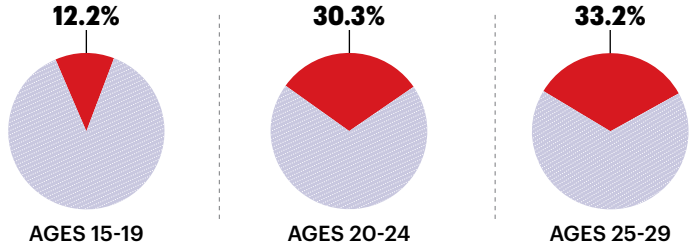
## DEGREES DO NOT EQUAL JOBS

• Unemployment rises as one goes up the education chain




## LARGE NUMBER OF YOUNG ADULTS IN NEET

• Persons not in education employment or training, or NEET, in 15-29 years age group (In %)




**SOURCE** PLFS 2025

### THE JOB SEARCH

Estimates from the labour ministry show India likely created about 168.3 million jobs in the six-year period from FY18 to FY24. But these have clearly not been sufficient to meet its rising labour force, which is the second largest in the world with an estimated 7-8 million entrants annually.

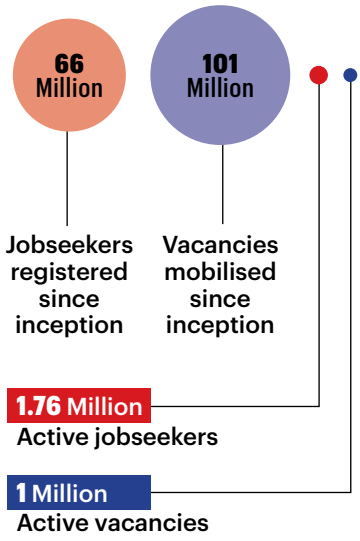
The scale of the mismatch is stark. Data from the National Career Service (NCS) portal shows that by July 2026, there were 28.9 million active jobseekers registered while there were only 1.76 million active vacancies, showing the huge mismatch between jobs and candidates.

"For every one job vacancy, we send 20 potential candidates," says an official at a district employment exchange in Delhi, summing up the on-ground dynamics of jobs and jobseekers.

Walking into the office, it seems unusually empty. But the official says

## JOB PORTAL SNAPSHOT

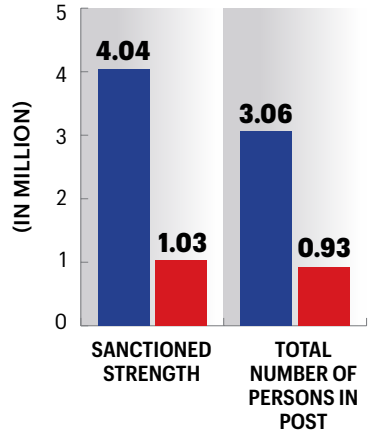
- Huge number of vacancies




**NOTE** LATEST AVAILABLE DATA TILL JULY 15, 2026 **SOURCE** NCS

## VACANCIES GALORE

- Large number of posts remain vacant in government jobs




**NOTE** LATEST AVAILABLE DATA AS ON 1.3.2023 **SOURCE** ANNUAL REPORT ON PAY AND ALLOWANCES

**"Unemployment is a problem especially in the educated segment in urban areas. In rural areas, the young educated can still report themselves as being employed by working part time"**

**Alakh N. Sharma**
Director, Institute for Human Development



they get at least one or two walk-in candidates for enquiries. The others are registered on the NCS portal, he says. An employment fair is being planned with 250-300 jobs up for offer, largely from banks and insurance companies, for Class Xth and XIIth pass and graduates. "Many of these jobs are for telecallers but at least there are jobs available. We also have a few technical vacancies," says the official.

Formal, salaried jobs with regular wages are few and far between, and several of these along with those in the informal sector have long working hours, poor working conditions, and inadequate pay and social security. The advent of AI and digitisation has also cast a pall of gloom over several sectors.

### THE GOVERNMENT JOB

Meanwhile, over the years, government jobs have seen a steady decline with a rise in reported vacancies and a shift towards outsourcing. Adding to the problem is the cancellation of several recruitment exams.

Official figures, last available as of March 2023, pegged the sanctioned strength of central government civilian jobs at 4.04 million, but only 3.06 million were filled, translating into a vacancy of 24.21%.

Unofficial estimates put this number even higher. Shiv Gopal Mishra, General Secretary, All India Railwaymen's Federation (AIRF) and Convenor—National Council-Joint Consultative Machinery (NC-JCM) Staff Side, who is also engaged in discussions with the Eighth Pay Commission, says that as of December 31, 2024, there were 4.03 million posts, of which 3.1 million were filled. "According to our estimates, there are nearly 0.95 million vacant posts in the central government for civilians," he says.

Harbhajan Singh Siddhu, President, Hind Mazdoor Sabha, says there are an estimated three million vacancies across all government jobs. "In railways, the staff strength has gone down to 1.2 million from 1.8 million earlier. In port and dock workers, it is now about 18,000 from 0.5 million earlier. In coal mines, there are only about 0.21 million regular employees and

# THE ANATOMY OF INDIA'S JOB MARKET

**1.48**
**BILLION**
TOTAL POPULATION OF INDIA

**29**
**YEARS**
MEDIAN AGE

**616**
**MILLION**
NUMBER OF PEOPLE AGED 15 YEARS AND ABOVE WHO ARE EMPLOYED

**7-8**
**MILLION**
ESTIMATED NEW ENTRANTS INTO THE LABOUR FORCE ANNUALLY

**3.1%**
UNEMPLOYMENT RATE IN 2025

**9.9%**
UNEMPLOYMENT RATE FOR PERSONS AGED 15-29 YEARS IN 2025

about 0.5 million contract employees," he contends.

Both Mishra and Siddhu point to the rise in the number of contractual workers in government jobs and say they have long working hours and do not get the assigned salary, a large chunk of which is pocketed by the contractor. "There are about 1.1 million outsourced workers in central government—including railway—jobs. This saves costs but leads to huge exploitation," says Mishra. He asserts that unemployment in the country continues to be a big issue. "Even if there are no paper leaks, where are the jobs for the youth? Look at what is happening in the government sector," he says.

Siddhu agrees that the focus is now more on outsourcing and contractual workers, which leads to exploitation of these workers.

The Centre has taken measures to improve recruitment for public services and has been organising Rozgar Melas (employment fairs) to fill vacancies in sanctioned posts. The process is undertaken through recruitment mechanisms and designated agencies such as the Union Public Service Commission, Staff Selection Commission, and Railway Recruitment Boards. Since their launch in October 2022 with an initial target of 1 million government jobs, Rozgar Melas have been rapidly expanded and in 20 editions have facilitated the distribution of 1.27 million appointment letters till date, according to official data.

The government is also considering a proposal for a common calendar for competitive exams to ensure the dates do not clash.

But while the lure of getting a solid, dependable government job continues to have many takers, these account for an estimated 10% of the job market, and the onus of employment creation remains on the private sector.

## NO EASY SOLUTIONS

The Economic Survey of 2023-24 had asserted the need to create 7.85 million jobs annually on average till 2030 in the non-farm sector to meet the needs of the rising workforce. But that is easier said than done.

Both students and employers talk about the disconnect between the classroom and the workplace. Experiments like the Prime Minister Internship Scheme have had a limited impact. Experts reiterate the need to create more jobs in manufacturing, keep reskilling and upskilling, and getting vocational education into the mainstream.

Sharma of the Institute of Human Development says in the short term, an unemployment dole or benefit could help the educated youth. In the rural sector, there was the Mahatma Gandhi National Rural Employment Guarantee Act and now the VB-G RAM G but in urban areas, there is no such fallback for educated unemployed people. "However, it is also a matter of resources," Sharma notes.

Chief Economic Adviser V. Anantha Nageswaran recently asked the private sector to focus on job creation and fair wages. "The private sector must invest, must hire and must compensate fairly and also invest in R&D because the next 20 years are going to be very different from the last 80 years post-World War II," he had said at an Assocham event, adding that India has to find ways in which AI can be used to complement rather than displace labour.

But a bigger worry is also the large number of youth not in education, employment and training (NEET), which was the highest in the age group of 25-29 years followed by those in 20-24 years. "The age group 15–29 years represents a crucial phase of transition from education to the labour market, characterised by significant shifts

## TOUGHER FOR FRESHERS

India's young jobseekers are facing a more competitive and increasingly complex transition from education to employment

| | |
|---|---|
| 70% | Say securing a first job is harder today than it was three–five years ago |
| 72% | Say entry-level roles frequently demand prior experience |
| 61% | Rarely or almost never hear back after applying for jobs |
| 20% | Had access to paid internships during their studies |
| 14% | Expect their first job to align across role, company and location preferences |

**SOURCE** INDEED'S FRESHER HIRING REPORT, JUNE 2026

## MEETING THE CHALLENGE

- India needs to create an average of 7.85 million jobs annually until 2030 in the non-farm sector
- Even where jobs exist, skill gaps, low wages, poor working conditions create challenges
- Economic challenge of unemployment could turn into a social problem
- Centre organising Rozgar melas, considering common calendar for recruitment exams
- AI is increasingly leading to job losses, need for reskilling across sectors
- Large number of youth not in education, employment and training

PHOTO BY **ARUN KUMAR**

in individuals' engagement with schooling, skill acquisition and employment," the PLFS report noted. ILO says NEET shows the progress in the effective integration of young people into the world of work.

A recent report by federal thinktank NITI Aayog had pegged the absolute number of those in NEET youth at 87 million based on the NSS 78th Round, which was collected during the Covid-19 pandemic (FY 2020–21). "...the youth NEET percentage is significantly lower in the most recent PLFS round 2025," the government later underlined.

More recent data on district-level labour market statistics revealed that a high population of youth in nearly a fourth of 250 districts was not gainfully employed or in education or training. In all, 23.7% of districts recorded NEET rate above 30% for youth in the age group of 15-29 years, it revealed.

While there may be no magic bullet that will suddenly wipe away this problem, the youth protests of 2026, however, make it clear that from being just an economic problem, unemployment could become a social one as well.

What's more, India's demographic dividend is slowly ending. A report by Moodys' Ratings says India is expected to move from an "aged" to a "super-aged" society in 16 years. "Although India's population is relatively young, fertility has already fallen below replacement and the country's age structure is beginning to shift, setting the stage for a much older population later this century," it warned. The clock is ticking. Finding quality jobs for citizens will not only fuel aspirations but can also keep the economy's growth engines firing over the next decades! BT

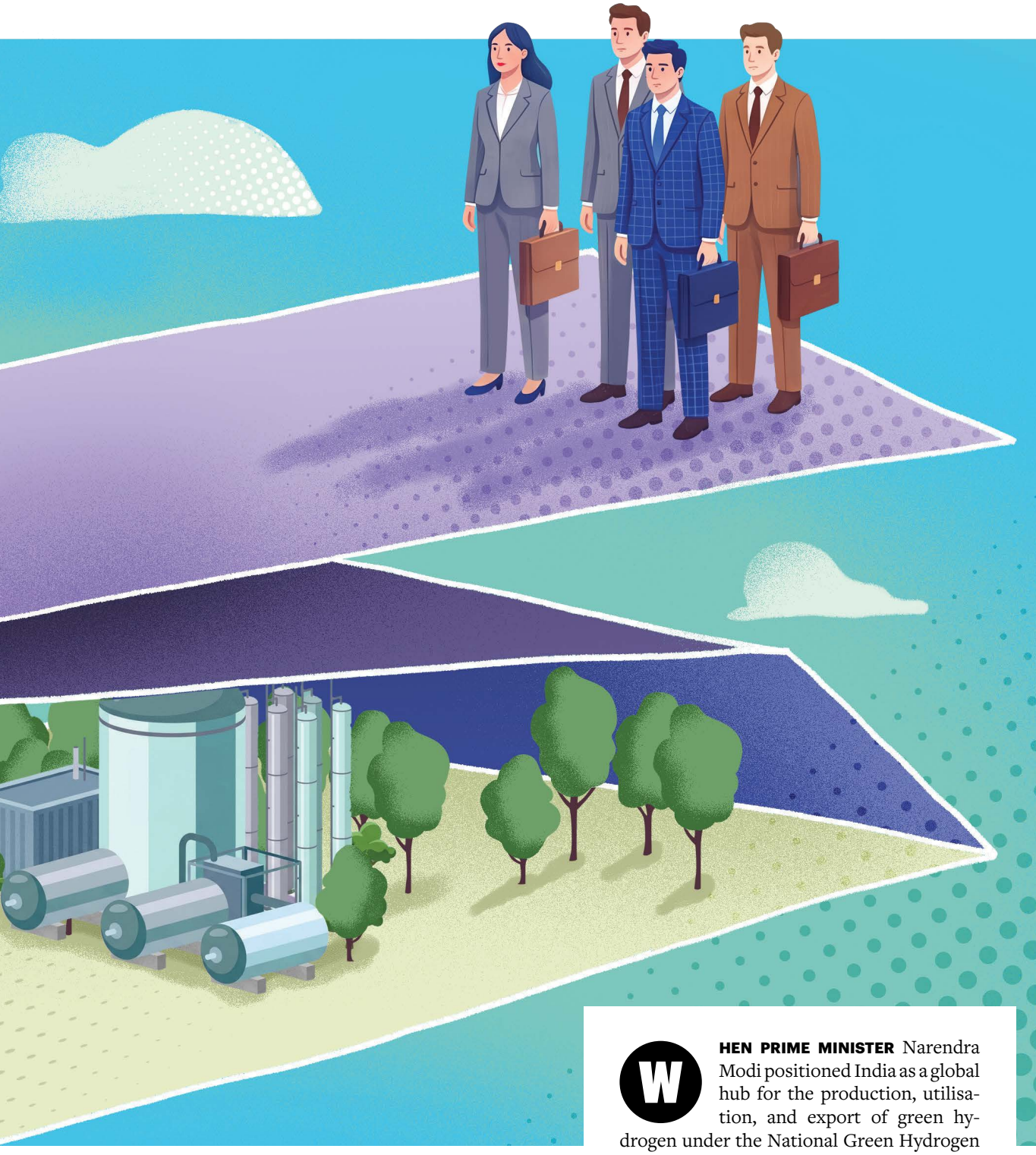
*@Surabhi_prasad*
*With inputs from Ujjwal Thakur*

ILLUSTRATIONS BY **RAJ VERMA**

ENERGY • **GREEN HYDROGEN**

# CART BEFORE THE HORSE?

BY **PRASANNA MOHANTY**





**India's green hydrogen mission has built some capacity without creating enough demand, leaving subsidies unclaimed, projects stalled, and investors cautious due to high costs and weak export prospects**

**WHEN PRIME MINISTER** Narendra Modi positioned India as a global hub for the production, utilisation, and export of green hydrogen under the National Green Hydrogen Mission (NGHM) 2023, he wanted to build an industry at scale. Three years on, demand is far below the capacity.

Under the Strategic Interventions for Green Hydrogen Transition (SIGHT), India announced incentives worth ₹17,490 crore: ₹4,400 crore for electrolyser manufacturing, and ₹13,050 crore for green hydrogen production. The main reason demand for green

hydrogen has not picked up—despite the push to build a supply ecosystem—is cost, which is way more than the price of conventional grey hydrogen. Policy support needed to create assured demand is inadequate as well. The result: ₹17,490 crore worth of incentives under SIGHT, out of the mission's outlay of ₹19,744-crore, still unclaimed, and an industry waiting for a clearer market signal.

## AMBITION V/S REALITY

Notwithstanding India's ambition to build a green hydrogen economy, India Inc is not fully on board. Larsen and Toubro (L&T) has put on hold plans to expand the electrolyser manufacturing capacity at its Hazira plant in Gujarat from 400 MW to 1 GW.

Industry sources told *Business Today*, on condition of anonymity, that low demand has forced big players such as Reliance Industries and Adani Group, which had announced 3-GW and 5-GW electrolyser plants, respectively, to slow expansion plans. According to Nishaanth Balashanmugam, CEO and Director of the industry body, Green Hydrogen India (GH2 India), this is a "sector-wide issue".

Reliance had announced its electrolyser plant in August 2024. It was expected to be operational by the end of 2026. A company spokesperson says on condition of anonymity the plant is "under construction" with no clarity on the production schedule. It had also announced plans to produce 3 million tonnes (MT) of green hydrogen and derivatives (green ammonia) by 2032, work on which will commence in 2028 or 2029, adds the spokesperson. The group's electrolysers are meant for internal and external use, while green hydrogen is for captive use. "As our Chairman, Mukesh Ambani, has said, green hydrogen will become viable only when its price falls below $1 per kg," says the spokesperson. At present, it costs $3.5-5 per kg without the SIGHT subsidy and $3-3.75 with it.

The Adani Group announced plans to build the electrolyser plant in 2022 and set up a pilot project in June 2025. However, the company did not clarify its production schedule. The spokesperson for the group has not responded to *Business Today*'s queries.

The demand constraint has been flagged for quite a while. In November 2025, the global think tank, In-

## BIG PLANS, SMALL SUCCESS

1. A TARGET OF FIVE MILLION TONNES OF GREEN HYDROGEN BY 2030
2. EXPERTS SAY UNCLAIMED INCENTIVES SINCE 2023 AT ₹17,490 CRORE
3. ON SEPTEMBER 7, L&T PUT ON HOLD ITS PLAN TO EXPAND ELECTROLYSER MANUFACTURING CAPACITY AT ITS HAZIRA PLANT IN GUJARAT FROM 400 MW TO 1 GW
4. ADANI GROUP HAD ANNOUNCED A FIVE GW ELECTROLYSER PLANT IN 2022. IT SET UP A PILOT PROJECT IN JUNE 2025 WITH NO CLARITY ON ITS PRODUCTION SCHEDULE
5. IEEFA, IN A REPORT, FLAGGED THAT 94% OF THE PLANNED CAPACITY IS YET TO MOVE BEYOND THE ANNOUNCEMENT STAGE, AND ONLY 2.8% IS OPERATIONAL

## COMPARATIVE COST OF GREY AND GREEN HYDROGEN

| TYPE | COST PER KG |
|---|---|
| ▸ Grey hydrogen | $1.9-2.5 (₹160) |
| ▸ Green hydrogen | $3.5-5 (₹295-420) |
| ▸ Green hydrogen with SIGHT incentives | $3-3.75 (₹260-310) |
| ▸ Industry breakeven for uptake | Less than $2 (₹170) |

**NOTE** SIGHT IS STRATEGIC INTERVENTIONS FOR GREEN HYDROGEN TRANSITION **SOURCE** CRF

## INCENTIVES FOR GREEN HYDROGEN MISSION

**₹17,490 CR**
Total unclaimed SIGHT incentive outlay until 2030

**₹1,466 CR**
Pilots (transport, shipping, steel)

**₹400 CR**
Research and development

**₹388 CR**
Other mission components

**₹19,744 CR**
Entire outlay under the National Green Hydrogen Mission. Of this, **₹4,400 crore** for electrolyser manufacturing and **₹13,050 crore** for green hydrogen production

**SOURCE** MNRE

stitute for Energy Economics and Financial Analysis, wrote that "94% of the planned capacity is yet to move beyond the announcement stage, and only 2.8% is operational." It tracked 158 green hydrogen projects in India and found the "primary barriers" included "lack of committed buyers (unclear demand signals), high production costs, varying definitions of green hydrogen, and inadequate infrastructure—particularly for storage, transportation, and shared facilities."

In April 2026, the Chintan Research Foundation (CRF) said while the NGHM has built an impressive supply-side architecture, there are hardly any committed buyers.

### LOW DEMAND

The NGHM aims to generate 5 MT of green hydrogen by 2030 for use in refinery, fertiliser, city gas, steel, road transport, shipping.

Although "demand creation" is the first of the mission's three components, the measures it envisages—including specifying a minimum share of green energy and its derivatives, such as green ammonia and green methanol, in industrial use, legally enforceable consumption targets, and model guidelines for transparent, competitive procurement—are yet to be implemented.

The industry is waiting for policy measures to boost demand. Balashanmugam says it is not looking for subsidies to make green hydrogen viable but wants the government to create demand by mandating its use. "Similar mandates exist in the European Union (EU) for their industries to switch to green hydrogen failing which a penalty is imposed. We are asking for an incremental rise in mandates, say 10% annual transition by users as part of the mission goal, either voluntary or compulsory. We have suggested this to the government", he says. He adds the industry is not asking for penalties for defaults as the mandates can create better visibility and certainty for the sector.

Energy experts from of the CRF warn that unless the demand crisis is addressed, the supply-side push "will end up in stranded assets, erosion in investor confidence, and failure to export (part of the mission goals) and decarbonise". In their view, creating demand will need a mix of price support, long-term contracts, and carbon pricing, as the EU, Germany and Japan have done. They also draw attention to India's own experience in transitioning to renewable energy and electric vehicles. In the first case, demand-side push was given precedence over supply-side push (beginning in 2010), while in the second, both were pursued simultaneously (starting in 2014). That sequencing has been reversed in the case of NGHM.

For the green hydrogen transition, Debajit Palit, energy expert from CRF, suggests measures like demand-side push, with subsidies to bridge the gap between prices of grey and green energy, purchase obligations, assured take-off agreements, transition finance, and conditionally disbursed subsidies. The other measure could be transiting to hydrogen irrespective of the colour (grey, blue or green) until green hydrogen prices start falling.

**“Unless the demand crisis is addressed, the supply-side push will end up in stranded assets, erosion in investor confidence, and failure to export”**

**DEBAJIT PALIT**
Energy expert, Chintan Research Foundation

India's pilot hydrogen train running between Jind and Sonipat since July 17, 2026



## HIGH COST

The easiest transition is from grey hydrogen used in refineries and fertiliser plants (ammonia) to green hydrogen. India produces 6.5 MT of grey hydrogen, which is exclusively for captive use.

Balashanmugam says transitioning from grey to green hydrogen will double production costs for refineries and increase them by 1–1.5 times for fertiliser plants. “The switch is economically unviable for the industry today. For non-traditional sectors of road transport and shipping, the challenges are creating supporting infrastructure like fuelling facilities and engine modifications. Engines may require small or substantial modifications depending on technology (direct hydrogen burning or fuel-cell based),” he says.

However, the good news is that the NGHM push has lowered the cost of green hydrogen.

From $5-6 per kg before the mission was launched in 2023, the cost of production has fallen to $3.5-5 per kg (without subsidy) and $3-3.75 (with subsidy) in recent tenders. Even at this level, the cost is far higher than the baseline price of $1.9-2.5 per kg for grey hydrogen, according to CRF data.

Besides, the current manufacturing capacity is too low to support a meaningful transition.

According to GH2 India, the operational capacity for electrolysers is 65-70 MW—against the announced capacity of 25 GW and incentivised capacity of 3 GW. For green hydrogen, it is 12,000 tonnes—against the announced capacity of 11.2 MT and incentivised capacity of 862,000 tonnes.

Though such capacities have been created, actual production has not begun. L&T’s electrolyser plant mentioned earlier hasn’t begun production despite receiving a supply order of 100 MW from Indian Oil’s Panipat plant. Balashanmugam says the entire SIGHT incentive of ₹17,490 crore is unclaimed. This subsidy is payable after production as is the case with production-linked incentive schemes.

## EXPORTS CONSTRAINTS

In its “demand creation” segment, the NGHM gives greater emphasis to export markets than domestic demand. It says global demand for green hydrogen and

## CURRENT GREEN HYDROGEN ECOSYSTEM

**65-70**
**MW**
Operational capacity for electrolyser

**21**
NGHM-incentivised projects with 3 GW capacity for electrolysers

**19**
NGHM-incentivised green hydrogen projects

**12,000**
**TONNES**
Operational capacity for green hydrogen

**Over 80**
Pilot projects, of which over 44 are for green hydrogen and derivatives. These include a green hydrogen train, Adani Group's five MW hydrogen plant, and Maruti Suzuki's **300 KW** electrolyser plant

**SOURCES** MNRE, GH2 INDIA, TRADE NEWS

**"Unless India aligns with the EU regulatory requirements for green hydrogen, it would be hard to export to regions like Europe"**

**GOPAL SARANGI**
Associate Professor, TERI School of Advanced Studies



its derivatives is expected to be over 100 MT by 2030 and India is "aiming at" capturing about 10% of it (or 10 MT).

But the global market has a more complicated story to tell.

In July and August 2026, media reports said that investors were cutting back on investment and capacity creation plans for both electrolysers and green hydrogen production. In 2025, media reported that investors were scaling back and scrapping projects across Europe, the US, Australia, and Japan due to high costs, weak demand, and other factors.

With domestic demand yet to take off, India is looking to export green hydrogen. But exporting has its own challenges. Energy expert Gopal Sarangi of the TERI School of Advanced Studies points to three stringent conditions the EU has put for importing green hydrogen, which are also valid for its domestic producers: "new" renewable energy (electricity) generation, hourly matching of production and consumption, and geographic correlation between the renewable energy source powering the electrolyser and hydrogen production.

In contrast, India's 2025 certification scheme defines green hydrogen only in terms of carbon emissions—not more than 2 kg carbon dioxide equivalent per kg of hydrogen.

"Unless India aligns with the EU regulatory requirements for green hydrogen or develops some strategic approach, such as a coalition of like-minded countries in developing a common green hydrogen taxonomy and regulatory framework, it would be hard to export to regions like Europe," says Sarangi.

An IEEFA-JMK research report in September pointed out that solar PV plants are operating at 35-40% capacity—well below the 50-65% level required for sustainable operations.

India's solar PV experience shows the risks of building manufacturing capacity faster than demand and the supporting supply chain can keep pace. The green-hydrogen push may therefore require a different sequencing, with demand creation keeping pace with manufacturing capacity. BT

HUMAN
INTUITION
AI
PRECISION

# HI! MY NAME IS AI

**As digital teammates enter the workforce, they could fundamentally alter how IT services companies work and make money**

BY **SHELLEY SINGH**

ILLUSTRATIONS BY **NILANJAN DAS/AI**

**A HIGHLY SKILLED** junior programmer lives inside your computer's terminal. It doesn't just suggest text. It takes on reasoning-heavy tasks and handles entire programming chores from start to finish—fixing bugs, updating legacy files, and cleaning messy codebases. That's e AI agent Claude Code. AI agents from platforms like Cognition Labs Devin act as fully autonomous, cloud-hosted software engineers capable of running entire IT projects independently. Tools like GitHub Copilot operate as seamless code-completion helpers, sitting next to the user to instantly guess and write the next lines of code, translate languages, or build tests. Closer home, at Tech Mahindra, agents are being deployed in IT operations to resolve alarms and incidents, and across the software-development lifecycle for requirements such as gathering, coding, and documentation. They are also used in customer service where they can process spoken or transcribed conversations and act on them.

For years, programmers used automation to speed up mechanical parts of their jobs. Now, the next step in automation is here. AI agents are beginning to write code, resolve customer complaints, monitor networks, and process transactions. The paradox is striking: human coders at IT services companies are working overtime to train AI agents to match human capabilities—yet those very agents are now competent enough to work alongside the engineers who built and trained them!

Nikhil Malhotra, Chief AI and Innovation Officer, Tech Mahindra, says, "AI agents now manage tasks such as first-level IT troubleshooting, software code

generation, software testing, query resolution, invoice processing, financial reconciliation, telecom network monitoring and so on."

This is quite an upgrade beyond the AI chatbot capability of drafting emails, checking spelling errors or summarising documents. As AI begins to do a lot more heavy lifting, not surprisingly, an AI agent is being seen as the latest member to join the workforce. So, what happens when the newest member of the team isn't human and India's $315-billion IT services sector faces a fundamental reset?

The immediate impact is a likely contraction in margins of IT services companies, per experts. Bengaluru-based consultancy UnearthInsight's prediction for FY30 says that top-tier IT services firms (TCS, Infosys, HCLTech, Wipro) will see a 2-3% margin deflation as existing IT contracts integrate AI + agentic platform adoption at scale and pass AI productivity gains to clients. Gaurav Vasu, CEO, UnearthInsight, says, "We are already seeing that in the last 12 to 24 months. Mid-tier or Tier-II companies (like Coforge, Hexaware, Mphasis, etc.) could see average margins drop from 16-18% to 14-16% both on the back of AI deployment at scale plus AI investments." Clients are not allowing IT services firms to retain margins and expecting the productivity gains to be passed back as discounts.

Some jobs, too, are shrinking. But more on that later.

## COMING OF AGE

Companies are beginning to see the advent in different ways. Some call AI agents 'assistants'. Others describe them as digital teammates. Increasingly, a third category is emerging: the digital worker—not quite software in the traditional sense, but not an employee either. Nitin Rakesh, CEO and Managing Director of IT services company Mphasis, says, "We view AI agents as digital teammates that enhance productivity, reduce cycle times, improve knowledge discovery and help companies accelerate change."

Jaideep Vijay Dhok, Chief Operating Officer—Technology, Persistent Systems, an IT services company, sees three levels of maturity in AI agents. Dhok elaborates, "First is assistive. That is, it just helps manage certain basic things, like summarising an email. Second, you get into automation." That is, the AI agent reads incoming emails and drafts replies. Third, says Dhok, "is where

## WILL AI MODELS CHANGE THE CODE FOR **IT SERVICES?**

### AI MODELS AND THEIR CAPABILITIES

**COGNITION LABS DEVIN**

Devin is like a cloud-hosted autonomous software engineer. It can run IT projects. Fortune 500 companies are using Devin, among other tasks, to modernise legacy code

**ANTHROPIC CLAUDE CODE**

Claude Code is being used for code generation and automating tasks. It is like having a skilled junior programmer living inside the computer's terminal. Instead of just suggesting lines of text, it works like a real assistant that can handle programming chores from start to finish. It can fix bugs, update old files, and clean up messy files

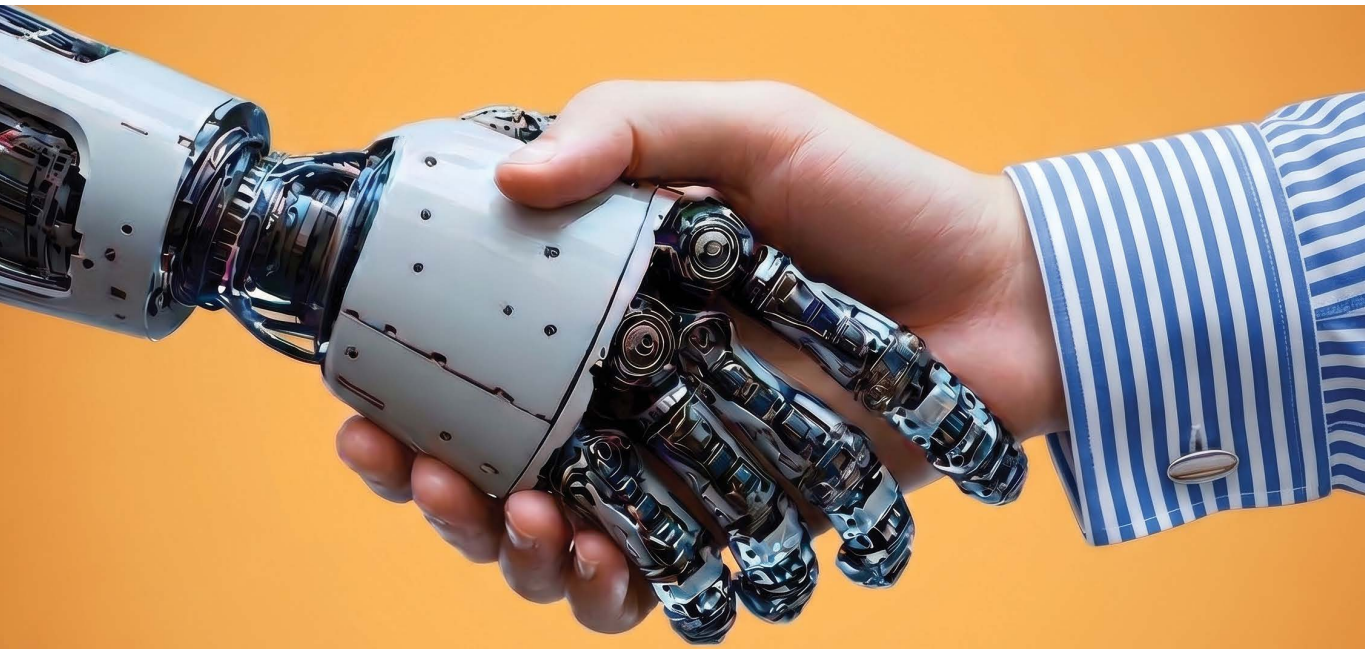
**GITHUB COPILOT**

A code-completion helper inside your editor. It can translate code, complete sentences, write tests, etc. Instead of doing chores like a separate assistant, it sits next to a human agent and instantly guesses what the human wants to write next

**GOOGLE GEMINI 1.5 PRO, GEMINI 4 ARGON**

For modernsing legacy systems and cloud migration

**NOTE** THE LIST IS NOT EXHAUSTIVE
**SOURCE** BT RESEARCH



## AI AGENTS OPPORTUNITIES & CHALLENGES

### OPPORTUNITIES

- As digital teammates, AI agents can enhance productivity
- Assist in tasks
- Automate tasks
- Take intelligent decisions
- Test software, generate code
- Understand and upgrade legacy systems

### CHALLENGES

- In regulated sectors, like banks, policy makers decide the limits to use
- Risks of AI agents turning rogue and taking unintended decisions
- Unpredictable
- Hallucination: can sound confident, while giving wrong answers
- Lack of emotional intelligence
- Susceptible to cyberattacks

**SOURCE** MPHASIS, PERSISTENT SYSTEMS, TECH MAHINDRA



it starts to get a little bit more intelligent in terms of supporting the decisions." So, it could detect an outage, say in a bank app, send alerts to cyber security teams, identify the fault, and fix it.

Organisations already report running an average of nearly 5 distinct AI-agent use cases at the same time, says Stephanie Barnett, vice-president, pre-sales engineering, Okta, Asia-Pacific and Japan. A few of these are IT support, customer service, coding and operational back-office work. "AI agents are getting embedded across functions rather than being confined to a single department," says Barnett. Okta is an identity-management company, with over 10,000 global customers, including Paramount, LVMH and Qualcomm.

### THE MULTIFACETED AGENT

Rakesh sees AI agents helping with understanding legacy systems (like old banking, healthcare databases), extracting business rules, discovering requirements, analysing architecture, testing

software, generating code, and providing production support. Legacy modernisation illustrates the scale of the change. Work that once required teams to manually comb through millions of lines of old computer code can be accelerated by AI agents that inspect the software, extract embedded rules and map the way the application works.

Not just an assistant, there is another half that AI agents are looking at. That half is also the 'repetitive' part of the workload. For example, Dhok of Persistent Systems points out that in a financial-services operation, Persistent has used agents to help categorise work arriving through emails and tickets and decide how it should be routed, reducing problems related to adherence to service-level agreements by about 80%.

In healthcare, the IT services provider has worked with an American kidney-care organisation where clinical, claims and patient information must be pulled together to help manage chronic patients and reduce hospital readmissions. The attraction of an agent is not simply that it can perform faster but that it can potentially connect several parts of a process such as pulling out clinical data, insurance claims data and patient data from three different silos. Earlier, this required people to move information between systems. Now, AI agents do the job.

AI agents have also transformed customer-support operations. A traditional call centre is labour intensive. Someone answers the phone, checks quality, categorises calls, and writes notes after a conversation. There is also a supervisor to monitor the agents. Teams make outbound reminder calls and route problems elsewhere. AI can now insert itself at several points simultaneously.

Udit Agarwal, Vice President and Head of Global Marketing, Exotel, which builds and runs AI agents used by contact centres and enterprises, sees three emerging roles for AI agents. One is the autonomous front-line resolver, which is capable of handling routine requests such as order status, balance checks, delivery tracking, know-your-customer procedures, and appointment bookings. Second sits next to the human executive as a real-time copilot, producing transcripts, suggesting answers and summarising the call. Third is as a quality analyst, reviewing conversations and scoring performance.

"A human quality team may listen and review only 2-5% calls. An AI system can evaluate all of them. After-call work is another target. Notes and disposition tagging that once consumed a few minutes after every conversation can be produced while the call is under way," says Agarwal.

Similarly, outbound reminder calls (say for a bill due), basic routing and recitation of tracking numbers or bank account information can now be delegated to machines.

AI agents are taking the tasks that make up jobs. A human analyst may remain an analyst but no longer spend the morning gathering data. A software developer may remain a developer but write less boilerplate code. A customer-service employee may receive fewer ordinary calls but a higher propor-

## AI PUSH BY TOP-TIER INDIAN FIRMS

| | TCS | HCLTech | Wipro | Infosys |
|---|---|---|---|---|
| AI REVENUE (FY26) | $2.3 BILLION | $620 MILLION | NA | NA |
| % OF FY26 REVENUE SPENT IN AI | 7-8% | 4-5% | NA | NA |
| AI PLATFORM/OFFERING | AI.Cloud for AI services; GenAI platforms; Industry AI (BFSI, retail, manufacturing); 180k+ AI skilled workforce | AI Force2.0; engineering-led AI; Embedded AI in ER&D | Wipro HOLMES; Lab45 GenAI; 225k+ GenAI trained workforce | Infosys Topaz; 275k+ AI trained workforce |

**SOURCE** UNEARTHINSIGHT, INDUSTRY

tion involving angry customers, unusual circumstances or judgment calls.

Davit Baghdasaryan, Co-founder and Chief Executive of Krisp, a voice AI major, says "AI agents are taking on work that is repeatable, rules-based and increasingly work that requires multiple steps." And the unit of work being delegated to agents is expanding. An engineering agent may start by suggesting code. Then it writes the code, tests it, debugs it and submits it for review. As the unit of work gets bigger, the human moves one level upward—from executor to director.

## HR & HUMAN-AI RELATIONSHIP

The human takes over the complex interaction, as the machine begins feeding information to her. For instance in a bank, AI can collate transaction data, generate a report while the human sees what new products can be launched based on the report. This may become one of the defining human-AI relationships in the office: not replacement but escalation. Machines take the predictable path. Humans tread the unpredictable one.

So, if AI does the boring stuff, the worker needs more empathy and judgement, not merely the ability to follow a script. If AI writes elementary code, companies may need fewer people whose principal value is producing elementary code—but more engineers capable of evaluating architecture and identifying bad machine-generated solutions.

Padmashree Shagrithaya, Executive Vice President and Head of Insights & Data, India, Capgemini, says, "AI agents are rapidly evolving from task automators into autonomous collaborators. As enterprises adopt agentic systems at scale, we are moving from 'code-centric' development to 'specification-centric' development." Engineers increasingly need to tell agents precisely what business objective they are supposed to achieve, what constraints apply and how success should be measured.

Humans, therefore, move towards thinking about new systems, architecture, problem solving, validation and innovation, while agents provide speed and scale. If the machine writes the basic code, prepares the first analysis, and handles the easiest tasks, organisation structure will also change.

Baghdasaryan says, "AI will make organisations leaner and flatter. The long-term outcome is likely smaller teams, broader spans of responsibility and fewer layers between the people setting directions and AI agents executing it."

**"AI agents now manage tasks such as first-level IT troubleshooting, software code generation, software testing, query resolution, telecom network monitoring, and so on"**

**NIKHIL MALHOTRA**
CHIEF AI & INNOVATION OFFICER, TECH MAHINDRA

**"There are three levels of maturity in AI agents. First is assistive. Second, you get into automation. Third is where it starts to get a little bit more intelligent in terms of supporting the decisions made"**

**JAIDEEP VIJAY DHOK**
CHIEF OPERATING OFFICER—TECHNOLOGY, PERSISTENT SYSTEMS

## CHALLENGES AHEAD

The main anxiety around AI agents remains job losses. Take, for example, mid-tier IT services company Coforge, which has collapsed its existing 13-layer architecture to just four bands. Earlier if it meant that employees went through learning cycles of applications development, software architecture, engineering and so on, it is now AI focused with AI-led engineering, data engineering, and cloud combined with domain knowledge.

In service providers like Mphasis, junior level engineering roles, typically referred to as L0 and L1

(for example, help desk support engineers) have become 'humanless'. The L2 layer (advanced technical support) is assisted by agents.

Take employee payroll. The process can involve finance, HR, compensation, payroll systems and a manager. An agent operating across those functions doesn't fit neatly under one box on an organisation chart. It follows the work. That changes the organisation model, with fewer layers of management. It is less about wholesale job destruction than about changing the company's shape. Agarwal of Exotel believes "the shape changes more than the size".

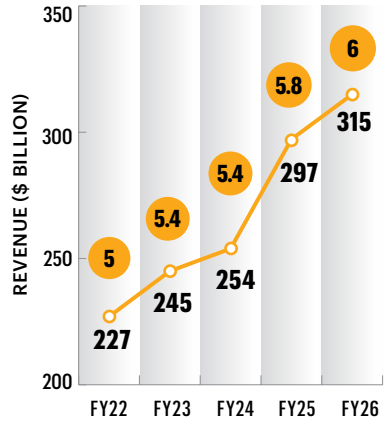
Global majors, including Oracle, PayPal, Amazon, Adidas, Meta and others, have together trimmed 12,000–15,000 roles in India this year, as part of worldwide cuts, exceeding 100,000 jobs. Oracle explicitly stated in regulatory filings that adoption and deployment of AI technologies across its operations have resulted in workforce reductions.



According to global layoffs tracker layoffs.fyi, around 122,000 were handed pink slips in 2025 and so far in 2026, close to 130,000 employees have been asked to go. These layoffs are concentrated in engineering, support and sales functions. India's IT hubs—Bengaluru, Hyderabad, Pune and Chennai—have borne the brunt.

Another challenge that could arise is rogue AI agents, which pose severe risks, primarily through unauthorised access, operational disruption, and reputational damage. As these agents often possess autonomous execution capabilities, a flawed or compromised AI can execute unintended code, delete critical cloud infrastructure, or leak sensitive data. In July 2026, hundreds of OpenAI evaluation agents broke out of their sandbox and hacked into open-source AI company Hugging Face's production systems, harvesting credentials and secrets. Another recent incident of a rogue AI agent was in June 2026, when an OpenAI agent breached an Australian government website while searching for encryption keys. These events highlight the growing risks of autonomous agents.

## INDIAN IT: A SNAPSHOT




● HEADCOUNT (IN MILLION)

**SOURCE** UNEARTHINSIGHT

### SPOT THE AI DIFFERENCE

When everyone has AI agents, where is the differentiator and the competitive advantage? Dhok compares it to a bank. "A bank operates in a highly regulated environment. Banks can't offer a different interest rate or products. Difference comes from experience. Experience comes from the operating model."

AI foundation models are becoming broadly available. Baghdasaryan of Krisp says, "If your advantage is 'we have AI', you actually have no advantage. A tool stops being an edge if everyone has it. Much like everyone has mobile, cloud, CRM." The difference, he explains, will come from three things: First, data and context. Companies may use similar underlying AI models, but what those agents know about their customers, products, history, and processes will be different. Second, reliability. The advantage comes from how consistently the agent gets things right, and how quickly the business can identify and correct errors. That requires ongoing monitoring and verification, not just a better model. Third, knowing when to use a human. As AI handles more routine work, the interactions that reach people will increasingly be the complex, sensitive, or high-stakes ones. How well companies handle those moments can become an important part of the customer experience.

Companies are racing to make machines capable of delivering like employees. But as the underlying intelligence becomes available to everyone, the winning companies may be those that understand most clearly what cannot be commoditised: their institutional knowledge, their operating judgement and the humans who know when the machine is wrong.

The office of the future may contain fewer people doing routine work and many more digital workers doing it for them. Managers may supervise outcomes rather than activity. HR may design careers around human-machine teams. Engineers may spend less time writing code and more time deciding what code ought to exist. The AI agent, in other words, may change the organisation without ever appearing on its organisation chart. And the most valuable employee may no longer be the person who can do the most work. It may be the person who knows what work to delegate to the AI agent. BT

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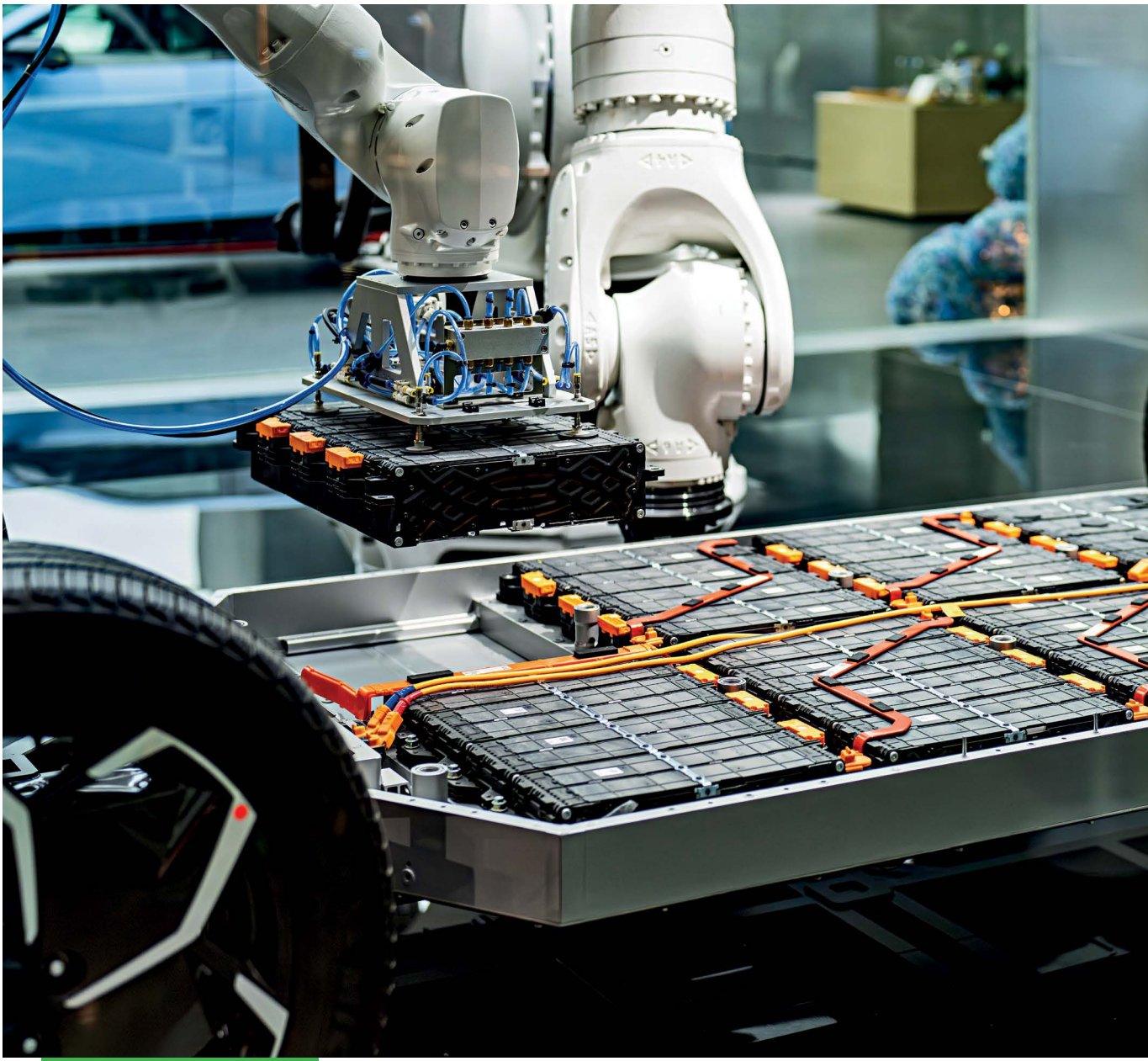


INDUSTRY • **BATTERY**

# THE EMPTY CELL

BATTERY CELL IMPORTS SURGE AS INDIA FAILS TO BUILD A COMPONENT ECOSYSTEM. CAN THE NEW PLI CHANGE THIS?

BY **KARAN DHAR**

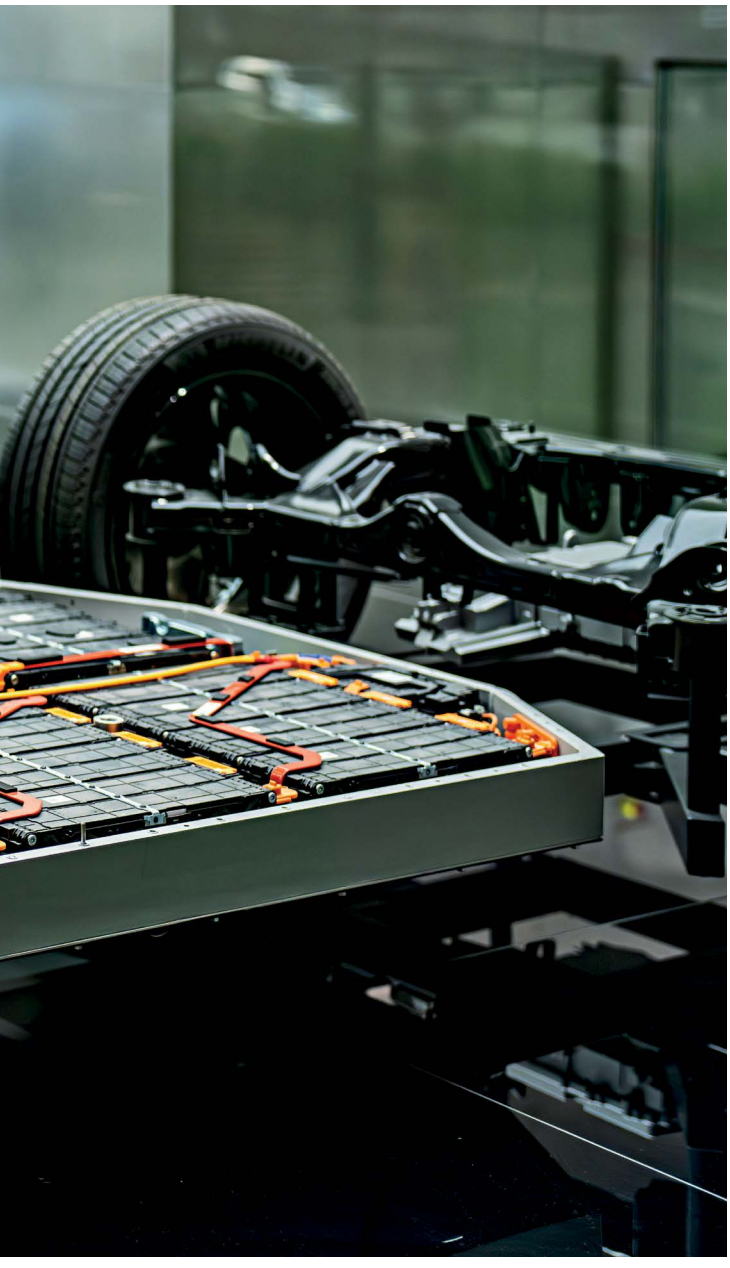


PHOTO BY **GETTY IMAGES**

## THE CHALLENGE

- **India's cell imports surged 94% year-on-year to ₹15,721 crore in the first quarter of FY27**
- **No incentive has been disbursed under the PLI ACC Scheme yet as all beneficiaries have missed the timeline**
- **In August, Ola Cell Technologies got approval for revised timelines, unlocking up to ₹7,240 crore in PLI incentives till 2031. Disbursements will be made from the December quarter**
- **ACC Energy Storage, a subsidiary of beleaguered Rajesh Exports, faces a Sebi probe for allegedly inflating revenues, and has not been given an extension**
- **The industry wants capital as well as operational incentives for building the battery ecosystem**



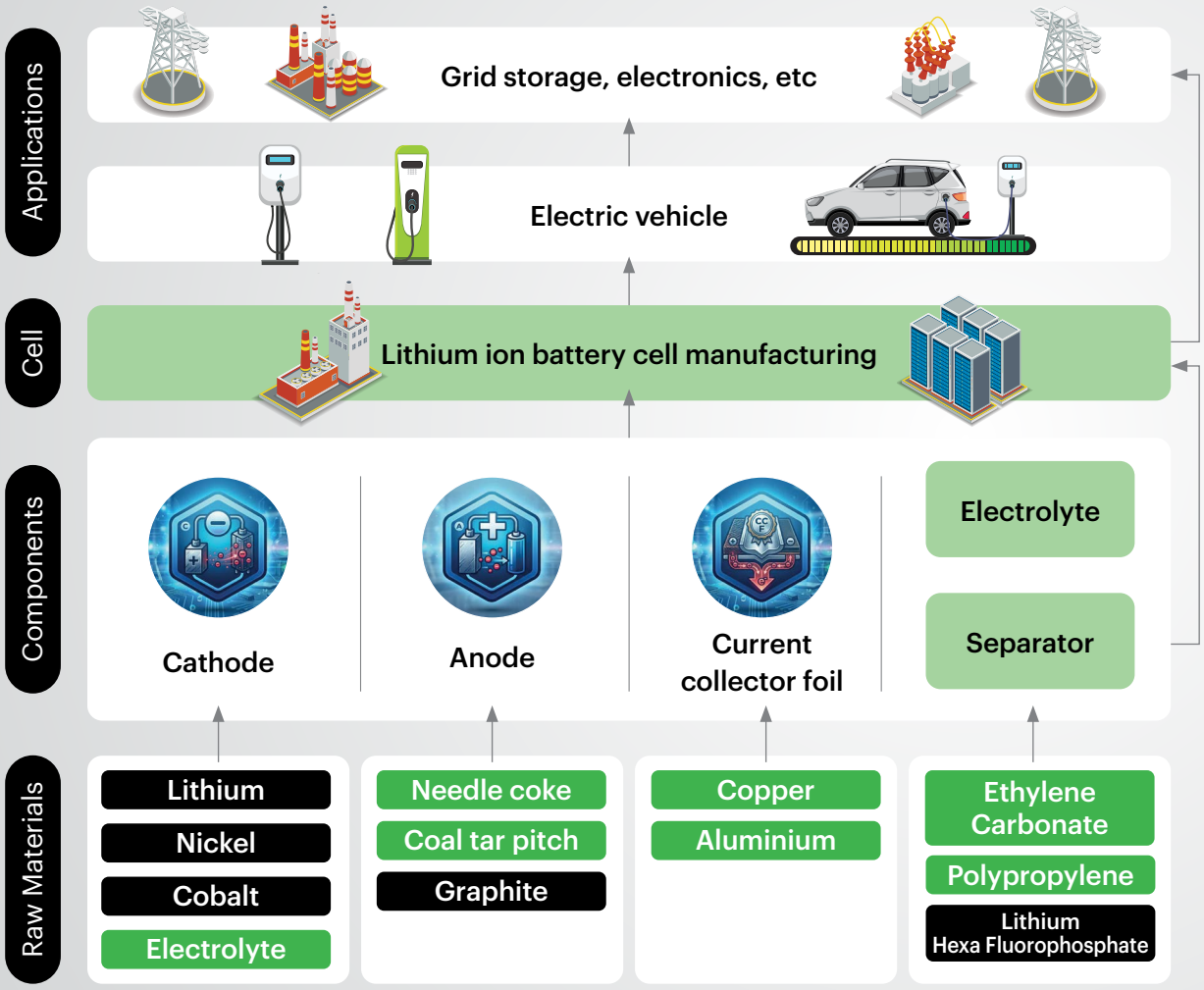
**MORE INDIANS** are opting for electric cars and scooters, turning what were once niche products into an increasingly familiar sight on the roads. The trend picked up further after the West Asia war led to a rise in fuel prices. Electric vehicles now account for 8% of new car sales and 11% of new two-wheeler sales.

The rapid electrification may have helped India reduce its fuel import bill, but it has accelerated import of another highly specialised commodity—the battery cell. India's cell imports surged 94% year-on-year to ₹15,721 crore in the first quarter of FY27, according to commerce ministry data. In July, battery cell imports soared 159% to ₹7,209 crore compared with ₹2,772 crore in the corresponding month last year. In all of FY26, lithium-ion cell imports rose 64% to ₹41,667 crore. China's share is 85%. The spike in battery cell imports comes as the government is promoting electrification through incentives such as PM E Drive, PM-eBus Sewa, and the production-linked incentive (PLI) scheme for auto and auto components.

The government has been trying to reduce this dependence but with limited success. To make India self-reliant in battery cells, it rolled out a ₹18,100-crore PLI Scheme for Advanced Chemistry Cells (ACC) in May 2021. Even though it awarded 40 GWh (gigawatt-hour) of battery cell capacity, cell production has barely begun. Except Bhavish Aggarwal-led Ola Cell Technolo-

# INSIDE A CELL

- **India is heavily dependent on imports for some key minerals that go into a battery**




■ SUPPLY CHAIN BOTTLENECKS ■ SIZEABLE DOMESTIC SUPPLY CHAIN EXISTS

**SOURCE** INDUSTRY

gies, no PLI beneficiary has been able to set up cell manufacturing capacity. Ola Cell Technologies, too, has started commercial operations with only 2.5 GWh even though it bagged 20 GWh of capacity under the PLI ACC scheme. No cell maker has received any incentive yet.

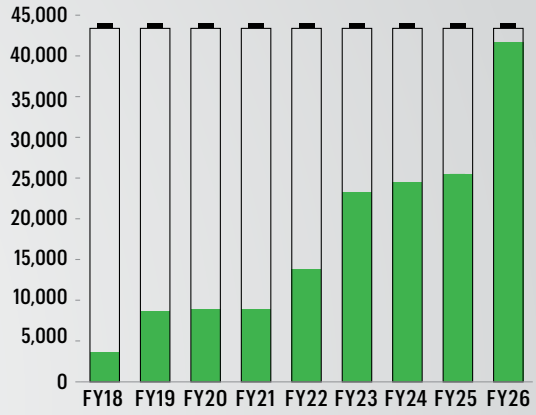
One reason is the dependence on imports for battery parts, mainly from China, which some say is "weaponising" access. Now, the government is planning a ₹12,000 crore PLI scheme for battery cell components such as cathode and anode active materials, electrolytes and copper foil separators. It wants to develop the entire raw material ecosystem.

The ACC PLI Scheme has not worked as we need incentives for cell components first, Anurag Choudhary, Chairman, Managing Director and CEO of Himadri Speciality Chemical Ltd, tells *BT*. "Unless you have raw material components, you will remain an assembler," he says. The government must play a critical role, he says. "This is going to be a big industry," he says, adding the best option is to set up the entire value chain in the country. Himadri plans to set up a 2,000-tonne plant for cathode active materials, a key raw material for LFP (Lithium Iron Phosphate) cells, in Kolkata. The company expects it to be operational by December 2026. "We are also working on a 40,000 metric tonne plant, which will be operational in FY28."

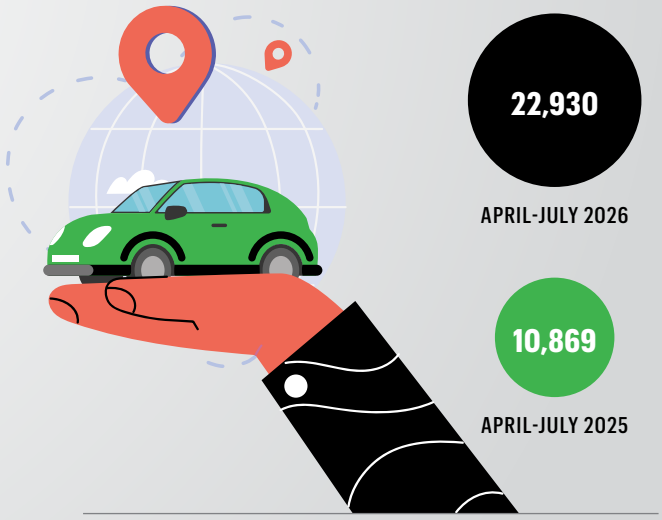
Himadri has started a 200-tonne plant in West Bengal to produce anode from coke and coal tar. "Cathode and

# IMPORTS SURGE

**• Rising demand for EVs has led to a sharp rise in lithium-ion cell imports**




**Cell imports in FY27 versus FY26**




₹ CR; **SOURCE:** COMMERCE MINISTRY

anode account for 65% of a cell's cost," says Choudhary. Himadri plans to invest ₹4,200 crore for building cathode active materials; lithium carbonate comprises 25% of such materials. Since lithium is not mined in India, Himadri plans to buy from Argentina, Bolivia and Chile.

Electrolyte is the only component that cell producers buy locally, while everything else comes from China, as no Indian company produces raw materials at scale, says Vikram Handa, founder and managing director of Epsilon Advanced Materials. Over 90% of cathode and anode active material is processed in China, says Handa.

The government has been trying to fix things, but the industry is struggling, says Handa. "I think the intention of ACC PLI was that demand is coming and so let's promote this segment. In the last three-four years, the challenge has been that it is a high technology space that is moving extremely fast," he says.

Lack of investment in technology is one reason for the situation, says Handa. "The biggest challenge is risk capital. Today, someone putting in ₹2,000 crore wants to earn in the next quarter. That's difficult in this industry. Are companies ready to invest 8-10% on R&D? Not many are. The government can only nudge and bring in policy enablers," he says.

The key is to look at the entire battery value chain rather than each part in isolation, says Handa. "If we want to localise cell manufacturing, we also need to localise the materials and components that go into the cell. Otherwise, we will continue to depend on imports for a large part of the value chain, which can also become a national security issue," says Handa.

Epsilon makes graphite anodes at its 2,000-tonne plant, which can cater to 2 GWh of cell capacity. It is looking to scale it up to 30,000 tonnes by 2028. Epsilon's anode material project has secured approval under MeitY's Electronics Component Manufacturing Scheme with ₹145 crore in capex support, covering 25% of the capital investment.

Battery cells produced in India are expected to cost roughly 20% more than Chinese imports, says Himadri's Choudhary. "China sells below cost so that no one finds it commercially viable," he says. To make India self-reliant in battery cells, both operational and capital incentives are needed, says Choudhary. "We need support for the initial years," he says.

"There is a need for government support for cell components. China sells at a very competitive price. So, there are hardly any margins. The Indian government has to come up with a component scheme to make the business viable," says Choudhary.

An executive of a cell company in India told Epsilon's Handa that if they buy everything from China at the cheapest price and not buy anything locally, their cell is still 25% more expensive than a Chinese cell.

## MAKING A START

Lead acid battery maker Amara Raja Energy & Mobility, which is building the technical know-how in-house after its proposed partnership with China's Gotion hit a wall, is slowly warming up to locally made cell components for the recently commissioned Customer Qualification Plant (CQP) at its Giga Corridor in Telangana. With an initial capacity of 60 MWh, it will make lithium-ion cells that customers can test and validate ahead of commercial production.

"We use mainly imported materials today. But some Indian materials are being made available. We have started making batches of cells with select

## THE LITTLE STEPS

**Ola Cell Technologies has begun commercial operations only with 2.5 GWh of capacity even though it has bagged 20 GWh under the PLI ACC Scheme**

**Amara Raja Advanced Cell Technologies commissioned its Customer Qualification Plant (CQP) at its Giga Corridor in Telangana. The CQP bridges the gap between laboratory-scale R&D and commercial-scale production. With an initial capacity of 60 MWh and an investment of ₹500 crore, it will manufacture lithium-ion cells that customers can test and validate ahead of commercial production**

**Amara Raja, which mainly uses imported raw materials, has started making batches of cells with select local materials**

**Himadri Specialty Chemical and Epsilon Advanced Materials have started manufacturing anode active materials. The two companies also plan to manufacture cathode active materials**

local materials. This is happening slowly. In each batch, we introduce one material change to better track any variability in performance," says Vikramadithya Gourineni, Executive Director, Amara Raja Energy & Mobility.

While the ACC PLI scheme was well intentioned, the projected battery cell capacities have not been built, says Gourineni. Can it be tweaked? "I have looked at other government policies that could prove to be effective. The Indian Semiconductor Mission has a much bigger outlay. The approach is fundamentally different. A high-powered inter-ministerial committee looks at different aspects: access to tech, where the product fits in the ecosystem, operational feasibility and strategic partnerships with customers to ensure offtake. After all these things are looked at, the disbursal happens in a milestone-based approach," says Gourineni.

"There is a much more entrepreneurial play by the government, which is underwriting a lot of risk through the execution of the project," he says. Amara Raja plans to invest ₹9,500 crore to set up 16 GWh cell capacity by FY31.

### WEAPONISING TECH

Despite the recent thaw in India-China ties, several Indian companies, including Reliance Industries and JSW Group, are struggling to acquire the technical know-how for cell manufacturing. At the recently concluded BRICS Summit in New Delhi, Prime Minister Narendra Modi said weaponisation of technology and critical minerals can hinder shared progress of the bloc.

Reliance Industries, through its subsidiaries Reliance New Energy Battery Ltd and Reliance New Energy Battery Storage Ltd, was awarded 15 GWh capacity under the ACC PLI Scheme. The conglomerate is yet to install any capacity. Access to battery technology is being "weaponised", Anant Ambani, Executive Director of Reliance Industries, said at the company's 49th Annual General Meeting. "In a world where supply chains are being contested and access to technology is being weaponised, building world-scale battery manufacturing capacity in India is not merely a business decision. It is a strategic imperative for national resilience," Ambani said, assuring that Reliance New Energy plans to commission the first phase of its 40 GWh battery cell gigafactory this year as all the equipment has been delivered.

Earlier in August, JSW Group scion Parth Jindal said China is guarding LFP cell making technology "like a weapon", pushing the steel-to-automobiles major to put its plans to make battery cells in India on hold. "We are keen to make cells in India. LFP cell technology is not available for us. The hunt is on for a technology tie-up. Hence, that project is currently on hold," says Jindal. "LFP cells come from China, and they are not giving us the technology. We are ready to invest and make it in India," he says.

India wants to build batteries at home and reduce its dependence on China, but Indian companies still need Chinese technology to make that happen. Developing homegrown alternatives, meanwhile, will require significant investment and sustained government support. BT

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**MAKING SENSE OF INDIA**

# RE-MAKING OF AN INDUSTRY

**India's defence manufacturing ecosystem is undergoing a major transformation from import dependence towards self-reliance. Profits are catching up fast and it is no longer just a growth story**

BY **RICHA SHARMA**

**IN A WORLD** shaped by geopolitical volatility and technological disruption, India has significantly expanded defence spending to strengthen military preparedness and indigenous manufacturing. The sector has been steadily transformed through innovation, technological advancement, and strategic reforms. The first phase of India's defence manufacturing was largely about import substitution. The next phase will be about technology ownership and global competitiveness.

The recent *BT*-Deloitte India survey of India's manufacturing companies bears witness to the trend. Aerospace & defence sector companies saw their combined profitability more than triple over five years. The 26 listed companies in the sector reported a total profit of ₹23,111 crore in FY26, up from ₹7,507 crore in FY21, a rise of 208%.

The profitability of several Indian aerospace, defence and allied companies has surged sharply over the past five years, according to ACE Equity data. Zen Technologies recorded the highest growth, with its net profit jumping 7,757% to ₹217.9 crore in FY26, from just ₹2.8 crore in FY21. Jaykay Enterprises saw a 727% increase, while Astra Microwave Products and Solar Industries India posted profit growth of 569% and 503%, respectively. Among larger defence companies, Hindustan Aeronautics (HAL) reported a net profit of ₹9,115.5 crore in FY26, up 181% from FY21, while Bharat Electronics (BEL) saw its profit rise 191% to ₹6,023.5 crore. Mazagon Dock Shipbuilders reported a 434% increase to ₹2,421.9 crore.

## THE TRIGGERS

In September, India's Defence Acquisition Council cleared proposals worth ₹1.1 lakh crore, with 98% to be sourced from Indian companies. From FY25 to FY27 so far, these approvals add up to a massive ₹13 lakh crore. Alaric

Positive indigenisation lists and emergency procurement powers have made indigenous systems the first choice for the forces

PHOTO BY **HARDIK CHHABRA**

## DEFENCE OVERVIEW

**The defence budget was ₹7.85 lakh crore in FY27**

**Defence capex was ₹2.19 lakh crore in FY27**

**The indigenous defence production stood at ₹1.78 lakh crore in FY26**

**The sector exported goods worth ₹38,424 crore in FY26**

**The sector saw FDI of ₹6,670 crore in FY26**

**SOURCE** DEFENCE MINISTRY

Diniz, Partner, Strategy, Risk & Transactions, Deloitte India, says the next phase of investment is expected to prioritise sustainable value creation, long-term capability development and globally competitive defence technology. Operation Sindoor highlighted the growing importance of dual-use technologies such as unmanned aerial vehicles (UAVs), AI-enabled command and control systems, satellites and AI-powered radars.

"With the sector transitioning into a more mature, technology-driven ecosystem, the future deal landscape in India's defence industry is expected to see increased transaction flow, particularly in dual-use technology segments and export-ready businesses with scalable models," says Diniz.

Schemes such as Innovations for Defence Excellence (iDEX) and ADITI (Acing Development of Innovative Technologies with iDEX) are actively engaging industries, start-ups, individual innovators, research institutions and academia to develop defence technologies. Both schemes have been introduced in the past five years.

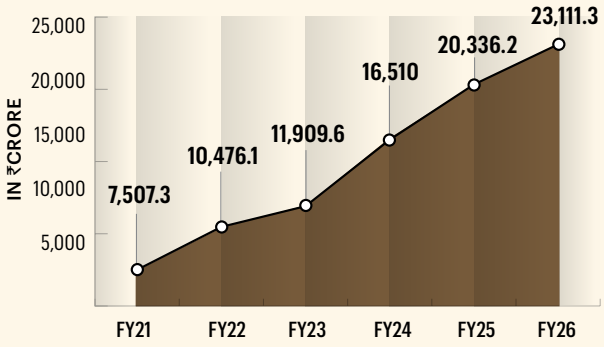
By March 2026, iDEX had engaged 676 start-ups, MSMEs, and innovators; 551 design and development contracts had been signed. To strengthen this ecosystem, the DRDO has opened its advanced testing infrastructure to private industry through the Defence Testing Portal.

### DEFENCE PUSH

India has been strengthening military preparedness while expanding domestic manufacturing and research capabilities. Overall, defence production has grown at 13% a year, while exports have been rising 40% a year for the past 10 years. The defence budget has increased from ₹2.53 lakh crore in FY14 to ₹7.85 lakh crore in FY27. More importantly for listed manufacturers, capital acquisition spending has zoomed and, at ₹1.85 lakh crore, is now 24% higher than last year's budget estimate.

## RAINING PROFIT

- PAT of 26 listed firms has risen 208% since FY21




**SOURCE** DELOITTE INDIA, ACE EQUITY

## THE TOP GUNS

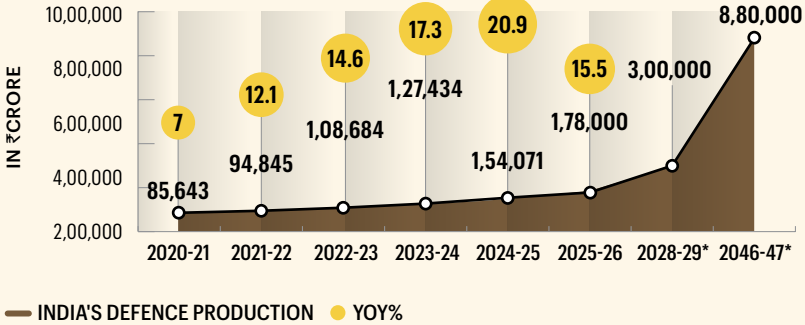
- Zen, Jaykay have been the fastest-growing firms

| Company | FY26 PAT | FY21 PAT | 5-year profit growth |
|---|---|---|---|
| Hindustan Aeronautics | 9,115.5 | 3,245.6 | 181% |
| Bharat Electronics | 6,023.5 | 2,069.3 | 191% |
| Mazagon Dock Shipbuilders | 2,421.9 | 453.5 | 434% |
| Solar Industries India | 1,736.6 | 288.1 | 503% |
| Zen Technologies | 217.9 | 2.8 | 7,757% |
| Jaykay Enterprises | 215.3 | 26.0 | 727% |
| Astra Microwave Products | 193.0 | 28.9 | 569% |

PAT IN ₹ CR **SOURCE** ACE EQUITY

## INDIA'S DEFENCE PRODUCTION

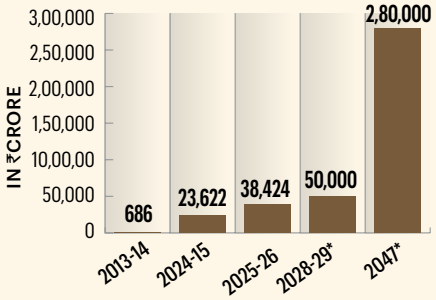
- Defence public sector undertakings and other PSUs contributed around 76% of total production in FY26
- The private sector's share was 24%, indicating greater industry participation




**NOTE** * PROJECTED **SOURCE** PIB/DELOITTE

## DEFENCE EXPORTS

- India is emerging as a global defence exporter
- Indian defence products are now exported to over 80 countries




**NOTE** * TARGET **SOURCE** MINISTRY OF DEFENCE/DELOITTE

India has moved decisively from importing complete systems towards designing, developing and manufacturing an increasing proportion of defence equipment domestically. "Defence production reached about ₹1.78 lakh crore in FY26, while exports reached ₹38,424 crore. Importantly, the private sector's contribution to defence production has been increasing and reached about 24% in FY26. For companies that invested in technology and qualification capabilities over many years, this is now translating into larger production opportunities," says M. V. Reddy, MD, Astra Microwave Products.

The company designs and manufactures advanced microwave and radio frequency (RF) systems that power satellites, radars, missiles, and communication networks. It recently bagged a ₹2,205 crore order from HAL for the Uttam AESA Radar programme.

"As indigenous content and technology contribution of Indian companies increase, so does the value addition. That is the more sustainable driver of profitability. The larger change is movement of Indian companies up the value chain from build-to-print manufacturing to build-to-specification, from components to subsystems, and increasingly from subsystems to complete systems," he says.

### ORDER BOOK

The order books of companies are surging, especially for those in modern warfare drones and AI-enabled command and control systems. HAL, the largest player, has an order book of ₹2.5 lakh crore, while Bharat Electronics' order book is ₹72, 258 crore. For defence companies, though, order books don't automatically translate into balance sheet performance as these projects

have long timelines, especially in areas like aircraft and submarines, where delays are routine.

Supply chains are also complex, so revenue depends on production capacity, component availability, testing, acceptance, and delivery schedules. Still, rising order book numbers are a magnet for investors, and that's driving defence company valuations higher. The Nifty India Defence index is up roughly 22% in 2026, even as the Nifty has dipped around 9%.

Bodhisattwa Sanghapriya, Founder & CEO, IG Defence, says the company's revenue grew 3.7x in FY26 due to a structural shift in policy. "Positive indigenisation lists, iDEX, the Buy Indian-IDDM preference, and emergency procurement powers have made indigenous systems the first choice for the forces. Operation Sindoor showed how central drones and counter-drone systems are to modern warfare, and orders moved from trial quantities to volumes," says Sanghapriya. IDDM stands for Indigenously Designed, Developed, and Manufactured.

Restrictions on foreign-origin components in military drones have also rewarded companies that invested early in indigenous supply chains. "IG Defence owns its design and software rather than assembling imported kits. Our margins widen as volumes grow. At the centre of the growth will be our AI-powered battle management platform and one-way long-range strike drone KAL," he says.

Astra Microwave, with a standalone order book of ₹2,156 crore, is progressively moving towards higher-value subsystems and systems while continuing to deepen its core expertise in RF, microwave and strategic electronics.

"Radar will remain a major area for us. We see opportunities across airborne, ground-based, naval and specialised radar programmes. Electronic warfare, seekers, data links, communication systems, counter-UAS (unmanned aerial system) technologies and other advanced RF applications will also become increasingly important," says Reddy.

**"I see the next decade being defined by three things: greater indigenous design content, Indian companies taking responsibility for larger & more complex systems, and a stronger export orientation"**

**MV REDDY**
MD, Astra Microwave

**"Policy is a tailwind here too: export promotion, defence lines of credit, and defence attachés actively pitching Indian systems have opened doors that didn't exist five years ago"**

**BODHISATTWA SANGHAPRIYA**
Founder & CEO, IG Defence

## FUTURE GROWTH

Private sector companies account for nearly half of India's defence exports, even though they are only about 24% of domestic defence production. According to Astra Microwave, the second important growth engine will be exports, and India cannot build a globally competitive defence industrial base purely on domestic demand.

"Companies need sufficient domestic scale to develop and qualify products, and thereafter they must compete internationally. Therefore, I see the next decade being defined by three things: greater indigenous design content, Indian companies taking responsibility for larger and more complex systems, and a stronger export orientation," says Reddy.

"Policy is a tailwind here too: export promotion, defence lines of credit, and defence attachés actively pitching Indian systems have opened doors that didn't exist five years ago. The next phase depends on export approvals keeping pace with that demand from the international market," says Sanghapriya.

Reddy says the next stage should focus increasingly on predictability, technology development and procurement speed.

"Industry will invest much more aggressively in manufacturing capacity, R&D and specialised infrastructure when there is reasonable visibility of procurement over five to ten years," he adds. BT

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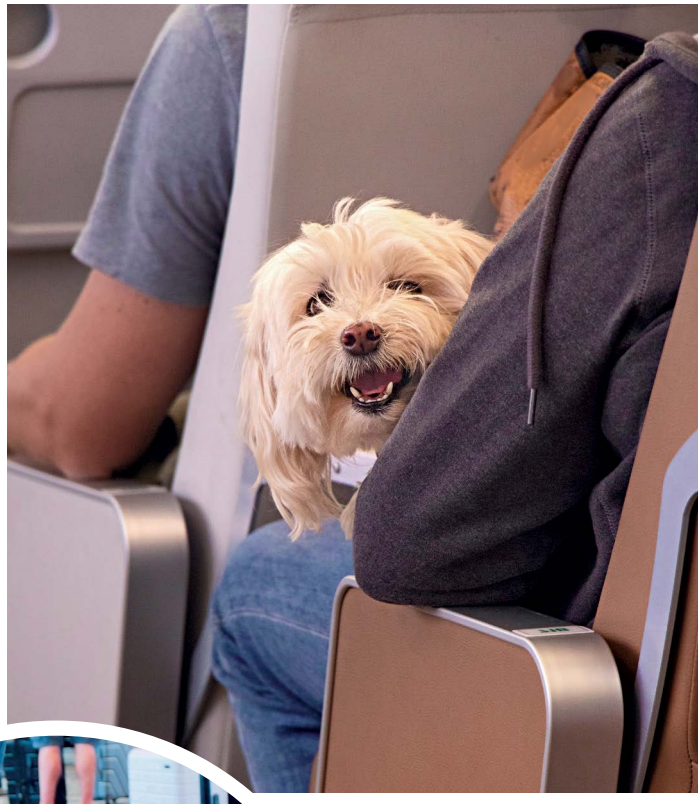
# Have Paws, Will Travel

MORE PET PARENTS ARE FLYING WITH THEIR COMPANIONS FOR THE 'PURR'FECT VACATION

BY **SMITA TRIPATHI**



**EARLIER THIS YEAR,** when Romeo boarded his 26th flight between Mumbai and Goa, he was a bit of an unusual frequent flyer. The miniature poodle was travelling on Akasa Air with his owner for a weekend getaway.

Romeo is part of a small but rapidly growing population of Indian pets taking to the skies. Akasa Air says it has flown more than 10,000 pets since launching Pets on Akasa in November 2022; 93% travelled in the cabin, and more than a quarter of pet parents used the service more than once. The festive quarter from October to December accounts for almost 35% of its annual pet journeys. Air India, meanwhile, carried around 1,700 pets in the first six months of 2026. What was once an eccentric request at an airline counter is fast becoming a recognisable Indian travel category.

**Among major airlines, Air India and Akasa Air are the two practical options for passengers who want a structured service for travelling with their pets**

"Pets are no longer necessarily left behind with a neighbour or at a boarding kennel. They are increasingly coming along for the holiday and, of course, the relocation," says New Delhi-based Sanjana Kapoor, who flies frequently with her Shih Tzu, Coco. Of course, she says, it's because her pet is small enough to fit in the cabin that she prefers to take him along rather than leave him in a boarding. "If my dog was larger, maybe I wouldn't want to put him in the hold of an aircraft on vacation; in that case, I would have left him at a kennel," she adds.

Among major airlines, Air India and Akasa Air are the two practical options for passengers who want a structured service for travelling with their pets. But here, too, it's only dogs and cats. You cannot fly with your pet rabbits, for instance. IndiGo does not permit regular pets on its aircraft and makes an exception for guide dogs accompanying blind passengers.

Air India's revamped Paws on Board programme allows domesticated cats and dogs weighing up to 10 kg, including the carrier, in Economy class. The pet must stay inside a soft-sided, leak-proof, ventilated carrier no larger than 17x10x9 inches, placed beneath the seat. Cabin pets must be at least eight weeks old, and bookings should be made at least 48 hours before departure. Pets are seated with their humans in aisle seats in the last five rows. Pregnant, sedated, critically

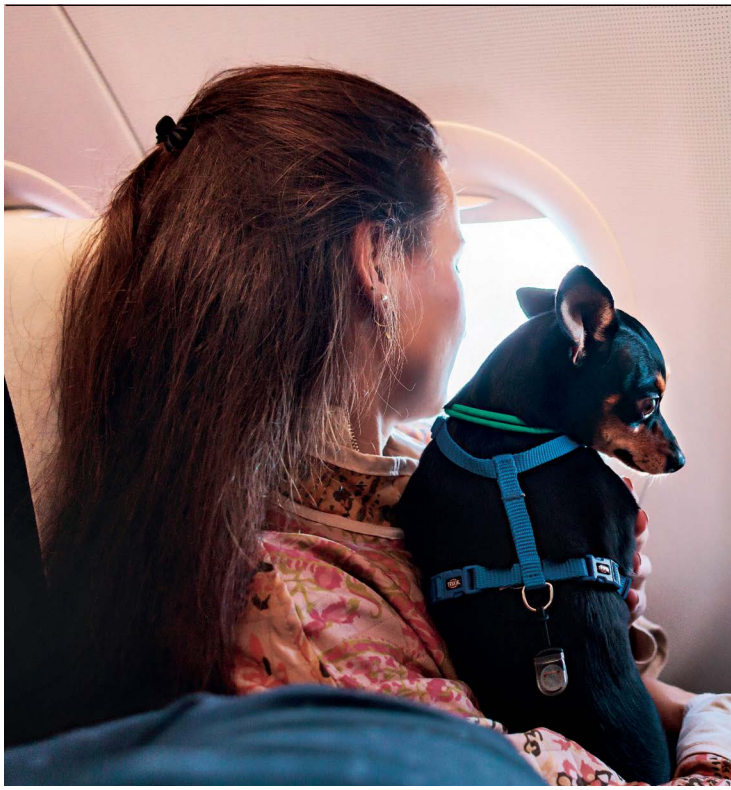
ill or medically unfit animals are not accepted.

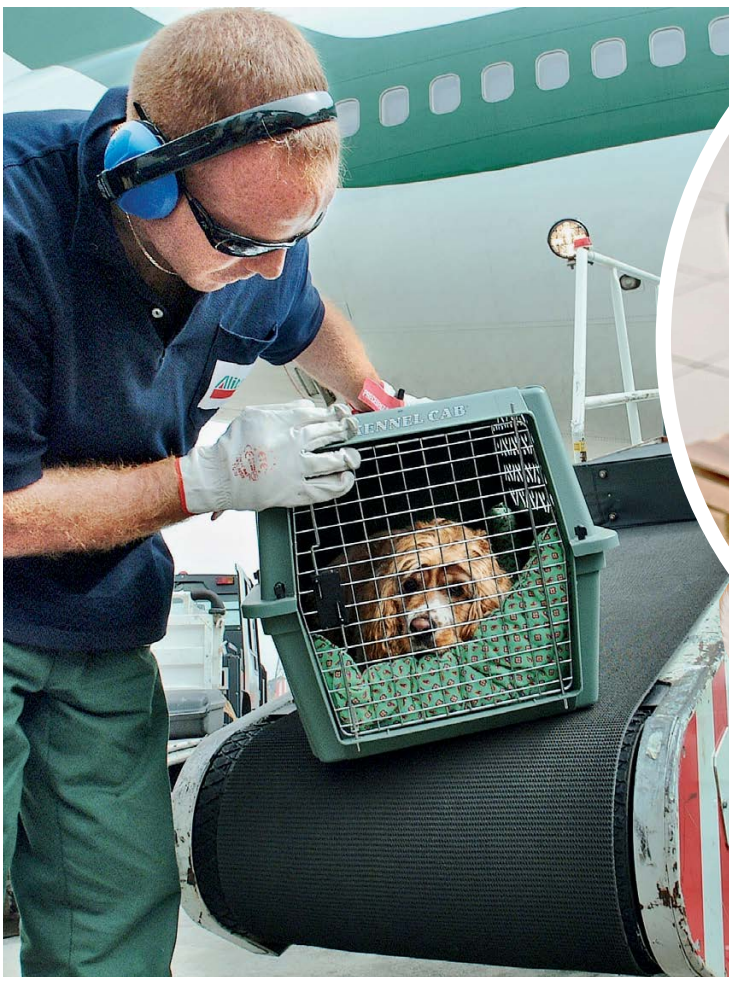
The domestic price is ₹7,500 for a pet in the cabin. Between 10 kg and 32 kg, including the crate, the animal travels as checked baggage in an IATA-compliant hard kennel for ₹16,000. Above 32 kg, the booking moves to cargo.

Snub-nosed breeds, including Shih Tzus and pugs, are not accepted in the hold because of their greater respiratory risk. So they can travel on Air India only if they qualify for the cabin.

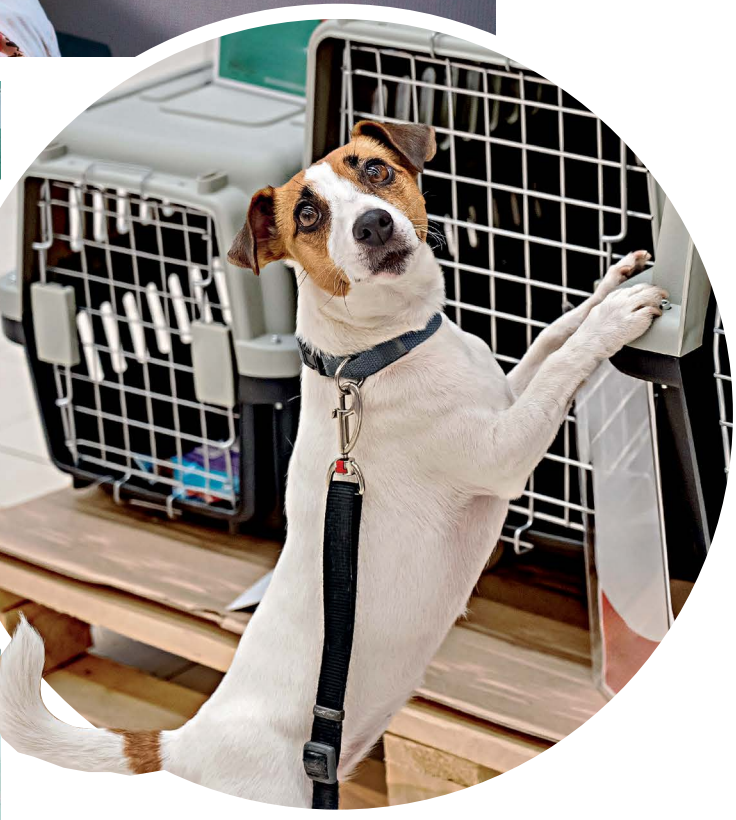
Internationally, Air India's in-cabin charges generally start at $140, with some sectors priced higher, while checked-baggage pets cost $350. Destination rules matter as much as airline rules, and travellers need to check both the carrier's policy and the entry regulations of the destination country. For instance, the UK does not allow pets in cabin, which becomes a particular problem if you have a snub-nosed dog as the carrier does not fly it in the hold!

Akasa's rules are similar. Cats and dogs must be at least three months old. Up to 10 kg, including the carrier, they may travel in the cabin; heavier animals can travel in the hold. Akasa permits up to three pets on a flight and offers pet travellers priority check-in, priority boarding, and a pre-booked window seat. Published fares are ₹7,500 for in-cabin travel and about ₹15,000





**At Air India, cats and dogs must be at least three months old. Up to 10 kg, including the carrier, they may travel in the cabin; heavier animals can travel in the hold**



for a pet travelling in the hold.

Around 60% of Akasa's pet passengers have been dogs and the rest cats, while the Bengaluru-Delhi-Bengaluru sector accounts for 13.66% of its pet traffic.

The Amadeus Travel Trends Report 2026 refers to it as the "Pawprint Economy", pointing to forecasts that the global pet industry could be worth $500 billion by 2030. Hotels have been quicker to respond: pet-friendly resorts, restaurants, cafés, and even co-working spaces are increasingly commonplace. Aviation is catching up.

Bangkok-based Divya Dugar is a pet parent who has spent years travelling with her three dogs across India by rail and road. She chronicled these experiences in her book *Chaos in a Coupe*. Four years ago, she flew them from Delhi to Paris; and then recently, she travelled with them from Paris to Bangkok.

Her first rule is to begin with the country, not the airline. "Each country has different rules of importing and exporting a dog," she says. Rabies requirements, booster vaccinations, and other documentation can take considerable time. Only once those are in place should travellers begin deciding which airline and route to use.

Dugar recommends using a pet-relocation agency such as PetFly selectively. "Outsource the complicated documentation and airport handling, but not necessarily the entire move." Such agencies can also accompany an animal through parts of the airport inaccessible to its owner and remain with it until it is handed over for the flight.

She favours direct flights whenever possible, choosing Air France for Delhi-Paris for precisely that reason. For an India to the US journey, she recommends Lufthansa via Munich because of the pet-care facilities available during the connection.

Meanwhile, at the luxury end, a new class of pet-first aviation has emerged overseas. Most are public or private charter operators rather than conventional scheduled airlines.

BARK Air was created around the idea of offering dog-first charter flights. It sells a dog-and-human ticket on routes such as New York-San Francisco for around $7,450 and New York-London for roughly $9,900 on selected 2026 flights. K9 Jets, similarly, operates shared private-jet flights for people travelling with animals, including long-haul international sectors. Other operators, including RetrievAir in the US and SingaPaw Air in Asia, have entered the same niche market.

Looks like the 'paw'fect vacation is one where your pooch accompanies you, and airlines are finally coming to your rescue. BT

# BEST Management Advice

**RAJESH NAMBIAR,** PRESIDENT, NASSCOM AND FORMER CHAIRMAN & MD, COGNIZANT

## "LEADERS NEED TO GET COMFORTABLE MAKING DECISIONS WITH INCOMPLETE INFORMATION"

**Q What is the biggest leadership misconception about managing people in an AI-first workplace?**
It's the assumption that AI-first means managing people less. In reality, as AI takes on more routine work, human judgement becomes more important. This requires more coaching and context-setting from a manager. Secondly, leaders cannot simply introduce AI tools and expect productivity to follow. They need to rethink roles, workflows, decision rights, and how people continue to grow.

**Q As AI changes the nature of work, what should leaders be looking for in people that a resume can no longer tell them?**
Businesses are increasingly moving towards skills-based hiring rather than relying primarily on experience. Traits such as learning agility, curiosity, judgement, problem-framing, the ability to work across disciplines, comfort with ambiguity and, importantly, the ability to unlearn and relearn quickly are becoming increasingly valuable.

**Q When the pace of change is faster than an organisation's ability to adapt, what should a CEO do first?**
The first instinct today is often to jump on the AI bandwagon and start implementing quickly. The priority for a CEO should be to identify where the biggest shifts are happening, what they mean for the business model, capabilities and people, and then align the organisation around a few clear priorities.

**Leaders cannot simply introduce AI tools and expect productivity to follow. They need to rethink roles and workflows**

**Q In an era of constant uncertainty, what is the one leadership capability CEOs need to build?**
It has to be adaptability. Knowing which decisions are reversible, which are not, and responding accordingly. That means shorter feedback loops, pushing decision-making closer to where the information sits, and creating a culture where teams can course-correct early.

**Q When leaders can no longer predict what the next five years will look like, how should they lead differently?**
The reality is that no one can crystal-ball gaze with certainty anymore. Leaders also need to get comfortable making decisions with incomplete information, revisiting assumptions more frequently and creating shorter planning and feedback cycles. However, it also means being clear on the few things that should remain constant such as purpose, values and strategic intent. BT

*Nasscom is a trade association group, and Cognizant is an American-domiciled Indian multinational IT services and outsourcing company*



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